

Seminars

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Abstract

This semester, I will just record all seminars I go to in an uncategorized manner. I will try to record the date, the speaker, and which seminar it was to maintain some semblance of organization.

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1 January 30: Multiplicity Formulae for Spherical Varieties

This talk was given by Toan Pham at Johns Hopkins University for the student automorphic representations seminar.

1.1 The Theorems

Fix a homogeneous spherical variety $X = G/H$ over a local field F . Also, choose a character $\chi: H(F) \rightarrow \mathbb{C}^\times$ and an irreducible representation π of $G(F)$. Then the local relative Langlands program is interested in the multiplicity

$$m(\pi, \chi) := \dim \operatorname{Hom}_{H(F)}(\pi, \chi).$$

Note that

$$m(\pi, \chi) = \dim \operatorname{Hom}_{G(F)}(\pi, C^\infty(X(F), \chi)).$$

Here are some starting examples.

Example 1 (Whittaker). An interesting case is when $H \subseteq G$ is a unipotent subgroup. For example, one can take the unipotent radical U of the Borel subgroup of GL_2 . Then we can take χ to be lifted from any additive character $F \rightarrow \mathbb{C}^\times$. One can generalize this example to work from any $\mathfrak{sl}_2$ -triple, and it produces Whittaker models of π in $C^\infty(X(F), \chi)$.

Example 2 (Gan–Gross–Prasad). Fix a quadratic extension E/F . Then one can take $X = \operatorname{SO}_n \backslash (\operatorname{SO}_n \times \operatorname{SO}_{n+1})$ or $X = \operatorname{U}_n \backslash (\operatorname{U}_n \times \operatorname{U}_{n+1})$. In general, there are more examples arising from so-called “GGP triples” (G, H, χ) .

Remark 3. It turns out that $m(\pi, \chi) \leq 1$ in Examples 1 and 2, but this is not always true.

Nonetheless, it becomes interesting to discover when $m(\pi, \chi) \geq 1$. Here are some answers.

Theorem 4. Fix everything as in Examples 1 and 2 and choose a Langlands parameter $\varphi: W_F \rightarrow {}^L G$. Then the Langlands packet Π_φ contains exactly one π for which $m(\pi, \chi) \neq 0$.

Theorem 5. Fix everything as in Example 2, and choose a Langlands parameter $\varphi: W_F \rightarrow {}^L G$. Then $\pi \in \Pi_\varphi$ has $m(\pi, \chi) \neq 0$ if and only if it maps to a specified unitary character.

Our method will be the so-called “local trace formulae,” developed originally by Waldspurger.

1.2 Local Trace Formulae

For the rest of the talk, we work in the context of Example 2. The type of result we are trying to prove rewrites $m(\pi, \chi)$ in terms of a sum of orbital integrals.

Example 6. For finite groups G , we can use the character Θ_π of π to see that

$$m(\pi, \chi) = \sum_{\text{conj. } [x] \subseteq H} \frac{1}{\#C_H(x)} \cdot \Theta_\pi(x) \overline{\chi}(h).$$

This right-hand side can be viewed as some twisted sum of sizes of conjugacy classes.

Our method to prove such formulae will rest on local trace formulae, which basically amount to finding two ways of expressing the trace of some $f \in C_c^\infty(G)$ acting on $L^p(H(F) \backslash G(F), \chi)$. On one hand, we may write

$$\begin{aligned} (Rf \cdot \varphi)(x) &= \int_{G(F)} f(g) \varphi(xg) dg \\ &= \int_{H(F) \backslash G(F)} K_f(x, y) \varphi(y) dy, \end{aligned}$$

where $K_f(x, y) = \int_{H(F)} f(x^{-1}hy) \chi(h) dh$. We morally then expect the trace of Rf to be the sum of K_f along the diagonal, as one finds with finite groups.

Proposition 7. Fix everything as above. Say that $f \in C_c^\infty(G(F))$ is strongly cuspidal if and only if

$$\int_{U(F)} f(um) du = 0$$

for any parabolic P with Levi decomposition $P = MU$. If f is strongly cuspidal, then the trace of Rf converges absolutely.

The local trace formula now amounts to expressing $J(f)$ either via a spectral or a geometric expansion.

- The spectral expansion is

$$J_{\text{spec}}(f) := \int_{\mathfrak{X}(G)} D(\pi) \hat{\Theta}_f(\pi) m(\pi) d\pi.$$

Here, $\mathfrak{X}(G)$ consists of the space of tempered representations (i.e., found in $L^2(G)$) arising from parabolic induction of elliptic representations. Then Θ_f is a weighted orbital integral of f , and $\hat{\Theta}_f$ is its Fourier transform. The $D(\pi)$ is unknown, but we have been reassured that it is not important.

The idea to show that $J(f) = J_{\text{spec}}(f)$ is to use the Plancherel formula

$$\langle f_1, f_2 \rangle = \int_{\hat{G}} J_\pi(f_1 \otimes f_2) d_X \pi,$$

where J_π is the natural composite

$$C_c^\infty(X \times X) \rightarrow \pi \otimes \tilde{\pi} \rightarrow \mathbb{C}.$$

It turns out that J_π is non-vanishing if and only if $m(\pi, \chi) \neq 0$. Yiannis claims that the forward direction is easy.

- The geometric expansion associates a quasicharacter $\theta_f: G_{\text{reg}}(F) \rightarrow \mathbb{C}$ defined on the regular semi-simple locus, and then we have

$$m_{\text{geom}}(\theta) := \lim_{s \rightarrow 0} \int_{\Gamma(G, H)} D^G(x)^{1/2} c_\theta(x) \Delta^{s-1/2} dx.$$

Here, $\Gamma(G, H)$ is the space of semisimple conjugacy classes of $G(F)$ which are represented by an element in $H(F)$, and c_θ is some extension of θ to the semisimple locus $G_{\text{ss}}(F)$. Our geometric expansion is then $J_{\text{geom}}(f) := m_{\text{geom}}(\Theta_f)$.

It now turns out that

$$J_{\text{spec}}(f) = J(f) = J_{\text{geom}}(f),$$

though both of these equalities are theorems. Though it requires an argument, this turns out to be equivalent to a multiplicity formula $m_{\text{geom}}(\pi) = m(\pi)$.

Conjecture 8. One has $m_{\text{geom}}(\pi) = m(\pi)$ for any spherical X .

2 January 31: Finiteness Theorems in Arithmetic Statistics

This talk was given by Fatemehzahra Janbazi for the JHUnion number theory days.

2.1 The Birch–Merriman Theorems

For today, K will be a number field, and S is a finite set of places, including the infinite ones.

Notation 9. For a ring R , we define $U_n(R)$ to be the set of binary forms f which are homogeneous of degree n .

Note that GL_2 naturally acts on U_n .

Definition 10. There is also a discriminant Δ on U_n . We say that $f \in U_n(\mathbb{Q}_p)$ has *good reduction* if and only if $\Delta(f) \in \mathcal{O}_p^\times$.

Remark 11. The discriminant satisfies

$$\Delta(\gamma \cdot f) = \det \gamma^{n(n-1)} \Delta(f).$$

We then have the following theorems.

Theorem 12 (Birch–Merriman). The set $U_n(\mathcal{O}_{K,S})$ with good reduction outside S breaks into infinitely many $\mathrm{GL}_2(\mathcal{O}_{K,S})$ -orbits.

Theorem 13. For any nonzero $D \in \mathcal{O}_K$, the set of forms in $U_n(\mathcal{O}_K)$ with discriminant D has finitely many equivalence classes for the action of $\mathrm{GL}_2(\mathcal{O}_K)$.

We are interested in analogous results for ternary forms.

Definition 14. For a ring R , we define $V_n(R)$ to be the set of homogeneous ternary forms of degree n . There is also a discriminant Δ of ternary forms, and we say that f has *good reduction* at p if and only if $\Delta(f) \in \mathcal{O}_p^\times$.

Remark 15. As usual, there is a natural GL_3 -action on V_n , and it satisfies

$$\Delta(\gamma \cdot f) = \det \gamma^{n(n-1)} \Delta(f).$$

Here are our main theorems.

Theorem 16. The set of elements in $V_n(\mathcal{O}_{K,S})$ with good reduction outside S breaks up into finitely many $\mathrm{GL}_3(\mathcal{O}_{K,S})$ -orbits.

Theorem 17. For any nonzero $D \in \mathcal{O}_K$, the set of forms in $V_n(\mathcal{O}_K)$ with discriminant D has finitely many equivalence classes for the action of $\mathrm{GL}_3(\mathcal{O}_K)$.

Here is another analogue.

Definition 18. For a ring R , we define

$$V_{2,2}(R) := \frac{R[x_1, x_2, x_3]_2 \otimes_R R[z_1, z_2, z_3]_2}{(\sum_i x_i z_i) \cap R[x_1, x_2, x_3]_2 \otimes_R R[z_1, z_2, z_3]_2}.$$

Remark 19. There is now a natural action of GL_3 on $V_{2,2}$ given by $(\gamma \cdot f)(x, z) := f(x\gamma, z\delta\gamma)$.

Definition 20. A $K3$ surface S is *generic* if and only if it is smooth and cover $\mathbb{P}^2$ via a natural projection from a flag variety. Some $f \in V_{2,2}$ has *good reduction* if and only if it products a generic $K3$ surface.

Theorem 21. The set of elements in $V_{2,2}(\mathcal{O}_{K,S})$ with good reduction outside S and Picard rank 2 breaks up into finitely many orbits for the action of $\mathrm{GL}_2(\mathcal{O}_{K,S})$.

Remark 22. Let's sketch the idea of Theorem 16. Let $T_n(K, S)$ be the set of f with discriminant in $\mathcal{O}_{K,S}^\times$. Each such f cuts out a curve C_f which has good reduction outside S , but Faltings's theorem implies that there are only finitely many such curves, which can be enumerated by some invariants $I_1(f), \dots, I_k(f)$. One can now chase around to prove the theorem.

Here is an application.

Example 23. Let $W_n(R)$ be the set of pairs (F, G) of elements in V_n . This has a natural action of $\mathrm{GL}_2 \times \mathrm{GL}_3$ given by

$$(\gamma_2, \gamma_3) \cdot (F, G) = (\gamma_3 F, \gamma_3 G) \gamma_2^T.$$

One can prove a similar result about finiteness of elements in $W_n(\mathcal{O}_K)$ with fixed discriminant. The proof uses both the finitenesses of U_n and V_n .

2.2 Geometric Reformulation

We will want more notation.

Notation 24. Let $U_n(K, S)$ be the set of elements in $U_n(\mathcal{O}_{K,S})$ and $\Delta(f) \in \mathcal{O}_{K,S}^\times$ and which split over K . We also let $\Omega_n(\mathbb{P}^1; S)$ to be the set of n points with good reduction outside S .

The point of introducing these sets is that there is a natural map $U_n(K, S) \rightarrow \Omega_n(\mathbb{P}^1; S)$, preserving the GL_2 -action, and finiteness theorems can be moved from one side to the other.

Thus, we see that we may be interested in analogous questions for curves other than $\mathbb{P}^1$.

Theorem 25. Fix a smooth projective curve C over K with good reduction at S . One can define $\Omega_n(C; S)$ in the same way. For any $n \geq 1$, the set of classes in $\Omega_n(C; S)$ up to the action by $\mathrm{Aut}_{\mathcal{O}_{K,S}}(C)$ is finite.

Example 26. For $C = \mathbb{P}^1$ and $n = 1$, finiteness is equivalent to finiteness of the class group of $\mathcal{O}_{K,S}$.

Example 27. If $C = \mathbb{P}^1$ and $n \geq 3$, we recover the Birch–Merriman theorems. Alternatively, one can move some points to $\{0, 1, \infty\}$, and then the remaining points are seen to be finite by the unit equation.

Example 28. If C has genus 1, then we have an elliptic curve; moving some point to the origin, we recover Siegel's theorem.

Example 29. Lastly, if C has genus 2, then at $n = 1$, we recover Faltings's theorem.

The proof of Theorem 25 more or less upgrades the above examples by doing some Galois cohomology.

3 January 31: Infinitely Many Supersingular Primes

This talk as given by Fangu Chen at Johns Hopkins University for the JHUnior number theory days.

3.1 Expectations for Supersingular Primes

Here are our definitions for elliptic curves.

Definition 30 (ordinary, supersingular). Fix an elliptic curve E over a finite field of characteristic p . Then E is *ordinary* if and only if $E[p] \cong \mathbb{Z}/p\mathbb{Z}$; it is *supersingular* if and only if $E[p] = 0$.

Remark 31. There are equivalent formulations based on the Frobenius action or a calculation of the endomorphism ring.

Remark 32. It turns out that every elliptic curve is ordinary or supersingular.

Question 33. Fix an elliptic curve E over $\mathbb{Q}$. For how many primes p are there with $E_{\mathbb{F}_p}$ admit supersingular reduction?

Here is a heuristic argument: one knows $|\mathrm{tr} \mathrm{Frob}_p| \leq 2\sqrt{p}$, so one can expect that p is supersingular with probability $\sim 1/\sqrt{p}$.

Conjecture 34 (Lang–Trotter). Fix an elliptic curve E over $\mathbb{Q}$ without complex multiplication. Then the number of primes $p < X$ for which $E_{\mathbb{F}_p}$ has supersingular reduction is asymptotic to $c_E \sqrt{x}/\log x$ for some constant c_E .

For example, Serre has proved that the set of supersingular primes has density 0; Elkies has proved that there are infinitely many.

We are interested in analogous questions for abelian varieties.

Remark 35. In the case of complex multiplication, it is not so hard to study supersingularity via the Newton polygon and the Shimura–Taniyama formula.

One expects that the non-ordinary abelian varieties (measured via the Newton polygon) should have density zero. Recent work has shown that the supersingular primes has density zero. We are interested in showing infinitude.

3.2 The Main Theorem

For our talk, we fix a totally real number field F as our base and a quaternion algebra B over F which is split at exactly one real place. Then let V be the subset of B with reduced trace equal to zero, and let Q_F be the reduced norm of V .

We will require all these to admit F -multiplication. In the case $[F : \mathbb{Q}] = 3$, Mumford has defined some abelian fourfolds with endomorphism algebra $\mathbb{Z}$ but admitting some extra Hodge classes in $H^4(A; \mathbb{Q})$. There is a Kuga–Satake construction which takes $K3$ surfaces to abelian varieties; it turns out that A^{64} is isogenous to an abelian variety coming from a $K3$ surface.

Remark 36. We have recently computed some of these fourfolds as coming from some explicit Jacobians.

In the case that F has narrow class number 1, the Shimura curve parameterizing Mumford’s abelian fourfolds admits a canonical model Sh .

Theorem 37. Assume that Sh is isomorphic to $\mathbb{P}_F^1$ as well as some technical conditions. If A is in Sh and has field of moduli F , then A admits infinitely many primes of supersingular reduction.

Remark 38. The extra conditions are as follows.

- For F : we want 2 to be inert in F and $F(\sqrt{-\varepsilon})$ to have class number 1 for all $\varepsilon \in \mathcal{O}_F^\times$ which is negative at exactly one real place F . This condition simplifies calculations at the archimedean place; some case-by-case analysis could relax this condition.
- For B : there are also some ramification conditions.

There are many examples of such F ; for example, there are cubic fields.

Remark 39. Such extensions of Elkies’s theorem have a long history. For example, there has been a lot of work on abelian surfaces with some quaternionic multiplication. Notably, all previous work has been done with Shimura curves of PEL type, isomorphic to $\mathbb{P}^1$. This work is notable because the underlying Shimura curve merely has Hodge type.

3.3 Proof Idea

Let’s review the proof idea for elliptic curves, due to Elkies.

Theorem 40 (Elkies). Fix an elliptic curve E over $\mathbb{Q}$. Then E admits supersingular reduction at infinitely many primes p .

Proof. Suppose E has finitely many supersingular primes in a set S .

1. We start with some CM points. Fix CM elliptic curves $\{E_1, \dots, E_h\}$ with $\mathbb{Z}[\sqrt{-\ell}] \subseteq \text{End}_{\overline{\mathbb{Q}}}(E_\bullet)$, where $\ell \equiv -1 \pmod{4}$ is prime. Then we set

$$P_\ell(x) = \prod_i (x - j(E_i)),$$

and it turns out to be in $\mathbb{Z}[x]$. We would like to construct ℓ for which $\nu_p(P_\ell(j(E))) > 0$ (i.e., E has supersingular reduction) and $\left(\frac{-\ell}{p}\right) \neq 1$ (i.e., $E_i \pmod{p}$ has supersingular reduction). Accordingly, let N be the numerator of $P_\ell(j(E))$, and then we have two cases.

- We may want $\ell \mid N$, in which case we choose $p = \ell$.
- We may want $\left(\frac{-\ell}{N}\right) = -1$, in which case we can still find some p .
- Lastly, we may want $p \notin S$, given from $\left(\frac{-\ell}{q}\right) = 1$ for all $q \in S$.

We will achieve these many congruence conditions by some explicit analysis at finite and real places.

2. Now, at finite places, we may want $\left(\frac{N}{\ell}\right) = +1$. The point is that CM liftings are paired, so one finds that $\deg P_\ell(x)$ is even. It follows that $P_\ell(x)$ is a square $(\text{mod } \ell)$.

3. At the real place, one shows that the roots of P_ℓ are paired, and then we get a non-square at N .

The proof is now completed upon some local calculations with quadratic reciprocity. ■

Our proof uses the following inputs.

1. CM points: these are parameterized by some conjugacy classes of embeddings of CM extensions into B . We then construct $P_\lambda(x)$ and do a quadratic reciprocity argument as before.
2. Finite places: we use something about integral models of Shimura varieties and do pairings as before.
3. Real places: we use something about Hecke equidistribution.

4 January 31: p -adic Higher Green's Functions

This talk was given by Hazem Hassan for JHUnion number theory days.

4.1 Higher Green's Functions

We begin with a story of complex multiplication. Let $\mathcal{H}$ be the complex upper-half plane, and let $j: \mathcal{H} \rightarrow \mathbb{C}$ is the j -invariant.

Definition 41. A CM point τ is a point in $\mathcal{H}$ such that $\mathbb{Q}(\tau)$ is an imaginary quadratic field.

Theorem 42. Fix a CM point $\tau \in \mathcal{H}$. Then $\mathbb{Q}(\tau, j(\tau))$ is an abelian extension of $\mathbb{Q}(\tau)$.

Remark 43. In fact, the theory of complex multiplication allows us to construct all abelian extensions of $\mathbb{Q}(\tau)$ by working only a little harder.

We now introduce some modular forms.

Example 44. Here are some modular forms of weight 0.

- The function $j - j(\tau)$ is a holomorphic modular form of weight 0. Here, "holomorphic" is on $\mathcal{H}$, not $\mathbb{P}_{\mathbb{C}}^1$.
- The function $\log(j - j(\tau))$ is a multivalued holomorphic modular form of weight 0.
- The function $\log |j - j(\tau)|$ is a real analytic function of weight 0.
- One can also take a logarithmic derivative $d \log(j - j(\tau))$, which is still holomorphic but now of weight 2.

There are also higher-weight versions of some of these.

Example 45. Fix some even $k \geq 4$, and set $n := k - 2$. Then

$$f_{k,\tau}(z) := \sum_{\gamma \in \text{SL}_2(\mathbb{Z})} \left(\frac{\gamma\tau - \gamma\bar{\tau}}{(z - \gamma\tau)(z - \gamma\bar{\tau})} \right)^{k/2}$$

is a meromorphic modular form of weight k with (simple) poles on $\text{SL}_2(\mathbb{Z}) \cdot \tau$. We cannot evaluate this at our CM points because of the poles, so we would like a primitive.

Remark 46. We want a notion of differentiation to say something about $d \log$. However, differentiation is not totally SL_2 -invariant: if f has weight $-n$, then one finds

$$d^{n+1}(f|_{-n}\gamma) = (d^{n+1}f)|_{-k},$$

where $\cdot|_{\bullet}$ is the twisted action associated with a modular form.

There is now a weight $-n$ modular form $\tilde{f}_{k,\tau}(z)$ which is a multivalued modular form of weight $-n$; it is constructed via some repeated integration process. We may now define higher Green's functions.

Definition 47 (higher Green's function). We define

$$G_{k/2}(\tau, \sigma) := \mathrm{Re} \, \delta_{-n}^{n/2} \tilde{f}_{k,\tau}(\sigma),$$

where $\delta_{\ell} := \frac{d}{dz} + \frac{\ell}{z-\bar{z}}$.

Theorem 48 (Gross–Zagier conjecture). Fix a normalized weakly holomorphic modular form f of weight $-n$ with Laurent expansion $f = \sum_m c(m)q^m$. Then for CM points τ and σ ,

$$\sum_{m>0} c(-m)m^{n/2}(T_m \cdot G_{k/2})(\tau, \sigma) \in \overline{\mathbb{Q}}.$$

These numbers turn out to be intersections of Heegner cycles, which is why we may expect them to be algebraic (due to the standard conjectures). This is now known to Bruinier–Li–Yang, using regularized θ -lifts.

4.2 Real Multiplication

We now try to tell a similar story for real multiplication, retelling the story of extensions of real quadratic fields via some geometric picture.

- The analogue of $\mathcal{H}$ is the p -adic upper half-plane $\mathcal{H}_p := \mathbb{P}^1(\mathbb{C}_p) \setminus \mathbb{P}^1(\mathbb{Q}_p)$.
- The analogue of $\mathrm{SL}_2(\mathbb{Z})$ is $\Gamma := \mathrm{SL}_2(\mathbb{Z}[1/p])$.
- Modular forms come from some sections $H^0(\mathrm{SL}_2(\mathbb{Z}); \mathcal{M}_k)$ of a line bundle. It turns out that the right analogue is $H^1(\Gamma; \mathcal{M}_k)$, where we go up in a degree to see some modular symbols.
- The Heegner points admit some archimedean intersection numbers $\log |j(\tau) - j(\sigma)|$. In the p -adic upper half-plane, one has some Stark–Heegner points, but they only conjecturally lie in the real quadratic field. There is something known about algebraicity.
- Heegner cycles have some interesting higher Green's functions. We are interested in putting something on the real multiplication side called “Stark–Heegner cycles.”

We cannot define Stark–Heegner cycles, but we can say something about their conjectural intersections, which should be p -adic higher Green's functions.

Theorem 49. Fix an RM point $\tau \in \mathcal{H}_p$ and some integer $k \geq 2$. Then there is a unique rigid meromorphic cocycle (up to analytic cocycles and scaling) $J_{k,\tau} \in H^1(\Gamma; \mathcal{M}_k)$ with poles supported on $\Gamma \cdot \tau$. Conversely, such cocycles span the “parabolic cohomology.”

Remark 50. In fact, the poles of $J_{k,\tau}$ have the same shape as the poles of the earlier defined $f_{k,\tau}$.

Example 51. By linearity, one can define $J_{k,D}$ for any divisor D with real multiplication. If it has “strong degree zero,” then we find

$$J_{k,D}(\gamma)(z) = \sum_{\substack{b \in \Gamma \\ \tau \in |D|}} i([r, \gamma r], [b\bar{\tau}, b\tau]) \left(\frac{b\tau - b\bar{\tau}}{(z - b\tau)(z - b\bar{\tau})} \right)^{k/2}.$$

Here, $i([r, \gamma r], [b\bar{\tau}, b\tau])$ is a signed intersection number of some geodesics (in $\mathcal{H}$!), where $r \in \mathbb{Q}$ is arbitrary. Notably, these intersection numbers are necessary to make the sum converge.

One now again takes some repeated integration, and we can state a Gross–Zagier conjecture. In fact, one can conjecture something about the prime factorizations of the special values, coming from some intersection numbers on Shimura curves.

5 January 31: Geometric and Arithmetic Siegel–Weil Formulae

This talk was given by Yu Luo at Johns Hopkins University for the JHUnion number theory days.

5.1 The Geometric Formula

For motivation, we may be interested in how an elliptic curve can map into an abelian variety. To geometrize, we let $\mathcal{A}_g$ denote the moduli space of principally polarized abelian varieties of dimension g .

Definition 52 (Noether–Lefschetz cycle). For each m , we define the functor $\text{NL}(m)$ as living over $\mathcal{A}_1 \times \mathcal{A}_g$ and parameterizing tuples (E, A, λ, h) such that $h: E \rightarrow A$ is some map, and λ is the polarization of A , and $\deg h^* \mathcal{L}_\lambda = m$.

Example 53. It turns out that $\text{NL}(1) \cong \mathcal{A}_1 \times \mathcal{A}_{g-1}$.

These cycles are interesting but difficult. We will be interested in similar questions with a little additional structure, coming from unitary Shimura varieties.

For today, F is an imaginary quadratic field, and we let V be a Hermitian space over F with signature $(n-1, 1)$ (where $n := \dim V$). We also fix some level structure $K \subseteq \text{U}(V)(\mathbb{A}_{\mathbb{Q}}^\infty)$.

Definition 54 (unitary Shimura variety). Fix everything as above. We define $\mathcal{M}_K$ to parameterize the data

$$(E, \iota_E, A, \iota, \lambda, \eta),$$

where (E, ι_E) is an elliptic curve with CM by F , (A, λ) is a principally polarized abelian variety, ι is an F -structure on A so that $\text{Lie } A$ has signature $(n-1, 1)$, and η is K -conjugacy class of a morphism $\widehat{V}(E) \rightarrow \widehat{V}(A)$.

Example 55. Choose a line $L \subseteq V$, and let K be the stabilizer of a lattice $\widehat{L}$. Then $\mathcal{M}_K(\mathbb{C})$ consists of copies of symmetric spaces which look like $\text{Aut}(L) \backslash D$, where $D \subseteq \mathbb{P}(V_{\mathbb{R}})$ consists of isotropic lines in $V_{\mathbb{R}}$.

Remark 56. There is a pairing structure on $\mathrm{Hom}_F(E, A)$ by sending a pair (x, y) to the composite

$$E \xrightarrow{x} A \xrightarrow{\lambda} A^\vee \xrightarrow{y^\vee} E^\vee \cong E.$$

In particular, this maps to $\mathrm{End}(E) \cong F$, so we receive a pairing analogous to the Rosati involution.

Having given our analogue of $\mathcal{A}_g$, we need to give an analogue of our special cycles.

Definition 57. Fix a positive definite Hermitian $r \times r$ matrix T with coefficients in F , and choose some $\mu \in K \setminus V(\mathbb{A}_f)^r$. Then we let $Z(T, \mu)$ parameterize triples (E, A, u) where $u \in \mathrm{Hom}_F(E, A)^r$ satisfies $(u, u) = T$ (and some level structure dictated by μ).

It turns out that $Z(T, \mu)$ lives in $\mathrm{CH}^r(\mathcal{M}_K)$, so we may take some linear combinations.

Notation 58. For any $\varphi_f \in S(V(\mathbb{A}_f)^r)^K$, we may define

$$Z(T, \varphi_f) := \sum_{\mu} \varphi_f(\mu) [Z(T, \mu)].$$

By definition of S , this is a finite sum.

Example 59. We continue from Example 55. For $x \in L$, we define $D(x)$ to be the lines perpendicular to x . Then $Z(t, 1_{\widehat{L}})$ on complex points is given by the projection

$$\mathrm{Aut}(L) \bigg/ \bigsqcup_{(x, x)=t} D(x) \rightarrow \mathrm{Aut}(L) \backslash D.$$

The geometric Siegel–Weil formula arises when $\mathrm{rank} T = r - 1$.

Theorem 60. Fix a positive definite Hermitian $r \times r$ matrix T of rank $r - 1$. Then

$$\deg Z(T, \varphi_f) \sim E_T(\tau, \varphi),$$

where $\sim$ means we are up to some explicit constant (coming from the Shimura variety), φ is an automorphic form constructed with finite part φ_f , and E_T refers to the T th Fourier coefficient for the Eisenstein series $E(\tau, \varphi)$ associated to φ .

5.2 The Arithmetic Formula

For the arithmetic Siegel–Weil formulae, we need to pass to some integral models. Accordingly, let S be a finite set of primes containing 2 and the ramified primes of the extension $F/\mathbb{Q}$. For simplicity, we will also assume that the lattice L of V is self-dual away from S , and we assume that $K^S = \mathrm{Stab}(\widehat{L}^S)$. In this situation, $\mathcal{M}_K$ admits an integral model over $\mathcal{O}_F[S^{-1}]$. If we derive, we also receive the special cycles $Z^{\mathbb{L}}(T, \varphi_f)$.

In this case, we will be interested in the case where $\mathrm{rank} T = r$, which produces a cycle of codimension 1 because $\mathcal{M}_K \rightarrow \mathrm{Spec} \mathcal{O}_F[S^{-1}]$ has relative dimension r .

Notation 61. We let $\mathrm{Diff}(T, V)$ be the set of nonsplit primes p for which $V_p \not\cong V(T)_p$, where $V(T)$ is the space with basis $\{e_1, \dots, e_n\}$ satisfies $(e_i, e_j) = T_{ij}$.

Theorem 62 (Kudla–Rapoport). The special cycle $Z(T, \varphi_f)$ is nonempty only if $\mathrm{Diff}(T, V)$ is a single prime $p \notin S$. In this case, the special cycle is supersingular over $\mathbb{F}_{p^2}$.

Theorem 63. Suppose $\text{Diff}(T, V) = \{p\}$. Then

$$\log(p^2 \deg Z^{\mathbb{L}}(T, \varphi_f)) \sim E'_T(\tau, \varphi).$$

One can view this as a natural generalization of the Gross–Zagier formula to unitary Shimura varieties. The moral is that we are relating an intersection number to a modular form, which can then be related to a special value. Here are some ingredients of the proof.

1. It turns out that

$$Z(\tau, \varphi) = \sum_{m>0} Z(m, \varphi) q^m$$

is a modular form. One can then take an arithmetic intersection with another special cycle to produce a genuine modular form. This produces a modular form whose coefficients are given by the left-hand side.

2. On the other hand, we can use the right-hand side to produce a modular form. It turns out that the coefficients are the same at all but finitely many coefficients, but the difference is a modular form, so full equality follows.

6 January 31: Higher Siegel–Weil Formulae

This talk was given by Mikayel Mkrtychyan at the Johns Hopkins University for the JHUnion number theory days.

6.1 Shtukas

We are interested in function field analogues of the geometric and arithmetic Siegel–Weil formulae. The analogue of (unitary) Shimura varieties are given by shtukas. For our setup, we fix some étale double cover $X' \rightarrow X$ of smooth proper curves over $k := \mathbb{F}_q$; let σ be the nontrivial automorphism.

Definition 64. A U_n -bundle is a pair $(\mathcal{F}, h)$ of a rank n bundle $\mathcal{F}$ on X' and an isomorphism $h: \mathcal{F} \rightarrow \sigma^* \mathcal{F}^\vee$. A *Hecke modification* is a diagram $x: \mathcal{F}_0 \rightarrow \mathcal{F}_1$, where $\mathcal{F}_0$ and $\mathcal{F}_1$ are U_n -bundles, $x \in X'$ is a point, and we are equipped with an isomorphism $\mathcal{F}_0|_{X' \setminus \{x, \sigma x\}} \cong \mathcal{F}_1|_{X' \setminus \{x, \sigma x\}}$. We also require there to be a natural short exact sequence

$$0 \rightarrow \mathcal{F}_0 \rightarrow \mathcal{F}_1 \rightarrow \delta_x \rightarrow 0,$$

where δ_x is the skyscraper.

Example 65. If $X' \rightarrow X$ is split, then a U_n -bundle is a vector bundle of rank n on X .

Definition 66 (shtuka). We define $\text{Sht}_{U_n}^r$ to be the moduli stack parameterizing diagrams

$$\mathcal{F}_0 \rightarrow \mathcal{F}_1 \rightarrow \cdots \rightarrow \mathcal{F}_r \cong \mathcal{F}_0^\tau,$$

where each $\mathcal{F}_\bullet$ is a U_n -bundle, each arrow is a Hecke modification, and $\mathcal{F}_0^\tau$ is the Frobenius pullback.

Here are some grounding remarks about shtukas.

Remark 67. There is a “leg morphism” $\text{Sht}_{U_n}^r \rightarrow (X')^r$ recording the points where the Hecke modifications occur. This is analogous to the structure morphism for a Shimura variety. It turns out that this map is smooth of relative dimension $r(n-1)$. Thus, $\dim \text{Sht}_{U_n}^r = rn$.

Example 68. One has that $\mathrm{Sht}_{\mathrm{U}_n}^0 = \mathrm{Bun}_{\mathrm{U}_n} k$, so functions on this space are basically unramified automorphic forms.

Remark 69. For $n > 1$, the space $\mathrm{Sht}_{\mathrm{U}_n}^r$ is not quasicompact.

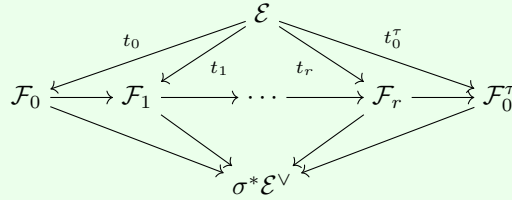
Remark 70. Taking the fibers over the various Hecke modifications, we produce “tautological” line bundles $\mathcal{L}_1, \dots, \mathcal{L}_r$ on $\mathrm{Sht}_{\mathrm{U}_n}^r$.

Remark 71. It turns out that $\mathrm{Sht}_{\mathrm{U}_n}^r$ vanishes for odd r , but it is not too hard to define a suitable analogue.

We will also want some special cycles. The following definition is due to Feng–Yun–Zhang.

Definition 72. Fix a pair $(\mathcal{E}, a)$, where $\mathcal{E}$ is a vector bundle of rank m on X' , and $a: \mathcal{E} \rightarrow \sigma^* \mathcal{E}^\vee$ is some Hermitian morphism. (This is an analogue of a lattice.) Then $(\mathcal{E}, a)$ is *non-degenerate* if and only if a is injective. We let $\mathcal{A}(m)$ be the moduli stack of non-degenerate pairs.

Definition 73. Fix a pair $(\mathcal{E}, a)$ as before. Then we define the special cycle $Z_{\mathcal{E}}^r(a)$ on $\mathrm{Sht}_{\mathrm{U}_n}^r$ to parameterize diagrams



where the total composite is a .

Remark 74. To see that this is an analogue, imagine that $\mathcal{E}$ is a CM elliptic curve, so we are asking for maps to abelian varieties which preserve some structure.

Example 75. If a is an isomorphism, then taking orthogonal complements shows that $\mathcal{E}$ embeds into the shtuka, so $Z_{\mathcal{E}}^r(a) \cong \mathrm{Sht}_{\mathrm{U}_{n-m}}^r$. Even if a is merely generically an isomorphism, then such an isomorphism holds generically.

Example 76. If $a = 0$, then there is a natural map $\mathrm{Sht}_{\mathrm{U}_n}^r \subseteq Z_{\mathcal{E}}^r[0]$ simply by setting the $t_\bullet$ s to vanish.

Remark 77. One expects that $\dim Z_{\mathcal{E}}^r(a) = r(n - m)$, though the dimension could be much larger. For example, when a is an isomorphism, equality holds, and when $a = 0$, equality does not hold. Feng–Yun–Zhang have defined virtual fundamental classes $[Z_{\mathcal{E}}^r(a)]$ in $\mathrm{CH}_{r(n-m)} Z_{\mathcal{E}}^r(a)$.

6.2 The Formulae

We will need to define some Eisenstein series.

Definition 78. There is a Siegel Eisenstein series $\text{Eis}(g, s)_m$ for $\text{U}(m, m)$ such that each pair $(\mathcal{E}, a)$ (of rank m) has a Fourier coefficient $\text{Eis}_{(\mathcal{E}, a)}(s)_m$ with a rather technical definition. Our Eisenstein series are normalized so that the center of the functional equation is at $s = 0$.

Remark 79. The Eisenstein series turn out to have explicit combinatorial expansions. For example, if a has length two, then $\text{Eis}_{(\mathcal{E}, a)} = q^s \pm q^{-2}$, where the $+$ sign is used if we are split over a split point $x \in X$.

Theorem 80 (Feng–Yun–Zhang). Fix everything as above, and take $m = n$ and a non-degenerate pair $(\mathcal{E}, a)$. Then for all $r \geq 0$, we have

$$\deg [Z_{\mathcal{E}}^r(a)] \sim \partial_{s=0}^r \text{Eis}_{(\mathcal{E}, a)}(s)_m.$$

Here, $\partial_{s=0}^r$ means that we take r derivatives and then evaluate at 0.

Theorem 81. Fix everything as above, and take $m = n - 1$ and a non-degenerate pair $(\mathcal{E}, a)$. Then for every even $r \geq 0$, we have

$$\deg [Z_{\mathcal{E}}^r(a)] \cap [\mathcal{L}_1] \cap \cdots \cap [\mathcal{L}_r] \sim \partial_{s=0}^r (q^{s \deg \omega_X} \text{Eis}_{(\mathcal{E}, a)}(s - 1/2)_{n-1} L(\eta, 2s)).$$

Remark 82. There is an analogous statement for $r \geq 0$.

Remark 83. There is some analogous thing known for number fields when $r = 1$.

Let's say something about the proofs. The main difficulty in the proof is calculating on the geometric side. One can view the function

$$(\mathcal{E}, a) \mapsto \deg [Z_{\mathcal{E}}^r(a)]$$

as a k -valued function on the moduli stack $\mathcal{A}(n)$, so we hope that it arises from a sheaf. (The same holds for the $m = n - 1$ case.)

Definition 84 (Hitchin). Let $\mathcal{M}(m, n)$ be the moduli stack of morphism composites $\mathcal{E} \rightarrow \mathcal{F} \cong \sigma^* \mathcal{F}^\vee$ so that the composite

$$\mathcal{E} \rightarrow \mathcal{F} \cong \sigma^* \mathcal{F}^\vee \rightarrow \sigma^* \mathcal{E}^\vee$$

is injective.

There is a natural projection $\mathcal{M}(m, n) \rightarrow \mathcal{A}(m)$, and it turns out that there is a self-correspondence

$$\mathcal{M}(m, n) \leftarrow \text{Hk}_{\mathcal{M}}^1(m, n) \rightarrow \mathcal{M}(m, n).$$

Thus, we can do cohomology to this correspondence, from which a Lefschetz trace formula for correspondences lets us compute $\deg [Z_{\mathcal{E}}^r(a)]$ as the trace of Frobenius acting via this correspondence. (Again, something similar happens for $m = n - 1$.) We now discuss $m = n$ and $m = n - 1$ separately.

- Now, if $m = n$, there is a pullback square

$$\begin{array}{ccc} \mathcal{M}(n, n) & \longrightarrow & \text{Lg} \\ \downarrow & & \downarrow \\ \mathcal{A}(n) & \longrightarrow & \text{Herm} \end{array}$$

where the right arrow can be controlled étale-locally. Here, Herm is the moduli space of Hermitian torsion sheaves, and Lg parameterizes Langrangians. This allows one to control the fibers of $\mathcal{M}(n, n) \rightarrow \mathcal{A}(n)$.

- The case of $m = n - 1$ is more difficult. Instead, one factors $\mathcal{M}(n - 1, n) \rightarrow \mathcal{A}(n - 1)$ through some $\mathcal{A}(n - 1) \times \text{Bun}_{\mathbb{U}_1}$, and this factored morphism is a $(\mathbb{P}^1)^d$ -fibration. It also turns out that this factored map has full support, allowing us to control the fibers.

7 January 31: Hypergeometric Motives with CM

This talk was given by Esme Rosen at Johns Hopkins University for the JHUnion number theory days.

7.1 Hypergeometric Motives

Our modular forms are classical ones. As usual, they have L -functions, which have analytic continuation, and they frequently have Euler products, and so on. We will be considering normalized Hecke eigenforms f , which also have associated Galois representations ρ_f , which are known to come from elliptic curves.

Definition 85 (hypergeometric data). Fix $z \in \mathbb{P}^1(\mathbb{Q}) \setminus \{0, 1, \infty\}$ and two lists of rational numbers denoted $(a_0, \dots, a_n)$ and $(b_0, \dots, b_n)$ with $b_0 = 1$. Set M to be their least common multiple. Then there is a hypergeometric motive defined over $\mathbb{Q}(\zeta_M)$ related to classical hypergeometric functions

$${}_{n+1}F_n \left[\begin{matrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \end{matrix} \middle| z \right] = \sum \frac{(a_0)! \cdots (a_n)!}{(b_0)! \cdots (b_n)!} z^\bullet.$$

The associated hypergeometric functions give rise to the periods of the motives.

Example 86. The data $((1/2, 1/2), (1, 1))$ gives rise to the Legendre family of elliptic curves $y^2 = x(1 - x)(1 - zx)$.

Example 87. The data $((1/2, 1/2, 1/2), (1, 1, 1))$ gives rise to a family of $K3$ surfaces, which are known to be modular at the CM points. These are symmetric powers of elliptic curves. One expects these to be associated to weight 3 modular forms.

Example 88. Let's construct the hypergeometric motives for ${}_2F_1$ s; the general case is not too much harder. Start with the normalization X_{HD} of the curve

$$Y^M = X^i(1 - X)^j(1 - zX)^k$$

for some $i, j, k < M$ with $i + j + k \nmid M$. There is a standard γ whose period is some quotient of Γ s along with a ${}_2F_1$ value. By diagonalizing the μ_M action on X_{HD} , we can decompose the motive X_{HD} . (For example, this certainly produces an absolute Hodge cycle.)

Remark 89. There are finite field versions of our hypergeometric series, basically obtained by replacing Γ s with some Gauss sums. Katz has shown that there are Galois representations whose traces are the correct hypergeometric sums over finite fields.

Classical work in this de Rham realization has found that certain periods of modular forms (which are essential special values of the L -function) can be expressed as certain hypergeometric series. There is also work in the étale realization, expressing certain traces of Galois representations.

7.2 Some Theorems

Here is a main theorem.

Theorem 90. There is explicit modularity for some hypergeometric motives of the type

$$((1/2, 1/2, r), (1, 1, s)),$$

where (r, s) are some explicit pairs of rational numbers.

For another application, we consider Fermat motives. Fermat motives have complex multiplication, which means that their Galois representations are induced from Hecke characters. The periods of motives with CM by an imaginary quadratic field $\mathbb{Q}(\sqrt{-D})$ are known by the Chowla–Selberg formula.

Remark 91. Fermat motives arise in our story because they are ${}_2F_1(1)$ motives, which notably appear in the cohomology of Fermat curves. The associated Hecke characters are some Jacobi sums; we then take induction to get the associated modular form.

Theorem 92. For many of these hypergeometric motives, one can descend the Galois representations from $\mathbb{Q}(\zeta_M)$ to $\mathbb{Q}(\sqrt{-D})$ for some explicit D .

In fact, the theory of complex multiplication now allows us to compute some special values.

8 January 31: Finiteness of Heights of Motives

This talk was given by Alice Lin at Johns Hopkins University for the JHUnior number theory days.

8.1 Our Statement

We are going to be interested in heights today. Here is one such example.

Theorem 93 (Faltings). Fix $g \geq 1$, a number field F , and a finite set S of finite places. Then there are finitely many isomorphism classes of g -dimensional abelian varieties A over F with good reduction outside S .

Sketch. The proof has two steps.

1. Show that there are only finitely many isogeny classes.
2. Show that each isogeny class has only finitely many isomorphism classes. This is done basically by understanding how the Faltings height changes along isogenies. ■

The second step has been better understood recently.

Theorem 94 (Kisin–Mocz). Assume the Mumford–Tate conjecture for an abelian variety A over a number field F . Then for any $c, f > 0$, the set of abelian varieties B over $\overline{\mathbb{Q}}$ isogenous to A (over $\overline{\mathbb{Q}}$) with Faltings height at most c .

Remark 95. Unlike in Faltings’s situation, the set of abelian varieties B (with unbounded Faltings height) is infinite.

Remark 96. Recall that the Mumford–Tate conjecture asserts that the Mumford–Tate group agrees with the (connected component of the) ℓ -adic monodromy group. Here, the Mumford–Tate group is the Tannakian algebraic group of $H_B^1(A(\mathbb{C}); \mathbb{Q})$ in the category of Hodge structures, and the ℓ -adic monodromy group is the Zariski closure of the Galois representation. The Mumford–Tate conjecture is wide open in the general case, but a lot is known.

Here is our result, with definitions to follow.

Theorem 97. Let M be a Koshikawa motive defined over F , and assume the following.

- The adelic Mumford–Tate conjecture holds for M .
- All Hodge classes are absolutely Hodge and compatible with étale–de Rham comparison maps.

Then for any $c > 0$, the set of motives M' equipped with an isogeny $M' \rightarrow M$ with height at most c has only finitely many possible heights.

Remark 98. The second hypothesis is true for abelian varieties, due to Deligne and Blasius.

Remark 99. If $\text{MT}(M)$ acts irreducibly, then we can achieve only finitely many isomorphism classes.

Corollary 100. Fix a point x on a Shimura variety $\text{Sh}(G, X)(F)$. Suppose that the Galois representation

$$\rho_x: \text{Gal}(\overline{F}/F) \rightarrow G(\mathbb{A}_f)$$

has open image (so that the adelic Mumford–Tate conjecture holds). Then for any $c > 0$, there are only finitely many $\overline{F}$ -points in the Hecke orbit with height at most c .

Notably, this result is known due to Kisin–Mocz in the Hodge type case.

8.2 Explanation of the Theorem

Let's start by defining our motives.

Definition 101 (Koshikawa motive). A Koshikawa (pure) motive M over a number field F with weight w is (roughly) a system of cohomological realizations, equipped with a $\widehat{\mathbb{Z}}$ -integral Galois stable structure on the adelic étale realization. More precisely, we take

- A system of cohomological realizations: all the Betti cohomology realizations for each embedding $F \hookrightarrow \mathbb{C}$, the de Rham cohomology, all the ℓ -adic cohomology realizations, and
- A lattice: such as the $\widehat{\mathbb{Z}}$ -coefficients étale cohomology.

Definition 102 (isogeny). An isogeny $M' \rightarrow M$ of Koshikawa motives is a Galois-stable inclusion of the underlying integral lattices.

Remark 103. This is analogous to the fact that an isogeny of abelian varieties amounts to the data of an embedding of the (adelic) Tate modules.

We also need to define our height.

Definition 104 (Kato–Koshikawa). Fix a Koshikawa motive M , and set

$$V := \bigotimes_{r \in \mathbb{Z}} \det_F(\mathrm{gr}^r M_{\mathrm{dR}})^{\otimes r}.$$

Then integral p -adic Hodge theory provides an $\mathcal{O}_{F_v}$ -lattice in $V \otimes_F F_v$ arising from the $\widehat{\mathbb{Z}}$ -lattice; these assemble into an $\mathcal{O}_F$ -lattice $\mathcal{L} \subseteq V$. Then we define the height

$$h(M) = \frac{1}{[F : \mathbb{Q}]} \left(\log \# \left(\frac{\mathcal{L}}{\mathcal{O}_F \alpha} \right) - \sum_{\sigma : F \hookrightarrow \mathbb{C}} \log \|\alpha\|_{\sigma} \right).$$

Example 105. When $M = H^1(A)$, we find $V = \det H^0(A; \Omega^1) = H^0(A; \Omega_{A/F}^g)$. The integral p -adic Hodge theory can be replaced with the Néron model $\mathcal{A}$ over $\mathcal{O}_F$, allowing us to take $\mathcal{L} = H^0(\mathcal{A}; \Omega_{\mathcal{A}/\mathcal{O}_F}^g)$. The height is then exactly the Faltings height.

Roughly speaking, the proof boils down to an isogeny formula for how our height changes in isogenies. The Mumford–Tate conjecture (and other inputs) are used in order to move the integral p -adic Hodge theory out into more controlled cohomology theories.

9 February 3: Fulton–MacPherson Compactifications

This talk was given by Denis Nesterov for the Harvard–MIT algebraic geometry seminar.

9.1 Compactifying Configuration Space

We work over $\mathbb{C}$ throughout.

Notation 106. Fix a smooth projective variety X over $\mathbb{C}$. Define the *configuration space* $\mathrm{Conf}_n(X)$ to be the collection of n -tuples $(x_1, \dots, x_n)$ of distinct points in X .

There are various compactifications of $\mathrm{Conf}_n(X)$.

Example 107. The projective scheme X^n is a compactification of $\mathrm{Conf}_n(X)$.

Example 108. There is another compactification of $\mathrm{Conf}_n(X)$ using Fulton–MacPherson compactifications $\mathrm{FM}_n(X)$. Here, $\mathrm{FM}_n(X)$ is the space of n distinct points on the Fulton–MacPherson degenerations. These degenerations amount to adding in extra $\mathbb{P}^{\dim X}$ s to X , sometimes referred to as “bubbles.”

One frequently takes a quotient $\mathrm{Conf}_n(X)$ by the natural action by the symmetric group Σ_n . Then we can use Hilbert schemes to compactify.

Notation 109. Fix a smooth projective variety X over $\mathbb{C}$. Then $\mathrm{Hilb}_n(X)$ is the space of zero-dimensional subschemes $Z \subseteq X$ of length n .

These compactifications have various relations. Here are some well-known ones.

Definition 110. There is a *Hilbert–Chow map* $\mathrm{Hilb}_n(X) \rightarrow X^n / \Sigma_n$ given by sending a closed subscheme $Z \subseteq X$ to its support.

Remark 111. If $\dim X = 2$, then it turns out that $\mathrm{Hilb}_n(X) \rightarrow X^n/\Sigma_n$ is a crepant resolution of singularities (i.e., it preserves the canonical bundle).

Conjecture 112. If $\dim X = 2$, then there is a ring isomorphism $H_q^\bullet(\mathrm{Hilb}_n(X)) \cong H^\bullet(X^n/\Sigma_n)$, where the former is a quantum adjustment of the ring structure taken at $q = -1$, and the latter is orbifold cohomology.

Proposition 113. There is an iterated blowup map $\mathrm{FM}_n(X) \rightarrow X^n$. Explicitly, one blows up along the diagonals

$$\Delta_k := \{x_{i_1} = \cdots = x_{i_k} : i_1 < i_2 < \cdots < i_{k+1}\} \subseteq X^n.$$

Explicitly, one blows up along Δ_n , then Δ_{n-1} , then Δ_{n-2} , and so on.

Remark 114. This proposition was the original definition of $\mathrm{FM}_n(X)$.

9.2 Harder Relations

Here are some harder relations.

Theorem 115 (Haiman). Suppose $\dim X = 2$. The ordered Hilbert schemes $\mathrm{Hilb}_n^{\mathrm{iso}}(X)$ are isomorphic to $\mathrm{Bl}_{\Delta^1} X^n$. Here, $\mathrm{Hilb}_n^{\mathrm{iso}}(X)$ is the reduction of the fiber product sitting in the following pullback square.

$$\begin{array}{ccc} \mathrm{Hilb}_n^{\mathrm{iso}}(X) & \longrightarrow & \mathrm{Hilb}_n(X) \\ \downarrow & & \downarrow \\ X^n & \longrightarrow & [X^n/\Sigma_n] \end{array}$$

Remark 116. This result allowed Haiman to realize MacDonald polynomials geometrically.

Remark 117. This result was extended to $\dim X \geq 3$ by Ekedahl–Skjelnes, where one has to take a large component of $\mathrm{Hilb}_n(X)$ instead.

Remark 118. By keeping track of the blow-ups, one can see that there is a map $\mathrm{FM}_n(X) \rightarrow \mathrm{Hilb}_n^{\mathrm{iso}}(X)$ when $\dim X = 2$, which is a resolution of singularities.

The existence of a map $\mathrm{FM}_n(X) \rightarrow \mathrm{Hilb}_n^{\mathrm{iso}}(X)$ tells us that we may gain access to Conjecture 112 by taking cohomology. It turns out that

$$H^\bullet(\mathrm{FM}_n(X)) = H^\bullet(X^n)[D_I],$$

where D_I consists of the points living on some specified bubbles indexed by $I \subseteq \{1, \dots, n\}$, and they satisfy some relations. One can also compute $H^\bullet(\mathrm{Hilb}_n(X))$ as being spanned by decorated partitions of n ; this gives it an action of a Heisenberg algebra.

For our main theorem, recall that one uses various blow ups to produce intermediate spaces between X^n and $\mathrm{FM}_n(X)$. We can do something similar for the map $\mathrm{FM}_n(X)/\Sigma_n \rightarrow \mathrm{Hilb}_n(X)$.

Definition 119. Fix a smooth zero-dimensional subscheme Z of a Fulton–MacPherson degeneration $\tilde{X}$ of X . For $\varepsilon > 0$, we say that a pair $(\tilde{X}, Z)$ is ε -weighted if and only if we have

- (a) $\text{mult}_x(Z) \leq 1/\varepsilon$ for all $x \in \tilde{X}$,
- (b) $\text{mult}(Z|_{\text{bubble}}) > 1/\varepsilon$, and
- (c) $|\text{Aut}(\tilde{X}, Z)| < \infty$.

We now define $\text{Hilb}_n^\varepsilon(X)$ to be the moduli space of ε -weighted pairs $(\tilde{X}, Z)$.

Example 120. If $\varepsilon = 1$, then we are just looking at n distinct points in $\tilde{X}$.

Theorem 121. The spaces $\text{Hilb}_n^\varepsilon(X)$ is a proper Deligne–Mumford stack.

Example 122. With $\varepsilon = 1$, we have $\text{Hilb}_n^\varepsilon(X) = \text{FM}_n(X)/\Sigma_n$. With $\varepsilon = 1/n$, we have $\text{Hilb}_n^\varepsilon(X) = \text{Hilb}_n(X)$.

One expects that the map $\text{FM}_n(X) \rightarrow \text{Hilb}_n^{\text{iso}}(X)$ suitably factors through $\text{Hilb}_n^\varepsilon(X)$. Here is the application: taking blow ups, one can interpolate between X^n/Σ_n and $\text{FM}_n(X)/\Sigma_n$. By the main theorem, we know how to interpolate between $\text{FM}_n(X)/\Sigma_n$ and $\text{Hilb}_n(X)$. In this way, we are able to reduce Conjecture 112 to an explicit calculation on the surface $\mathbb{C}^2$.

10 February 6: Representation-Theoretic Background

This talk was given by Mitch at Johns Hopkins University for the student automorphic representations seminar.

10.1 Function Spaces

Recall that our spectral expansion is

$$J_{\text{spec}}(f) = \int_{\mathfrak{X}(G)} D(\pi) \hat{\Theta}_f(\pi) m(\bar{\pi}) d\pi.$$

We would like to explain what $\mathfrak{X}(G)$ and $\hat{\Theta}_f(\pi)$ are. In short, $\mathfrak{X}(G)$ will be a space of virtual elliptic representations of $G(F)$, and $\hat{\Theta}_f(\pi)$ is some weighted character. Namely, if π is an induced representation of some σ from a Levi subgroup $M \subseteq G$, then $\hat{\Theta}_f(\pi)$ is the trace of an intertwiner constructed from f .

Let's start by fixing our nice functions.

Definition 123. Fix a manifold M and a vector space E (of possibly infinite dimension). Then a function $f: M \rightarrow E$ is C^∞ if and only if φf is smooth for every continuous $\varphi: E \rightarrow \mathbb{R}$.

Definition 124. Given a reductive Lie algebra $\mathfrak{g}$, a smooth function f on $mfg(F)$ is *Schwartz* if and only if each $N \geq 1$ and $u \in \text{Sym } \mathfrak{g}(F)$ has

$$\sup_{X \in \mathfrak{g}(F)} \|X\|_{\mathfrak{g}}^N |\partial(u)f(X)| < \infty.$$

Approximately speaking, we are requiring all mixed partial derivatives to decay faster than any polynomial at 0 and ∞ .

Definition 125. Choose a minimal parabolic P and a maximal compact $K \subseteq G(F)$ so that we have an Iwasawa decomposition $G(F) = P(F)K$. Select e_K in the parabolic induction $i_P^G(1)$ which is constantly 1 on K . Then we define the pairing

$$(f, f') := \int_K f(k) \overline{f'(k)} dk$$

for $f, f' \in i_P^G(1)$. Then we define $\Xi^G: G \rightarrow \mathbb{C}$ by $\Xi^G(g) := (ge_K, e_K)$.

Remark 126. The definition of Ξ^G depends on many choices, but such choices only change the result by some scalars.

Remark 127. The function Ξ^G is supported on $K \backslash G/K = A^+$, and there is some machinery of matrix coefficients which tells us that it grows about as much as the square root of the modular character.

Definition 128. A function $f \in C^\infty(G(F))$ is *Harish-Chandra–Schwartz* if and only if $P_d(L_u R_v f)$ is finite (where L_u and R_v are translations with $u, v \in U_{\mathfrak{g}}$), where

$$P_d(f) := \sup_{g \in G} \frac{f(g) \sigma(g)^d}{\Xi^G(g)}$$

for any function f and positive integer d . Here, $\sigma(g)$ is defined to be $\|X\|$, where $g = k \exp(X)$ where $k \in K$.

Example 129. If $G = \mathbb{G}_m$, then Ξ^G is trivial (there is no parabolic anyway, and K is a point), so σ is just the logarithm. Upon identifying $\mathbb{R}^+$ with $\mathbb{R}$ by taking the logarithm, we recover just the usual Schwarz space.

Remark 130. In practice, one can replace Ξ^G by the $K \backslash G/K$ function which is $\delta^{1/2}$ on A^+ . It is basically “volume control” because we want certain functions to be integrable on A^+ which may not otherwise look like it.

10.2 Adjectives for Representations

Here are some more adjectives.

Definition 131 (tempered). A representation π is *tempered* if and only if $|(\pi(g)v, v')| \ll \Xi^G(g)$ for all v and v' .

Remark 132. It follows that the matrix coefficients all live in $L^{2+\varepsilon}(G/Z)$, for any $\varepsilon > 0$. Intuitively, we are just putting some (volume-controlled) bound on the matrix coefficients.

Definition 133 (square integrable). A representation π is *square integrable* if and only if π is irreducible and the matrix coefficients for $g \in G(F)/A_G(F)$ are in L^2 . The collection of such representations is denoted $\Pi_2(G)$.

Remark 134. Writing $P = MU$ and K for a maximal compact, one has that $i_P^G \sigma \cong i_{K_P}^K(\sigma|_{K_P})$, where $K_P = K \cap P(F)$. Thus, twists by unramified characters should not adjust bounds on the space.

Notation 135. Let $\mathcal{A}_G^*$ or $\mathfrak{a}_G^*$ denote the (rational) characters from G . We let them act on representations with the additional scaling by

$$\exp(\langle \lambda, H_G(g) \rangle),$$

where $\langle \lambda, H_G(g) \rangle$ is the pairing extended linearly from being $\log |\lambda(g)|$ on genuine characters.

Now, for fixed levi M , we choose two parabolics P and P' with Levi component M . For some representation σ_λ of M , we can define an intertwiner $J_{P'|P}(\sigma_\lambda): i_P^G(\sigma_\lambda) \rightarrow i_{P'}^G(\sigma_\lambda)$, which is normalized to $R_{P'|P}(\sigma)$. If one fixes a Levi $L \supseteq M$ and parabolic $Q \supseteq L$, one can use a technique called a (G, M) -family to produce an operator $R_L^Q(\sigma, P)$. We are now allowed to define

$$J_L^Q(\sigma, f) := \text{tr} \left(R_L^Q(\sigma, P) \circ (i_P^G(\sigma); f) \right).$$

These are our weighted characters.

10.3 The Space $\mathfrak{X}(G)$

Our next task is to define the space $\mathfrak{X}(G)$.

Notation 136. The set $\mathfrak{X}_{\text{temp}}(G)$ consists of the isomorphism classes of the parabolic inductions $i_M^G(\sigma)$ for any Levi M and $\sigma \in \Pi_2(M(F))$.

Remark 137. It turns out that $i_M^G(\sigma)$ and $i_{M'}^G(\sigma')$ if there is a conjugation of the internal data.

Thus, one can build some space $\tilde{\mathfrak{X}}_{\text{temp}}(G)$ which is a disjoint union of some orbis of representations $\sigma \in \Pi_2(M)$ over the Levi subgroups M (considered up to conjugacy). One can give the space $\tilde{\mathfrak{X}}_{\text{temp}}(G)$ a topology; it is Euclidean if $F = \mathbb{R}$ and a compact torus if F is p -adic. As such, $\mathfrak{X}_{\text{temp}}(G)$ is a quotient of this space and is thus realized as an orbifold. It admits a bundle $\mathcal{E}(G)$ whose sections over a given π is the space of endomorphisms $\text{End}(\pi)^\infty$.

The space of sections $C^\infty(\mathfrak{X}_{\text{temp}}(G); \mathcal{E}(G))$ admits a natural subspace $\mathcal{C}(\mathfrak{X}_{\text{temp}}(G); \mathcal{E}(G))$ of “Schwarz functions.” We won’t describe this subspace in the real case, but it amounts to the compactly supported functions in the p -adic case.

Theorem 138. There is an isomorphism

$$\mathcal{C}(G(F)) \cong \mathcal{C}(\mathfrak{X}_{\text{temp}}; \mathcal{E}(G))$$

of topological vector spaces, given by sending $f \in \mathcal{C}(G(F))$ to the section $\pi \mapsto \pi(f)$. The inverse is given by sending a section T to the function $g \mapsto \int_{\mathfrak{X}_{\text{temp}}(G)} \text{tr}(\pi(g)^{-1} T_\pi) \mu(\pi) d\pi$.

Remark 139. This is a strong Plancheral formula, in the sense that composing the two maps recovers the formula

$$f(1) = \int_{\mathfrak{X}_{\text{temp}}(G)} \text{tr} \pi(f) \mu(\pi) d\pi.$$

11 February 10: Bezrukavnikov’s Equivalence

This talk was given by Jialiang Zou at MIT for the categorical local Langlands seminar.

11.1 Local Langlands

Today, F is nonarchimedean, $\mathcal{O}_F$ is its ring of integers, ϖ is a uniformizer, and k is the residue field. We choose a split reductive group G with Borel subgroup B and torus T . We let W_F be the local Weil group.

Our goal is to classify the irreducible complex representations of $G(F)$ by certain L -parameters.

Definition 140 (Weil–Deligne). A Weil–Deligne representation of W_F in $\widehat{G}(\mathbb{C})$ is a pair (ρ, e) , where $\rho: W_F \rightarrow \widehat{G}(\mathbb{C})$ is a representation, and $e \in \text{Lie } \widehat{G}(\mathbb{C})$ is nilpotent, and

$$\rho(w)e\rho(w)^{-1} = |w|e,$$

where $|\cdot|$ is a suitable norm on W_F . An L -parameter of G is a Weil–Deligne representation (ρ, e) where ρ is semisimple.

Conjecture 141 (Local Langlands). Fix everything as above. Then there is a parameterization of $\text{Irr}_{\mathbb{C}} G(F)$ by triples (ρ, e, χ) , where (ρ, e) is an L -parameter, and χ is an irreducible representation of the component group $\pi_0 Z_{\widehat{G}}(\rho, e)$.

Remark 142. The data of χ is some kind of local system on the moduli space.

Remark 143. One expects this parameterization to be a bijection if we add in the pure inner forms of G .

A lot is known about this conjecture: it's known for some classes of representations, and it's known for classical groups. However, the general case is still open.

Let's prove a special case. Now, suppose that G admits an integral model, so we may let $K = G(\mathcal{O}_F)$ be a maximal compact, and let $I \subseteq F$ be the Iwahori subgroup, which is the preimage of the Borel $B(k) \subseteq G(k)$.

Definition 144 (unramified). An irreducible representation π is *unramified* if and only if $\pi^K \neq 0$.

Theorem 145 (Unramified local Langlands). Fix everything as above. Then there is a parameterization of $\text{Irr}_{\mathbb{C}}^{\text{unr}} G(F)$ by triples $(\rho, 1, 1)$, where $(\rho, 1)$ is an L -parameter, ρ is trivial on the inertia subgroup I_F .

Idea. The idea is that unramified irreducible representations are identified with simple module for the Hecke algebra $\mathcal{H}(G, K) := C_c^\infty(K \backslash G / K)$. Then the Satake isomorphism can identify $\mathcal{H}(G, K)$ with $\mathbb{C}[X_*(T)]^W$, which equivalently is $\mathbb{C}[X^*(\widehat{T})]^W$, which in turn parameterizes $\widehat{G}(\mathbb{C})$ -representations, which can be turned back into semisimple elements. ■

Remark 146. One can parameterize the triple $(\rho, 1, 1)$ by merely understanding the image $\rho(\text{Frob}) \in \widehat{G}(\mathbb{C})$, which is a semisimple element considered up to conjugacy.

11.2 Iwahori Local Langlands

Here is another special case.

Definition 147 (Iwahori). An irreducible representation π is *Iwahori* if and only if $\pi^I \neq 0$. This collection is $\text{Irr}_{\mathbb{C}}^{\text{IW}}(G(\mathbb{C}))$.

On the Galois side, we take tame ramification.

Definition 148 (tamely ramified). A Weil–Deligne representation (ρ, e) is *tamely ramified with unipotent monodromy*, abbreviated TRUM if and only if $\rho|_{I_F} = \text{id}$.

Remark 149. Note that (ρ, e) has allowed e to be arbitrary! However, the data of ρ still only amounts to the data of a semisimple element $s := \rho(\text{Frob})$ in $\widehat{G}(\mathbb{C})$. One can check that the condition $\rho(w)e\rho(w)^{-1} = |w|e$ is equivalent to having $ses^{-1} = qe$, meaning that we are looking at a pair (s, e) of a semisimple element s and a nilpotent element e which commute up to the scalar q .

Theorem 150 (Iwahori local Langlands). Fix everything as above. Then there is a parameterization of $\text{Irr}_{\mathbb{C}}^{\text{IW}} G(F)$ by triples (s, e, χ) , where $ses^{-1} = qe$ and χ is an irreducible representation of $\pi_0 Z_{\widehat{G}}(s, e)$.

This parameterization is not expected to be bijective: one needs to put some restriction on χ . The proof idea uses the Hecke algebra $C_c^\infty(I \backslash G/I)$.

Let's say something about the proof. On the automorphic side, it's worth noting that our representations are not too scary. These representations are not too scary.

Theorem 151 (Borel). An irreducible representation π is Iwahori if and only if π is a quotient of some unramified principal series $I_B^G \chi$ (meaning that $\chi|_{T(\mathcal{O}_F)} = 1$).

Let's use this to prove the Iwahori local Langlands conjecture for $G = \text{SL}_2(F)$.

Example 152. Take $G = \text{SL}_2(F)$ with $T = F^\times$, and our unramified character looks like $|\cdot|^s$ where $s \in \mathbb{C}$. Then the induction $\pi(\chi) := I_B^G(\chi)$ is unramified unless χ is $|\cdot|^{\pm 1}$ or $|\cdot|^{\pi i / \log q}$. (The latter is a quadratic character.) Working with $|\cdot|$ discovers the Steinberg representation, and working with the quadratic character produces an unramified representation and a new ramified representation.

Example 153. Let's classify (ρ, e, χ) for $\widehat{G}(\mathbb{C}) = \text{PGL}_2(\mathbb{C})$. There are two choices for our nilpotent, up to conjugacy.

- With $e = 0$, s can be arbitrary, written $s = \text{diag}(t, 1)$. Then $Z_{\widehat{G}}(s, e)$ is $\widehat{G}$ if $t = 1$, it is $\widehat{T} \rtimes \mathbb{Z}/2\mathbb{Z}$ if $t = -1$ (because we can add in the Weyl group!), and it is T otherwise. Thus, $\pi_0 Z_{\widehat{G}}(s, e)$ is 1 if $t \neq -1$ and is $\mathbb{Z}/2\mathbb{Z}$ if $t = -1$.
- With $e = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, then we see that s must be $\text{diag}(\sqrt{q}, 1/\sqrt{q})$, and one finds that $Z_{\widehat{G}}(s, e) = 1$.

One can now match the above two classifications by hand.

- The representation for $(s, 0, 1)$ is the unramified subquotient of $I_B^G |\cdot|^s$.
- The representation for $(\text{diag}(-1, 1), 0, \chi)$ (with χ nontrivial) is π_μ , coming from the quadratic character.
- Lastly, the representation for the last parameter is Steinberg.

Let's now say something about the general case. One can identify Iwahori irreducible representations with the simple H_I -modules.

Theorem 154 (Ginzburg). Simple H_I -modules can be identified with simple modules of $K^{\check{G} \times \mathbb{C}^\vee}(\text{St}_{\check{G}})$.

One can then embed simple modules $K^{\check{G} \times \mathbb{C}^\vee}(\text{St}_{\check{G}})$ into the enhanced Langlands parameters.

11.3 The Steinberg Variety

Of course, we are owed a definition of St_G .

Definition 155 (Steinberg variety). Let G be a reductive group over $\mathbb{C}$. Then we let $\mathcal{B}$ be the space of Borel subgroups, and we let $\mathcal{N} \subseteq \mathfrak{g}$ be the set of nilpotent elements. Then we let the *Steinberg variety* St_G to be $\tilde{N} \times_N \tilde{N}$ to consist of triples (e, b_1, b_2) where $b_1, b_2 \in \mathcal{B}$ and e is in the intersection $U_1 \cap U_2$.

Remark 156. Recall that there is a Springer resolution $\tilde{N} \rightarrow \mathcal{N}$ given by the tautological bundle on $\mathcal{B}$.

Notation 157. For each $w \in W$, we let $\text{St}_{G,w}$ be the pre-image of the subset Y_w , where $\{Y_w\}_w$ is a suitable decomposition of $\mathcal{B} \times \mathcal{B}$.

The geometry of these Steinberg varieties are rather interesting.

Remark 158. It turns out that $\text{St}_G = \bigsqcup_{w \in W} \text{St}_{G,w}$.

Remark 159. One can check that $\text{St}_{G,w}$ is the cotangent bundle of $\mathcal{B} \times \mathcal{B}$ to Y_w . Thus, $\text{St}_{G,w}$ is a Lagrangian subspace of $\tilde{N} \times \tilde{N}$. Computing its closure is in general difficult.

Example 160. For $G = \text{SL}_2(\mathbb{C})$, one finds that $\mathcal{B}$ is $\mathbb{P}^1$. The nilpotent cone $\mathcal{N}$ is two-dimensional. One can compute the Steinberg variety by hand. One finds that Y_1 is the closed subset of points (e, e) (which is the diagonal), and the Steinberg has points (e, e, e) . Then Y_2 is the complement.

Theorem 161. Fix everything as above. Then $H_{\text{top}}^{\text{BM}}(\text{St}_G)$ is isomorphic to $\mathbb{C}[W]$.

The moral is that we have a Springer correspondence sending irreducible representations W to pairs (e, χ) where e is nilpotent, and χ is an irreducible representation on $\pi_0 Z_G(e)$. Namely, consider the homology $H_{\text{top}}^{\text{BM}}(\mathcal{B}e)$, where $\mathcal{B}e$ is the Springer fiber over e . Then note that $H_{\text{top}}^{\text{BM}}(\mathcal{B}e)$ admits an action by $Z_G(e)$, so one receives a decomposition

$$H_{\text{top}}^{\text{BM}}(\mathcal{B}e) = \bigoplus_{\chi \in \text{Irr } \pi_0 Z_G(e)} E_\chi \otimes \chi,$$

and it turns out that this E_χ is an irreducible representation of W . The above theorem basically promises that varying over all e will recover the irreducible representations of W .

Example 162. For $G = \text{GL}_n$, one finds that $Z_G(e)$ is always trivial, so we are parameterizing irreducible representations of S_n by conjugacy classes of nilpotent elements. But such conjugacy classes are classified by partitions!

Remark 163. Set $\mathcal{B}_e^s := \{b \in \mathcal{B}e : sb = b\}$. If one restricts the Iwahori local Langlands correspondence to take χ appearing in $H_{\text{top}}^{\text{BM}}(\mathcal{B}_e^s)$, then it is a bijection!

11.4 Bezrukavnikov's Equivalence

We would like to build an analogue for the local Langlands correspondence which is less "mixed characteristic." Let k be an algebraically closed field, and G is a reductive group over $k((t))$. One can, as before, define an Iwahori subgroup $I \subseteq G$.

- On the automorphic side, there is the affine flag variety $\mathrm{Fl} := G((t))/I$, and we let $D(I \backslash G((t))/I)$ to be the derived category of quasicoherent sheaves on Fl .
- On the spectral side, we consider the derived category of $\check{G}$ -equivariant quasicoherent sheaves on the derived Steinberg variety $\mathrm{St}_G^L := \tilde{N} \times_{\mathfrak{g}_v}^L \tilde{N}$.

One statement of the equivalence is that the two categories defined above are equivalent. One can then say something about irreducible representations by passing to Grothendieck groups.

12 February 10: Euler Systems and Relative Cohomology

This talk was given by Christopher Skinner at MIT for the BC–MIT number theory seminar.

12.1 Euler Systems

Let's start with some motivation on Euler systems.

Theorem 164 (Dirichlet class number). Fix a prime q such that $q \equiv 1 \pmod{4}$, set $F := \mathbb{Q}(\sqrt{q})$, and let χ be the corresponding quadratic Galois character. Then

$$L(1, \chi) = \frac{h_F \log \varepsilon_F}{\sqrt{q}},$$

where ε_F is the fundamental unit.

Theorem 165 (Cyclotomic class number). Fix a prime q such that $q \equiv 1 \pmod{4}$, set $F := \mathbb{Q}(\sqrt{q})$, and let χ be the quadratic character $\pmod{q}$. Then

$$L(1, \chi) = -\frac{\tau(\chi)}{q} \sum_{a=1}^{q-1} \chi(a) \log |1 - \zeta_q^a|.$$

Remark 166. Taking χ to be the quadratic character gives a formula for an interesting unit in $\mathcal{O}_{\mathbb{Q}(\sqrt{q})}$. It turns out to generate a subgroup of index $2h_F$.

For this talk, the goal will be to take an explicit special value formula (like the second theorem), an algebraic result (like the remark), and then we hope to prove a special value result like the analytic class number formula. There is a similar story for elliptic curves, and it is the general story for Euler systems, which provide the algebraic input for when we already have an explicit formula for an L -value (e.g., the Gross–Zagier formula).

Let's be more explicit about what our Euler systems are.

Notation 167. Fix a nonarchimedean local field K over $\mathbb{Q}_p$, let V be a finite-dimensional vector space over K which admits an action by $G_{\mathbb{Q}}^S$, which is the absolute Galois group unramified away from a finite set of places S . Further, we choose a Galois-stable lattice $T \subseteq V$.

Example 168. One can take $V = \mathbb{Q}_p(\chi)$ with $T = \mathbb{Z}_p(\chi)$ for some character χ .

Example 169. One can take T to be the Tate module of an elliptic curve.

Definition 170 (Euler system). Fix a collection $\mathcal{L}$ of primes, and let $\mathcal{N}$ be the collection of products from $\mathcal{L}$ which are squarefree away from some prime p . Then an *Euler system* is a collection of classes $\{c_m\}_{m \in \mathcal{N}}$ where $c_m \in H^1(\mathbb{Q}[m]; T)$, satisfying some corestriction conditions

$$\text{cor}_{\mathbb{Q}[\ell m]/\mathbb{Q}[m]} c_{\ell m} = \begin{cases} P_\ell(\text{Frob}_\ell^{-1}) & \text{if } \ell \neq p, \\ c_m & \text{if } \ell = p \text{ and } p \mid n, \end{cases}$$

where P_ℓ is the characteristic polynomial of Frobenius acting on V^* (hence the term “Euler system”).

Remark 171. Kolyvagin’s approach requires producing ramified classes. This is difficult in general, so here the idea is to find ramified classes in highly ramified fields $\mathbb{Q}[m]$.

Kolyvagin then shows that $\mathcal{L}$ and $\text{im } \rho$ being big (for Chebotarev reasons) allows one to bound the size of certain Selmer groups, of the form

$$\#H_p^1(\mathbb{Q}; T^* \otimes \mathbb{Q}_p/\mathbb{Z}_p) \leq \# \left(\frac{H^1(\mathbb{Q}; T)}{c_1 \mathcal{O}} \right)$$

for the choice of class c_1 . If one takes a Bloch–Kato logarithm of c_1 , then one frequently finds a special value of a p -adic L -function.

12.2 Where to Find Them

We are interested in finding more Euler systems. For our purposes, recall that $H^1(F; V)$ classifies extensions

$$0 \rightarrow V \rightarrow \mathcal{E} \rightarrow K \rightarrow 0.$$

For our applications, V comes from algebraic geometry, and K has weight zero, so $\mathcal{E}$ is mixed and will need to come from somewhere interesting. Here is our idea: choose a subvariety $Y \subseteq X$ of codimension 1. Then we have a composite

$$H^{d-1}(Y; \mathbb{Q}_p) \rightarrow H^d((X, Y); \mathbb{Q}_p) \rightarrow H^d(X; \mathbb{Q}_p).$$

We hope that V can be found as a quotient of $H^{d-1}(Y; \mathbb{Q}_p)$, and $\mathbb{Q}_p$ can be found inside $H^d(X; \mathbb{Q}_p)$ (possibly up to twisting), meaning that we can find our extensions from $H^d((X, Y); \mathbb{Q}_p)$. These will be behaved because frequently $H^d(X; \mathbb{Q}_p)$ is often associated to nice automorphic forms (where X is a Shimura variety), so periods appear.

Remark 172. Intuitively, we are replacing previous constructions of Euler systems (found from units in K -groups) with something more motivic (e.g., Nori’s construction of mixed motives from relative cohomology).

Here are some examples.

Example 173. We can take X to be the modular curve minus the cusps at 0, and Y to be the cusps at ∞ . Then classes in $H^1(X; \mathbb{Q}_p)$ can come from certain differences of Eisenstein series, and there is a quotient map $H^0(Y; \mathbb{Q}_p) \rightarrow V$ given by evaluation. This recovers the Euler system for abelian characters of $\mathbb{Q}$.

Example 174. Replacing Y with CM points in the previous example, we recover elliptic units.

Theorem 175. Fix a modular form f . Then there is an Euler system for the symmetric square of f and an explicit value formula.

Proof. Rerun the argument for the embedding of Shimura varieties coming from $\mathrm{Sp}_2 \times \mathrm{Sp}_2 \rightarrow \mathrm{Sp}_4$. Then the idea is to use the doubling method to express $L(\mathrm{Sym}^2 f \otimes \chi, s)$ as an integral. ■

This method can also recover Kato's Euler system, an Euler system for the triple product, and infinitely many more.

13 February 13: More Background

This talk was again given by Mitch and then Xin at Johns Hopkins University for the student automorphic representations seminar.

13.1 Elliptic Representations

This part of the talk was given by Mitch. As usual, we are working with a reductive group G over a local field F .

Definition 176 (regular). An element $g \in G$ is *regular* if and only if its centralizer has minimal dimension, which is the dimension of a maximal torus.

Definition 177 (elliptic). A regular semisimple element $g \in G$ is *elliptic* if and only if its centralizer (modulo center) has no split part.

Example 178. Generic elements in $\mathrm{SO}(2, \mathbb{R}) \subseteq \mathrm{SL}(2, \mathbb{R})$ are elliptic. For example, this shows that the collection of elliptic elements is a semi-algebraic set in $\mathrm{SL}(2, \mathbb{R})$ of positive measure.

Remark 179. It turns out that an element $g \in G$ is elliptic if and only if it is contained in no proper parabolic.

We now turn to representations. Recall the following theorem.

Theorem 180 (Harish-Chandra). Fix a smooth representation π of G . Then the character Θ_π (which is a priori only a distribution) is represented by a locally integrable function on the regular semisimple elements.

Definition 181 (elliptic). An irreducible unitary representation π of $G(F)$ is *elliptic* if and only if π is non-vanishing on the collection of elliptic elements.

Example 182. The Steinberg representation is elliptic.

Remark 183. The character of a representation induced from a parabolic subgroup automatically has character vanishing on the elliptic elements.

Let's use our new adjective to classify representations. The irreducible tempered representations of G can always be found in a parabolic induction $I_P^G(\sigma)$, where σ is induced from a Levi M . Above a fixed σ , one can break the tempered representations of Π_σ into those which are finite and related to the representation theory of some finite group $\tilde{R}_\sigma$, which is a central extension of some subquotient of the Weyl group $W(M)$. (In particular, it is a quotient of the subgroup of $W(M)$ stabilizing σ .)

Now, elements of $\tilde{R}_\sigma$ are considered “regular.” Its characters which don’t vanish on the regular elements correspond to elliptic elements. Now, given a triple (M, σ, r) with $r \in \tilde{R}_\sigma$, one can produce a virtual distribution character Θ .

The space $\mathfrak{X}_{\text{ell}}(G)$ of triples where r is regular parameterizes elliptic elements. Then $\mathfrak{X}(G)$ can be constructed as the virtual space of representations induced from $\mathfrak{X}_{\text{ell}}(M)$ as M varies over the Levi subgroups. In particular, $\mathfrak{X}(G)$ is a subset of the free vector space on the tempered representations of G . The moral is that we have found a manifold which (roughly speaking) parameterizes representations.

13.2 The Weyl Integral Formula

Fix an additive character ψ on a local field F , which induces a duality $F \rightarrow F^*$. Further, fix a bilinear form B on $\mathfrak{g}(F)$ (where $\mathfrak{g}$ is the Lie algebra of G).

Definition 184 (Fourier transform). Fix a function f on $\mathfrak{g}(F)$. Then we define the *Fourier transform*

$$\hat{f}(X) := \int_{\mathfrak{g}(F)} \psi(Y) \psi(B(X, Y)) dY.$$

Notation 185. Let $\Gamma(G)$ be the collection of semisimple conjugacy classes of G .

Remark 186. Let $\mathcal{T}(G)$ be the collection of maximal tori in G . There is a way to put measures on all of them so that for a function $f \in C_c(\Gamma(G))$, we have

$$\int_{\Gamma(G)} f(x) dx = \sum_{T \in \mathcal{T}(G)} \frac{1}{\#W(G, T)} \int_{T(F)} f(t) dt.$$

These measures on the tori should be chosen compatibly with the chosen Haar measure on G .

In life, one wants to integrate on $G(F)$. The following decomposition is occasionally helpful.

Theorem 187 (Weyl integral formula). One has

$$\int_{G(F)} f(g) dg = \int_{\Gamma(G)} D^G(x)^{1/2} J_G(x, f) dx,$$

where

$$D^G(x) = |\det(1; \text{Ad}_x; \mathfrak{g}/\mathfrak{g}_x)|,$$

and

$$J_G(x, f) = D^G(x)^{1/2} \int_{G_x \backslash G} f(g^{-1} x g) dg.$$

Remark 188. Similarly, one has

$$\int_{\mathfrak{g}(F)} f(X) dX = \int_{\Gamma(\mathfrak{g})} D^G(X)^{1/2} J_G(X, f) dX,$$

where D^G and J_G are defined similarly.

Remark 189. It turns out that the functions D^G are locally integrable, the functions $J_G(x, -)$ are tempered distributions, and the functions $J_G(-, f)$ are locally bounded.

Idea. This formula is basically proven by passing from an integral over G to an integral over the covering $(G/T \times T)/W$. ■

13.3 Orbital Integrals

We are thus motivated to compute some orbital integrals.

Notation 190. For some conjugacy-stable subset $\mathcal{O}$ in the nilpotent cone, we define

$$J_{\mathcal{O}}(f) := \int_{\mathcal{O}} f(X) dX.$$

Remark 191. For a scalar $\lambda \in F^\times$, define $f_\lambda(x) := f(\lambda^{-1}x)$, we have

$$J_{\mathcal{O}}(f_\lambda) = |\lambda|^{-\dim \mathcal{O}/2} J_{\mathcal{O}}(f).$$

Here, if $\mathcal{O}$ is a regular element, then $\dim \mathcal{O}$ is $\dim G - \dim T$.

Remark 192. Any G -invariant distribution on $\mathfrak{g}(F)$ which is supported on the nilpotent cone and homogeneous in the sense of the previous remark, then it is a linear combination of some $J_{\mathcal{O}}$ s, where $\mathcal{O}$ varies over orbits in the regular nilpotent cone. In the p -adic case, this is not so hard because the nilpotent cone has only a finite number of orbits.

These orbital integrals also have some nice duality: the Fourier transform is represented by a smooth function!

Theorem 193. Fix a nilpotent orbit $\mathcal{O}$. Then

$$J_{\mathcal{O}}(\hat{f}) = \int_{\mathfrak{g}(F)} \hat{j}_{\mathcal{O}}(Y) f(Y) dY,$$

where $\hat{j}_{\mathcal{O}}$ is some smooth function on $C^\infty(\mathfrak{g}_{\text{reg}})$. In fact, $\hat{j}_{\mathcal{O}}(\lambda X) = |\lambda|^{-\dim \mathcal{O}/2} \hat{j}_{\mathcal{O}}(X)$.

Remark 194. Here is a global version of this statement: on G , we have

$$J_G(X, \hat{f}) = \int_{\mathfrak{g}(F)} \hat{j}(X, Y) f(Y) dY,$$

where $\hat{j}$ is some smooth locally integrable function on $\mathfrak{g}_{\text{reg}} \times \mathfrak{g}_{\text{reg}}$.

For certain applications (e.g., to the trace formula), we need to further weight our orbital integrals.

Notation 195. Fix a Levi subgroup M of G , and let K be a maximal compact subgroup of $M(F)$. For a choice of parabolic P with Levi subgroup M , define $\mathcal{A}_M := \text{Hom}(X^*(M), \mathbb{R})$, and then the theory of (G, M) families produces a volume term $\nu_M(g)$, which is the volume of convex hull of $H_p(g) \in \mathcal{A}_M / \mathcal{A}_G$ (where H_p is a certain function $H_P: G(F) \rightarrow \mathcal{A}_M$).

Remark 196. In fact, there is a more refined weight $\nu_L^Q(g)$, where L is a Levi containing M , and Q is a parabolic whose Levi contains L . One has

$$J_M(x, f) = D^G(x)^{1/2} \int_{G_x \backslash G} \nu_M(g) f(g^{-1}xg) dg$$

for any $x \in M$ which is regular in G , and the refined weight ν_Q^L allows us to define a weighted orbital integral J_L^Q . (It turns out that J_G^G is the full orbital integral on the regular elements of G .)

Now, our trace formula will not be defined for all functions. (Indeed, on a continuous family of representations, the trace is generally infinite.) To fix this, we need a cuspidality assumption.

Definition 197 (strongly cuspidal). Fix a function f . For a parabolic $P = MU$, we define

$$f^P(m) = \delta_P(m)^{1/2} \int_U f(mu) du$$

and

$$\varphi^P(X) = \int_U \varphi(X + N) dN.$$

Then f is *strongly cuspidal* if and only if $f^P = 0$ and $\varphi^P = 0$ for all parabolics P .

Remark 198. It is not hard to see that many orbital integrals vanish if f is strongly cuspidal, which will allow our trace formula to converge. Let's be more precise about this. On the spectral side, the "leading terms" in the trace all come from some continuous spectrum which are induced representations (the more induced, the higher the term is). We are interested in the "constant term," which is some invariant distribution, and we will only be able to see it by getting all the induced pieces to vanish. Thus, we will want to plug in strongly cuspidal functions. (A similar story holds on the geometric side with our orbital integrals.)

Remark 199. For elements $x \in M$ which are regular in G , one has $J_L^Q(x, f) = 0$ unless $L = M$ and $Q = G$.

Remark 200. The weighted orbital integral $J_M^G(x, f)$ does not depend on K , and it vanishes unless x is an elliptic element in M .

13.4 Quasicharacters

A number of the items $(\Theta_\pi, \hat{j}, \text{ and } \hat{j}_\mathcal{O})$ defined today turn out to have many properties in common. For this subsection, F is p -adic.

Definition 201 (quasicharacter). Fix a G -invariant open subset $\omega \subseteq \mathfrak{g}(F)$. Then we define a *quasicharacter* on ω to be a G -invariant function θ on the regular elements of ω satisfying the following: for all semisimple x and open neighborhood ω_X of X such that $\omega_X^G \subseteq \omega$, there are coefficients $c_\mathcal{O}$ for which

$$\theta(Y) = \sum_{\mathcal{O}} c_\mathcal{O}(X) \hat{j}_\mathcal{O}(Y),$$

where $\mathcal{O}$ varies over the nilpotent orbits of $\mathfrak{g}_X$. In other words, θ is locally a linear combination of Fourier transforms of nilpotent orbital integrals.

Remark 202. For example, on regular semisimple elements, we are required to locally look the Fourier transform of δ_0 , which is constant. Thus, we are smooth on the regular semisimple elements. Away from regular semisimple elements, we are allowed certain singularities.

Remark 203. There is a similar definition in the Lie group case.

Remark 204. There is a definition in the real case, which requires one to take into account some derivatives as well.

Example 205. We can see that $\hat{j}_{\mathcal{O}}$ is a quasicharacter for each nilpotent orbit $\mathcal{O}$.

Example 206. For irreducible admissible representations π , it turns out that Θ_π is a quasicharacter.

14 February 27

This talk was given by Matthew at Johns Hopkins University for the automorphic forms learning seminar.

14.1 Parabolic Induction on Unitary Groups

Here is a lemma to get us started. Let (G, H, ξ) be a GGP triple; for example, if H lives in codimension 1, then ξ is trivial.

Lemma 207. Let π be in $\mathfrak{X}_{\text{temp}}(G)$, which means that it is the parabolic induction of some σ which is from L^2 . For any $S, T \in \text{End}(\pi)^\infty$, one has $SL_\pi T$ is smooth, and

$$\mathcal{L}_\pi(S)\mathcal{L}_\pi(T) = \mathcal{L}_\pi(SL_\pi T).$$

Here, $\mathcal{L}_\pi$ is some intertwiner built from matrix coefficients of π , and $L_\pi: \pi^\infty \rightarrow \overline{\pi^{-\infty}}$ sends a vector e to $\mathcal{L}_\pi(e, -)$.

Proof. We won't say anything about $SL_\pi T$ being smooth. Now, $L_\pi S$ and $L_\pi T$ are endomorphisms of $\overline{\pi^{-\infty}}$, and they have image in $\text{Hom}_H(\overline{\pi^{-\infty}}, \xi)$. This space has dimension at most 1 if π is irreducible, in which case one can calculate

$$\begin{aligned} \mathcal{L}_\pi(SL_\pi T) &= \text{tr } L_\pi SL_\pi T \\ &= \text{tr}(L_\pi S) \text{tr}(L_\pi T) \\ &= \mathcal{L}_\pi(S)\mathcal{L}_\pi(T). \end{aligned}$$

Now, irreducibles are dense in $\mathfrak{X}_{\text{temp}}(G)$, and both sides are continuous in π , so we are done! ■

We now want to understand how $\mathcal{L}_\pi$ interacts with induction. Suppose that $P \subseteq \text{U}(V)$ is the stable of some flag of (totally) isotropic subspaces

$$0 = X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_k \subsetneq V.$$

In this situation, the opposite parabolic is the stabilizer of the opposite flag $\{\tilde{X}_i\}$, where $X_i \oplus \tilde{X}_i$ is always a non-degenerate hyperbolic quadratic space. Then the intersection of P with its opposite is the Levi subgroup.

Observe that we may extend our isotopic flag to some flag

$$0 = X_0 \subseteq X_1 \subsetneq \cdots \subseteq X_{k-1} \subsetneq X_k \subsetneq X_{k-1}^\perp \subsetneq \cdots \subsetneq X_1^\perp \subsetneq X_0^\perp = V.$$

A choice of opposite flag allows us to split these quotients as follows, which one can check by chasing around the splittings of the quadratic spaces.

- The quotients X_i/X_{i-1} are given by $X'_i := X_i \cap \tilde{X}_{i-1}^\perp$.
- The quotients $X_{i-1}^\perp/X_i^\perp$ are given by $X'_{-i} := \tilde{X}_i \cap X_{i-1}^\perp$.
- Lastly, the quotient $X_k^\perp/X_k$ is given by $X'_{k+1} := X_k^\perp \cap \tilde{X}_k^\perp$.

Notably, we see that the spaces X'_i and X'_{-i} are all totally isotropic, and by construction, $X'_i \oplus X'_{-i}$ are mutually orthogonal. It turns out that the Levi subgroup is thus exactly given by

$$\mathrm{GL}_E(X'_1) \times \cdots \times \mathrm{GL}_E(X'_k) \times \mathrm{U}(X'_{k+1}).$$

For example, we see that $\mathrm{U}(V)$ is quasi-split if and only if it contains a Borel, which means that there is a toral Levi subgroup, which then corresponds to being able to find a maximal isotropic flag $\{X_i\}_{i=1}^{\lfloor \dim V/2 \rfloor}$.

Now, take $G := \mathrm{U}(W) \times \mathrm{U}(V)$, and let $Q := Q_W \times W_V$ be some parabolic with Levi subgroups L_W and L_V , which we expand out as

$$L_W = \mathrm{GL}_E(X_1) \times \cdots \times \mathrm{GL}_E(X_a) \times \mathrm{U}(\tilde{W}) \quad \text{and} \quad L_V = \mathrm{GL}_E(Y_1) \times \cdots \times \mathrm{GL}_E(Y_b) \times \mathrm{U}(\tilde{V}).$$

Lemma 208. If W and V are compatible Hermitian spaces (meaning that W includes into V , and the other direct summand is an odd-dimensional space Z where $\mathrm{U}(Z)$ is quasi-split), and W can be decomposed as a sum of $\tilde{W}$ and some hyperbolic space, then $\tilde{W}$ and V are compatible.

Proof. One applies Witt's cancellation theorem. ■

Applied to the above situation, one finds that $\tilde{W}$ and $\tilde{V}$ are compatible, so one gets a smaller GGP triple from these Hermitian spaces.

Proposition 209. Let σ be a tempered representation of $L(F)$ of the form $\sigma_{\mathrm{GL}} \boxtimes \tilde{\sigma}$, where σ_{GL} is a $\boxtimes$ -product of irreducible tempered representations of GL_n factors, and $\tilde{\sigma}$ is tempered. Letting π be the parabolic induction, one has that $\mathcal{L}_\pi$ is nonzero if and only if $\mathcal{L}_{\tilde{\sigma}}$ is nonzero.

Proof. By induction on stages, one can reduce to the case where $a, b \leq 1$. By further introducing new GGP triples, one reduces to the case $a = 0$ and $b = 1$ and W has codimension 1 in V . In this case, the operator $\mathcal{L}_\pi$ is absolutely convergent, and one can check everything. ■

14.2 The Main Result

Let's now sketch the following theorem.

Theorem 210. If $m(\pi_0)$ is nonzero, then $\mathcal{L}_{\pi_0} \neq 0$.

Proof. Because $m(\pi_0)$ is nonzero, there is a nonzero intertwiner ℓ in $\mathrm{Hom}_H(\pi_0^\infty, \xi)$. Now, for any $e \in \pi_0^\infty$ and function f , we see that

$$\ell(\pi_0(f)e) = \int_{G(F)} f(g) \ell(\pi_0(g)e) dg,$$

which can be rearranged into

$$\int_{H(F) \backslash G(F)} \ell(\pi(g)e) \int_{H(F)} (R(g)f)(h) \xi(h) dh dg.$$

If f is a Paley–Weiner function for G , meaning that the map $\pi \mapsto \pi(f)$ is smooth and compactly supported, then the inner integral is

$$\int_{\mathfrak{X}_{\text{temp}}(G)} \mathcal{L}_{\pi}(\pi(R(g)f)) \mu(\pi) d\pi$$

by the Plancherel theorem. Note that $\pi(R(g)f) = \pi(f)\pi(g^{-1})$, so we see that

$$\ell(\pi_0(f)e) = \int_{H(F) \backslash G(F)} \ell(\pi_0(g)e) \int_{\mathfrak{X}_{\text{temp}}(G)} \mathcal{L}_{\pi}(\pi(f)\pi(g^{-1})) \mu(\pi) d\pi dg.$$

We now apply various test functions f . For example, if T is a smooth and compactly supported section of the bundle $\mathcal{E}(G)$ of smooth endomorphisms over $\mathfrak{X}_{\text{temp}}(G)$, then we could define f to have $\pi(f) = T_{\pi}$ by some argument with Fourier transforms. Then

$$\ell(T_{\pi_0}e) = \int_{H(F) \backslash G(F)} \ell(\pi_0(g)e) \int_{\mathfrak{X}_{\text{temp}}(G)} \mathcal{L}_{\pi}(T_{\pi}\pi(g^{-1})) \mu(\pi) d\pi dg \quad (14.1)$$

Now, π_0 is found in some induction $\text{Ind}_Q^G \sigma$, where Q has Levi quotient L . We now choose $e \in \pi_0^{\infty}$ so that $\ell(e) \neq 0$, and we may choose our section T^0 so that $T_{\text{Ind}_Q^G \sigma}^0 e = e$ (in particular, $T_{\pi_0}^0 e = e$ as well) and has support contained in the component $\{\text{Ind}_Q^G \sigma_{\lambda} : \lambda \in i\mathfrak{a}_L^*\}$ of σ . In this situation, the above equation tells us that there is some λ for which $\mathcal{L}_{\text{Ind}_Q^G(\sigma_{\lambda})} \neq 0$.

We now decompose σ as $\sigma_{\text{GL}} \boxtimes \tilde{\sigma}$ and $\sigma_{\lambda} = \sigma_{\text{GL}, \lambda} \boxtimes \tilde{\sigma}$. Thus, by Proposition 209, we see that $\mathcal{L}_{\text{Ind}_Q^G(\sigma)} \neq 0$. This is almost $\mathcal{L}_{\pi_0}$! To finish, choose our section T^1 so that

$$\mathcal{L}_{\text{Ind}_Q^G(\sigma)}(T_{\text{Ind}_Q^G(\sigma)}^1) \neq 0.$$

Then we define a section T^2 to have $T_{\pi}^2 := \mathcal{L}_{\pi}(T_{\pi}^1) T_{\pi}^0$. Now plugging in T^2 to (14.1), we see that $\ell(T_{\pi_0}^2 e)$ is nonzero (because this is the same thing as for $\text{Ind}_Q^G(\sigma)$, so we can reapply (14.1) with the representation $\text{Ind}_Q^G(\sigma)$ replacing π_0) and equals

$$\mathcal{L}_{\pi_0}(T_{\pi_0}^1) \ell(T_{\pi_0}^0 e) = \int_{H(F) \backslash G(F)} \ell(\pi_0(g)e) \int_{\mathfrak{X}_{\text{temp}}(Q)} \mathcal{L}_{\pi}(T_{\pi}^1) \mathcal{L}_{\pi}(T_{\pi}^0 \pi(g^{-1})) \mu(\pi) d\pi dg,$$

where we have passed to $\mathfrak{X}_{\text{temp}}(Q)$ due to the support of T^0 . But now the inner integrand is

$$\mathcal{L}_{\pi}(T_{\pi}^1 L_{\pi} T_{\pi}^0 \pi(g^{-1})),$$

which was shown earlier. Calling this $\mathcal{L}_{\pi}(T_{\pi_0}^3 e)$, we can go backwards and say that this is $\ell(T^3 \pi_0 e)$. But T_{π}^3 has L_{π} as a factor, so it is nonzero, so $\mathcal{L}_{\pi_0}$ is nonzero. ■

Corollary 211. Suppose that $\pi = \text{Ind}_Q^G(\sigma)$, where $\sigma = \sigma_{\text{GL}} \boxtimes \tilde{\sigma}$. Then $m(\pi) = m(\tilde{\sigma})$.

Corollary 212. Choose a component $C \subseteq \mathfrak{X}_{\text{temp}}(G)$. Then there is T for which $\mathcal{L}_{\pi}(T_{\pi}) = m(\pi)$ for all $\pi \in C$.

15 March 3: Geometry of the Automorphic Side

This talk was given by Kaiyi Chen at MIT for the tame local Langlands seminar.

15.1 The Kazhdan–Lusztig Isomorphism

This first part of the talk was given by Jialiang.

Theorem 213 (Kazhdan–Lusztig). There is an isomorphism

$$\mathcal{H}(G, I) \cong K^{\check{G} \times \mathbb{C}^\times}(\mathrm{St}_{\check{G}}).$$

Proof. Here, $\mathcal{H}(G, I) = C_c^\infty(I \backslash G / I)$ is the Iwahori Hecke algebra. It turns out to be a q -deformation of the affine Weyl group $X_*(T) \rtimes W$ and thus depends only on the root datum. Thus, there is a Bernstein presentation as follows: there are operators $\{T_w\}_{w \in W}$ satisfying some classical braid relations, there are generators θ_λ for $\lambda \in X^*(T)$ (which satisfy additivity relations), and there are commutativity relations

$$T_{s\alpha}\theta_\lambda - \theta_{s(\lambda)}T_{s\alpha} = (v - v^{-1}) \frac{\theta_{s\alpha(\lambda)} - \theta_\lambda}{1 - \theta_{-\alpha^\vee}}.$$

Note that sending $v \rightarrow 1$ recovers the extended Weyl group $X_*(T) \rtimes W$.

The idea of the proof is to embed both sides as certain endomorphism algebras of polynomial rings.

- On one hand, $\mathcal{H}(G, I)$ acts on $\mathcal{M} := \mathcal{H}_{\mathrm{ext}} \otimes_{\mathcal{H}_f} \det$, where $\det$ is a certain determinant representation given by $T_w \mapsto \det w$. This $\mathcal{M}$ is alternatively isomorphic to $\mathbb{C}[v, v^{-1}][X_*(T)]$.
- On the other hand, recall that $\mathrm{St}_{\check{G}} = \tilde{N} \times_N \tilde{N}$, where $\tilde{N}$ is the cotangent bundle of the flag variety $\check{G}/\check{B}$, which is the Springer resolution of N . Thus, our right-hand side admits a natural action on $K^{\check{G} \times \mathbb{C}^\times}(\tilde{N})$, which by a Thom isomorphism is identified with $K^{\check{G} \times \mathbb{C}^\times}(\check{G}/\check{B})$, which by induction is $K^{\check{B} \times \mathbb{C}^\times}(\mathrm{pt})$, which is then isomorphic to $\mathbb{C}[v, v^{-1}][X^*(T^\vee)]$.

It remains to show that the two representations are injective and are the same. Local methods shows that the representations are faithful. One can show that the image of the first one is contained in the image of the second one by tracking the action of certain generators. This produces a map from the left to the right, and one checks that it is an isomorphism with a rank calculation. ■

Remark 214. Bezrukavnikov's equivalence is a certain categorification of this argument.

Remark 215. One can show that $\mathcal{M}$ in the above proof is $C_c^\infty((U, \psi) \backslash G / I)$ as an Iwahori Hecke module.

15.2 Motivation

Fix a local field F with absolute Galois group Γ_F , and let G be a quasi-split reductive group. We will also let $\check{F}$ denote the unramified closure.

Recall that the classical local Langlands conjectures wants to associate irreducible smooth representations of $G(F)$ and Langlands parameters $W_F \rightarrow {}^L G$, where W_F is the Weil group. In general, this map is only expected to be surjective (after passing to certain families of inner forms of G). To make the map injective, we should enhance the Langlands parameter by attaching to $\varphi: W_F \rightarrow {}^L G$ an additional representation ρ of $C_{\check{G}}(\varphi)$.

This suggests that there is something stacky going on the spectral side.

Definition 216. We define the *stack of Langlands parameters* $\mathrm{Log}_{G, F}$ to consist of those Langlands parameters $\varphi: W_F \rightarrow {}^L G$ up to $\check{G}$ -conjugacy.

Thus, enhanced Langlands parameter forms are coherent sheaves on $\mathrm{Loc}_{G, F}$.

Now that there is some geometry on the spectral side, we may hope for some geometry on the automorphic side. For example, one now expects there to be an embedding $\mathrm{Rep} G(F) \rightarrow \mathrm{Coh}(\mathrm{Loc}_{G, F})$.

Conjecture 217. There is a reasonable equivalence

$$\bigcup_{L \text{ inner form}} \text{Rep } G'(F) \cong \text{Coh}(\text{Loc}_{L,G,F}),$$

where the left-hand side varies over inner forms of Levi subgroups of G .

Example 218. Let's explain why we may want to include Levi subgroups: even for $G = \text{GL}_2$, the point on the right-hand side coming from a principal series will have huge centralizer, thereby producing something like $[\text{pt}/\mathbb{G}_m]$ on the right-hand side. There is only one representation of G here, but we need tons of coherent sheaves, which are found by taking other inner forms coming from the Levi subgroup (which is the torus).

Descending to π_0 ought to produce the required bijection.

15.3 Isocrystals

The issue with our conjecture is that it is highly unclear how to take a union of categories. To this end, we will consider sheaves on the stack of G -isocrystals.

Definition 219 (isocrystal). A G -isocrystal is a G -torsor $\mathcal{E}$ on $\text{Spec } \check{F}$ equipped with an isomorphism $\mathcal{E} \rightarrow \sigma^* \mathcal{E}$.

Example 220. If $G = \text{GL}_n$, then we are looking at a vector space V over $\check{F}$ together with a bijective, Frobenius semi-linear endomorphism $\varphi: V \rightarrow V$. (Here, Frobenius semi-linear means that $\varphi(\lambda v) = \sigma(\lambda)\varphi(v)$.) This is the classical notion of an isocrystal.

Let's review the classical notion of isocrystal. For example, there is a classification.

Theorem 221 (Dieudonné–Manin). All isocrystals V over F are direct sums of simple isocrystals, and they are classified by their slope. Here, for each $\lambda = d/n$ with $\gcd(d, n) = 1$, the simple isocrystal V_λ is given by

$$V_\lambda := \check{F}^n,$$

where φ is defined by $\varphi(e_i) = e_{i+1}$ for $i < n$, and $\varphi(e_n) = \varpi^d e_1$.

One can make some similar theorems for G -isocrystals, which we will say something about later.

Proposition 222 (Kottwitz). The groupoid of G -isocrystals is the quotient of $G(\check{F})$ by $G(\check{F})$ with the adjoint action with $\text{Ad}_\sigma(g)(h) := gh\sigma(g)^{-1}$.

Sketch. All G -torsors on $\check{F}$ are trivial because $\check{F}$ has global dimension 1. Upon choosing an isomorphism (which is well-defined up to the adjoint action), we then use φ to produce an element of $G(\check{F})$, and the result follows. ■

Definition 223 (Kottwitz set). The Kottwitz set $B(G)$ is π_0 of the quotient $[G(\check{F})/\text{Ad}_\sigma G(\check{F})]$. For each $b \in B(G)$, we let $G_b(F)$ be its automorphism group

$$G_b(F) := \{g \in G(\check{F}) : gb\sigma(g)^{-1} = b\}.$$

While we're introducing definitions, here are some moduli spaces.

Definition 224 (loop group). The loop group LG is the functor

$$LG(R) := \text{Hom}(D_R^\times, G),$$

where $D_R^\times$ is the punctured disk. When $\text{char } F > 0$, then $D_R^\times := \text{Spec } R((t))$; for example, $D_k^* := \text{Spec } \check{F}$. When $\text{char } F = 0$, then we use the Witt vectors and have $D_R := \text{Spec } W_O(R)$ and $D_R^\times := \text{Spec } W_O(R)[1/\varpi]$. (Because we are using Witt vectors, we should only use perfect algebras R .)

Definition 225 (moduli of isocrystals). The moduli stack of isocrystals is $\text{Isoc}_G := [LG/\text{Ad}_\sigma LG]$.

Example 226. On a perfect algebra R , one finds that $\text{Isoc}_G(R)$ is the G -torsors $\mathcal{E}$ on $D_R^\times$ equipped with an isomorphism $\varphi: \mathcal{E} \rightarrow \sigma^* \mathcal{E}$, which trivializes étale-locally on R . The reason for the trivialization is that the quotient is taken with respect to the étale topology.

Example 227. If K is an algebraically closed extension of the residue field, then

$$\text{Isoc}_G(K) \cong [G(\check{F})/\text{Ad}_\sigma G(\check{F})].$$

For example, with $G = \text{GL}_n$, one can use the Dieudonné–Manin classification. The main content is that the stabilizers on both sides remains the same, which may be surprising if K is large. Roughly speaking, the idea is that even for large fields K of positive characteristic, its Frobenius-fixed points continue to be $\mathbb{F}_p$.

15.4 The Kottwitz Map

Let's return to trying to classify our G -isocrystals. In our above language, this amounts to trying to understand the set $B(G)$.

Theorem 228 (Kottwitz). There is a canonical embedding

$$B(G) \rightarrow X_*(T)_{+, \mathbb{Q}}^{\Gamma_F} \times \pi_1(G)_F.$$

The two coordinates are called the Newton point $\nu_\bullet$ and Kottwitz point $K_G(\bullet)$, respectively.

Remark 229. There is a description of the image of this map, but it is complicated. Heuristically, the Newton point is like the slope, so there is some integrality condition present, as in the Dieudonné–Manin classification.

Example 230. Consider $G = \text{GL}_n$. Then an isocrystal looks like

$$V = \bigoplus_i V_{d_i/r_i}^{\oplus m_i},$$

where $\sum_i m_i r_i = n$. Then we can form a Newton point to be the Newton polygon $(d_1/r_1, \dots, d_1/r_1, \dots)$, where d_i/r_i is required to appear $m_i r_i$ times. The Kottwitz point is just $\sum_i m_i d_i$, which is an integer.

Remark 231. We can give a partial ordering on $B(G)$ by saying that $b \leq b'$ if and only if $\nu_b \leq \nu_{b'}$ and $K_G(b) = K_G(b')$.

Remark 232. For each $b \in B(G)$, one sees that G_b is an algebraic group over F . In fact, it is an extended pure inner form of the Levi subgroup $M = C_G(\nu_b)$.

A partial ordering allows us to distinguish some elements.

Definition 233 (basic). A point $b \in B(G)$ is *basic* if and only if ν_b is central, meaning that G_b is an extended pure inner form of G . The locus of basic points is $B(G)_{\text{basic}}$.

Remark 234. It turns out that the elements $B(G)_{\text{basic}}$ are the minimal ones for the partial ordering.

Example 235. Consider $G = \text{GL}_n$ again. By examining the simples, we see that

$$G_b = \prod_i \text{GL}_{m_i}(D_{d_i/r_i}),$$

where $D_{d_i/r_i} = \text{Aut } V_{d_i/r_i}$ is a simple central division algebra of F ; it turns out to be the one corresponding to $d_i/r_i \in \text{Br } F$ (note we may identify $\text{Br } F = \mathbb{Q}/\mathbb{Z}$). Thus, basic points are exactly those where V is isotypic.

A partial ordering also allows us to stratify Isoc_G into $\text{Isoc}_{G, \leq b}$. Here is the main theorem on isocrystals for our purposes.

Theorem 236. Let $\text{Isoc}_{G, \leq b}$ consist of those points below b .

- (a) The topological space $|\text{Isoc}_G|$ is the partially ordered set $B(G)$ equipped with the partial ordering.
- (b) The map $\text{Isoc}_{G, \leq b} \rightarrow \text{Isoc}_G$ is a perfectly finitely presented closed embedding.
- (c) The map $\text{Isoc}_{G, b} \rightarrow \text{Isoc}_{G, \leq b}$ is an affine open embedding.
- (d) The stratum $\text{Isoc}_{G, b}$ is the pro-étale classifying space of $G_b(F)$.

Remark 237. This theorem tells us that Isoc_G is indeed able to glue together its extended pure inner forms of its Levi subgroups. Indeed, (d) tells us that each individual groupoid $\text{Isoc}_{G, b}$ has coherent sheaves which are representations of $G_b(F)$.

15.5 Shtukas

Let's say something about how our isocrystals relate to shtukas. The classical motivation is Deligne–Lusztig theory. Fix a reductive group G over a finite field k . It turns out that G is quasi-split by some version of Lang's theorem, so we may give it a Borel subgroup B . Deligne–Lusztig theory amounts to a long composite

$$B \backslash G / B \leftarrow [G / \text{Ad}_\sigma B] \xrightarrow{\text{Nt}} [G / \text{Ad}_\sigma B] \cong \mathbb{B}G(k).$$

The first map is an isomorphism, the second map is the Newton map, and the last map is an identification. The moral is that there is push-pull map

$$\text{Ch}: \text{Shv}(B \backslash G / B) \rightarrow \text{Rep } G(k).$$

For example, each $w \in W$ produces an embedding $B \backslash BwB / B \rightarrow B \backslash G / B$. Pushing forward the constant sheaf on this point recovers the classical Deligne–Lusztig representations R_w^* .

We now repeat this idea for the affine case. Accordingly, we want to write down some composite

$$I \backslash LG/I \xleftarrow{\delta} [LG/\mathrm{Ad}_\sigma I] \xrightarrow{\mathrm{Nt}} [LG/\mathrm{Ad}_\sigma LG].$$

The middle object will be the moduli of local shtukas.

Definition 238 (local Hecke stack). The *local Hecke stack* Hk is the quotient $I \backslash LG/I$, where I is the Iwahori subgroup. Given an algebra R , $\mathrm{Hk}(R)$ parameterizes pairs of I -torsors $\mathcal{E}_1$ and $\mathcal{E}_2$ on D_R , equipped with an isomorphism φ between them on the punctured disk $D_R^\times$.

Definition 239 (local shtuka). The *local shtuka stack* is the quotient $[LG/\mathrm{Ad}_\sigma I]$. Given an algebra R , $\mathrm{Sht}(R)$ parameterizes I -torsors $\mathcal{E}$ on D_R along with an isomorphism $\mathcal{E}|_{D_R^\times} \rightarrow \sigma^* \mathcal{E}|_{D_R^\times}$.

Remark 240. The fibers of Nt look like LG/I , which is an affine flag variety. It follows that Nt is “ind-proper.”

15.6 March 3: Gluing Triples

This talk was given by Piotr Achinger at MIT for the number theory seminar.

15.7 Motivation

For motivation, given a prime p and a variety X over $\mathbb{Q}_p$, we are interested in understanding the models $\mathcal{X}$ of X over $\mathbb{Z}_p$. For our purposes, we are just hunting for schemes $\mathcal{X}$ with the correct generic fiber. Frequently, we will refine this question to require $\mathcal{X}$ to be flat or proper or similar, but we will not do so yet.

Example 241. The theory of stable curves would like to understand degenerations of curves.

Example 242. The theory of Néron models finds models for abelian varieties.

Example 243. The theory of integral models of Shimura varieties is very rich.

Example 244. There is some Bruhat–Tits theory which is interested in finding models for algebraic groups.

We will approach this problem with nonarchimedean geometry. Here is why this may be a reasonable thing to do.

Example 245. The scheme $\mathbb{A}_{\mathbb{Q}_p}^1$ of course admits an integral model as $\mathbb{A}_{\mathbb{Z}_p}^1$, which amounts to a choice of some integral points $\mathbb{Z}_p \subseteq \mathbb{Q}_p$, which is some analytic open subset. But different models roughly correspond to linear transforms of our integral model, which amount to a different choice of rigid analytic open disk.

Example 246. The integral standard integral model of the modular curves $Y(N)$ more or less corresponds to the points $\{E : |j(E)| \leq 1\}$.

Let's now describe the general setup for the rest of the talk. Let S be a Noetherian scheme, $S_0 \subseteq S$ is a closed subscheme cut out by some ideal sheaf $\mathcal{I}$, and we set $S_n := V(\mathcal{I}^{n+1})$ for each n . We let $\hat{S}$ be the formal completion, $S^\circ := S \setminus S_0$ be the generic part, and $\hat{S}_{\text{rig}}$ is the associated rigid analytic space which looks like $\hat{S} \setminus S_0$. Thus, we have a commutative diagram as follows.

$$\begin{array}{ccc} \hat{S}_{\text{rig}} & \longrightarrow & \hat{S} \\ \downarrow & & \downarrow \\ S^\circ & \longrightarrow & S \end{array}$$

Here is the classical gluing result.

Theorem 247. There is an equivalence of categories

$$\text{Coh}(S) \rightarrow \text{Coh}(S^\circ) \times_{\text{Coh}(\hat{S}_{\text{rig}})} \text{Coh}(\hat{S}).$$

We would like to have a similar theorem for schemes over S . Of course, schemes are themselves sheaves, but they are probably not coherent. One can try to build a more general version of this theorem, but it is not so easy.

15.8 Where to Find Gluing Triples

Continue with the setup of the previous subsection. Further, let X be a scheme which is locally of finite type, let $X^\circ := X_{S^\circ}$ be its generic fiber, let $\hat{X}$ be the formal completion. Then we have rigid spaces $(X^\circ)^{\text{an}}$ which is the (rigid) analytification and $\hat{X}_{\text{rig}}$ which is the rigid generic fiber.

One may hope that our "gluing triple" simply amounts to identifying the two rigid spaces, but this cannot be the case.

Example 248. When $X = \mathbb{A}_{\mathbb{Z}_p}^1$, one finds that $(X^\circ)^{\text{an}}$ is the full analytification of $\mathbb{A}^1$, but $\hat{X}_{\text{rig}}$ is merely an open disk.

Nonetheless, there is a local isomorphism $j_X: \hat{X}_{\text{rig}} \rightarrow (X^\circ)^{\text{an}}$.

Remark 249. One can check that X being separated over S is equivalent to $X^\circ \rightarrow S^\circ$ and $\hat{X} \rightarrow \hat{S}$ being separated; in this case, j_X is an open embedding. If X is further proper, then those two maps are proper, and j_X is an isomorphism.

Definition 250 (gluing triples). A (separated) *gluing triple* on (S, S_0) is a triple $(U, \mathcal{X}, \eta)$, where U is a scheme locally of finite type and separated over S° , a formal scheme $\mathcal{X}$ which is separated and locally of finite type over $\hat{S}$, and an open embedding $j: \mathcal{X}_{\text{rig}} \rightarrow U^{\text{an}}$.

We would like to understand when all gluing triples actually come from genuine schemes.

Theorem 251. The functor $t: \text{Sch}_S^{\text{sep, loc. fin. type}} \rightarrow \text{Trp}$ (which the latter is the category of gluing triples) given by

$$t(X) := (X^\circ, \hat{X}, j_X)$$

is fully faithful.

Of course, the interesting question is essential surjectivity. It turns out to not be essentially surjective. To fix it, we need algebraic spaces.

Definition 252 (algebraic space). An algebraic space over S is an fppf sheaf $X : \text{Sch}_S \rightarrow \text{Set}$ admitting an étale surjection from a scheme. We further require that the diagonal $X \rightarrow X \times_S X$ is representable and quasicompact.

Remark 253. Algebraic spaces are occasionally more flexible to work with than schemes because there are Artin criteria to test if some fppf sheaf is an algebraic space.

Example 254. Choose a prime $p \geq 7$. Then there is a smooth proper algebraic space X over $\mathbb{Z}_p$ (which is not a scheme) such that $X_{\mathbb{Q}_p}$ and $X_{\mathbb{F}_p}$ are projective K3 surfaces.

Example 255. There is a K3 surface X over $\mathbb{Q}$ such that $X_{\mathbb{Q}_p}$ has a smooth proper model $\mathcal{X}$ over $\mathbb{Z}_p$ but for which X does not extend to a smooth proper scheme over $\mathbb{Z}_{(p)}$.

Thus, we see that our gluing triples may come from algebraic spaces instead of merely schemes!

Theorem 256. The functor t is an equivalence if one extends the source to separated algebraic spaces which are locally of finite type.

Remark 257. There is even a notion of a coherent sheaf on a gluing triple, and one can show that $\text{Coh}(X)$ is equivalent to $\text{Coh}(t(X))$. In particular, X is now uniquely determined from $t(X)$ because spaces can be reconstructed from their tensor category of coherent sheaves.

16 March 10: Sheaves on Isocrystals

This talk was given by Zeyu at MIT for the categorical Local Langlands seminar.

16.1 Some Category Theory

Let F be a p -adic local field, and we choose a reductive group G over F . We let $\tilde{F}$ denote the completion of the unramified closure of F , and we choose a model $\mathcal{G}$ of G over $\mathcal{O}_F$. Last time, we defined $B(G) = G(\tilde{F}) / \text{Ad}_\sigma G(\tilde{F})$, and we defined loop groups L^+G and LG , which allowed us to define a stack Isoc_G of isocrystals as $LG / \text{Ad}_\sigma LG$.

Theorem 258. For any algebraically closed extension K of κ_F , there is a bijection $\text{Isoc}_G(K) \rightarrow BG$.

Thus, the stratification of BG (using Newton and Kottwitz points) gives a stratification for Isoc_G .

Today, we would like to define a category of sheaves on Isoc_G . Unfortunately, we will have to use some ∞ -category theory. Let Pr^L be the ∞ -category of presentable ∞ -categories (with morphisms given by colimit-preserving functors), where presentable means that it admits all colimits (of some bounded cardinality). We also fix a coefficient field $\Lambda \in \{\overline{\mathbb{Q}_\ell}, \mathbb{Q}_\ell, \mathbb{Z}_\ell, \mathbb{F}_\ell\}$, which allows us to define LinCat_Λ to be the category of Λ -modules of Pr^L .

Definition 259 (compact). An object x in a category $\mathcal{C}$ is compact if and only if the functor $\text{Hom}(x, -)$ is continuous. We let $\mathcal{C}^\omega$ denote the full subcategory of compact objects. The category $\mathcal{C}$ is compactly generated if and only if any object in $\mathcal{C}$ is a filtered colimit of objects in $\mathcal{C}^\omega$.

Here are some categorical constructions.

Definition 260. Given a category $\mathcal{C}$, we define the category $\text{Ind } \mathcal{C}$ to consist of objects which are filtered systems $\{\mathcal{F}_i\}$ (denoted $\text{colim}_i \mathcal{F}_i$) with morphisms given by

$$\text{Hom}(\text{colim}_i \mathcal{F}_i, \text{colim}_j \mathcal{G}_j) = \lim_i \text{colim}_j \text{Hom}(\mathcal{F}_i, \mathcal{G}_j)$$

It is worthwhile to have some explicit constructions of limits and colimits in LinCat_Λ .

Definition 261. Given a collection of Λ -linear categories $\{\mathcal{C}_i\}$, we define the category $\lim \mathcal{C}_i$ as having objects which are infinite tuples $(c_i)_i$ such that $c_i \in \mathcal{C}_i$ and are equipped with identifications between the $c_i \in \mathcal{C}_i$ and $c_j \in \mathcal{C}_j$. The morphisms are defined by some limit of morphisms.

Definition 262. Given a collection of Λ -linear categories $\{\mathcal{C}_i\}$ for which the transition maps $L_{ij}: \mathcal{C}_i \rightarrow \mathcal{C}_j$ admit right adjoints (which is equivalent to L_{ij} preserving compact objects), we may define $\text{colim } \mathcal{C}_i$ as the corresponding limit of the right adjoints.

Remark 263. If the $\mathcal{C}_i$ s are compactly generated, then $\text{colim } \mathcal{C}_i$ is also compactly generated. If the index set is filtered, then one has an explicit description of the morphisms.

16.2 The Category of Sheaves

We are now ready to define our category of sheaves. Here are sheaves on schemes.

Definition 264. Fix a perfect finitely presented scheme X , we define $\text{Shv } X := \text{Ind Shv}_c X$ to be the inductive limit of the (derived) category of constructible sheaves on X . For general quasicompact quasiseparated schemes X , we may write X as a limit of perfect finitely presented schemes $\{X_i\}$, so we define

$$\text{Shv } X := \text{colim Shv } X_i,$$

where the internal maps are given by $(-)^!$.

Remark 265. Because $(-)^!$ preserves compact objects, it admits a right adjoint $(-)_b$, which allows us to express the colimit category as a limit.

And here are sheaves on stacks.

Definition 266. Fix a prestack X , which can be expressed as a colimit of quasicompact quasiseparated schemes $\{X_i\}$. Then we define

$$\text{Shv } X := \lim \text{Shv } X_i,$$

where the internal maps are again given by $(-)^!$.

Example 267. Suppose that K is a limit of finite groups $\{K_i\}$ so that $\text{Shv } K = \lim_i \text{Shv } K_i$. Let $p: K \rightarrow \text{pt}$ be the canonical projection, which yields adjoint functors $(p^!, p_*)$. For example, there is an object $\omega_K := p^! \Lambda$. We claim that $p_* \omega_K$ is $C^\infty(K)$. Indeed, we may compute

$$\text{Hom}(\Lambda, p_* \omega_K) = \text{Hom}(\omega_K, \omega_K).$$

This latter space can be expanded out in terms of the inductive limit, and eventually it unwinds correctly.

Example 268. We claim that $\mathrm{Shv} \mathbb{B}_{\mathrm{pro\acute{e}t}} K$ is $\mathrm{Rep}^{\mathrm{sm}} K$. (We remark that when K is prime to ℓ , the compact objects of $\mathrm{Rep}^{\mathrm{sm}} K$ should be given by perfect Λ -complexes.) Indeed, let $p: K \rightarrow \mathrm{pt}$ be the projection, and let $i: \mathrm{pt} \rightarrow \mathbb{B}_{\mathrm{pro\acute{e}t}} K$ be the embedding; these fit into the following pullback square.

$$\begin{array}{ccc} K & \longrightarrow & \mathrm{pt} \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{pt} & \longrightarrow & \mathbb{B}_{\mathrm{pro\acute{e}t}} K \end{array}$$

Then one finds that $\mathrm{Shv} \mathbb{B}_{\mathrm{pro\acute{e}t}} K$ admits an adjoint pair $i^!$ and i_* with Mod_Λ , so this category can be seen to be given by $i^! i_*$ -comodules. This is then equivalent to $C^\infty(K)$ -comodules, which gives smooth representations.

Example 269. Similarly, note that there is a pullback square as follows.

$$\begin{array}{ccc} K \backslash G_b(F) / K & \xrightarrow{p} & \mathbb{B}_{\mathrm{pro\acute{e}t}} K \\ p \downarrow & & \downarrow i \\ \mathbb{B}_{\mathrm{pro\acute{e}t}} K & \xrightarrow{i} & \mathbb{B}_{\mathrm{pro\acute{e}t}} G_b(F) \end{array}$$

Arguing as in the previous example, one finds that $\mathrm{Shv} \mathbb{B}_{\mathrm{pro\acute{e}t}} G_b(F)$ is equivalent to the category of $i^! i_*$ -comodules, where $i^!$ is compact induction, which is an adjoint for i_* (which can be thought of as restriction). Such comodules are smooth representations of $G_b(F)$.

For formal reasons relating to the geometry of our stratification, the two embeddings $j_b: \mathrm{Isoc}_{G,b} \rightarrow \mathrm{Isoc}_{G,\leq b}$ and $i_{<b}: \mathrm{Isoc}_{G,<b} \rightarrow \mathrm{Isoc}_{G,\leq b}$ admit $(-)^!$ functors automatically. In fact, $j_b^! = j_b^*$ in this case, which admits a right adjoint j_{b*} . Similarly, the functor $i_{<b,!} = i_{<b,*}$.

Remark 270. It turns out that we in fact have left adjoints $j_{b,!}$ to $j_b^!$ and $i_{<b}^*$ to $i_{<b,!}$, but this is not purely formal.

Remark 271. Let's explain why $\mathrm{Shv}(\mathrm{Isoc}_G)$ is compactly generated. The moral is that Isoc_G admits a covering by $\mathrm{Sht}^{\mathrm{loc}}$ (where this covering is ind-pfp and proper), and then $\mathrm{Sht}^{\mathrm{loc}}$ admits a covering by the loop group LG (which is a composite of a pfp map and a pro-unipotent map). Note that LG is a union of $LG_{\leq w} S$, where w is in the affine Weyl group, so we can compute its category of sheaves rather explicitly. One can then descend the chain to compute the category of sheaves on Isoc_G ; in particular, we find that $\mathrm{Shv} \mathrm{Isoc}_G$ is compactly generated.

Remark 272. It turns out that a sheaf $\mathcal{F} \in \mathrm{Shv}(\mathrm{Isoc}_G)$ is compact if and only if $i_b^* \mathcal{F} = 0$ for all but finitely many b , and $i_b^* \mathcal{F}$ is a compact smooth representation of $G_b(F)$ for all b .

16.3 Duality of Sheaves

It turns out that $\mathrm{Shv}(\mathrm{Isoc}_G)$ admits a duality, and explaining this is our next goal.

Definition 273 (dualizable). A Λ -linear category $\mathcal{C}$ is *dualizable* if and only if there is a category $\mathcal{C}^\vee$ which admits a unit $\mathrm{Mod}_\Lambda \rightarrow \mathcal{C} \otimes \mathcal{C}^\vee$ and a counit $\mathcal{C} \otimes \mathcal{C}^\vee \rightarrow \mathrm{Mod}_\Lambda$, satisfying various compatibilities. We let the pair of unit and counit be denoted $\mathbb{D}$.

Example 274. Let X be a pfp scheme, and set $\mathcal{C} := \mathrm{Shv}(X)$. Then there is a counit $\mathrm{Shv}(X) \otimes \mathrm{Shv}(X) \rightarrow \mathrm{Mod}_\Lambda$ given by $\Gamma(- \otimes^! -)$. To define the unit, we would like to take $\Delta_* \omega_X$, but this is a sheaf on $X \times X$, not in the tensor product. To fix this, we simply pass through the right adjoint $\boxtimes^R$ of $\boxtimes$: $\mathrm{Shv}(X) \otimes \mathrm{Shv}(X) \rightarrow \mathrm{Shv}(X \times X)$.

Remark 275. If $\mathcal{C}$ is compactly generated with dual category $\mathcal{C}^\vee$, then there is an equivalence

$$\mathbb{D}^\omega : (\mathcal{C}^\omega)^{\mathrm{op}} \rightarrow (\mathcal{C}^\vee)^\omega$$

satisfying $\mathrm{Hom}(x, y)$ is equal to the counit of $\mathbb{D}^\omega(x) \otimes y$.

Example 276. Let's extend Example 274 to work with Isoc_G . The main issue is that there is not an obvious candidate for the functor Γ , which should be pushforward to a point. But this sort of pushforward requires an argument to exist. Well, we can simply define Γ on the cover $\mathrm{Shv}(\mathrm{Sht}^{\mathrm{loc}})$.

17 March 12: The Descent Obstruction

This talk was given by Yutong Chen at MIT for the STAGE seminar.

17.1 The Selmer Subgroup

Let's review what we did last time. Fix an (affine) algebraic group G over a field k . Given a G -torsor $f: Z \rightarrow X$, note that any rational point $x \in X(k)$ produces a fiber $Z_x \rightarrow \{x\}$ living in the pullback square

$$\begin{array}{ccc} Z_x & \longrightarrow & Z \\ \downarrow & \lrcorner & \downarrow \\ x & \longrightarrow & X \end{array}$$

which is then manifestly another G -torsor.

Notation 277. Fix a G -torsor $f: Z \rightarrow X$. For $\tau \in H^1(k; G)$, we define $X(k)^\tau$ to be the collection of rational points $x \in X(k)$ for which the class $[Z_x]$ agrees with the class of τ .

Remark 278. We thus have a partition

$$X(k) = \bigsqcup_{\tau \in H^1(k; G)} X(k)^\tau.$$

Remark 279. By some twisting, we see that $Z_x = f^\tau(Z^\tau(k))$ because a torsor over k is trivial if and only if it admits a rational point.

We are interested in detecting when $X(k)^\tau$ may admit a rational point. For example, with k being a number field, we see that $k \hookrightarrow \mathbb{A}_k$ implies that $f^\tau(Z^\tau(k)) \hookrightarrow f^\tau(Z^\tau(\mathbb{A}_k))$, so the former being nonempty implies that the latter is nonempty.

Definition 280 (Selmer set). Fix a G -torsor $f: Z \rightarrow X$. Then we define

$$\mathrm{Sel}_2(k; G) := \{ \tau \in H^1(k; G) : Z^\tau(\mathbb{A}_k) \neq \emptyset \}.$$

The point is that our hunt for rational points may only focus on $\text{Sel}_2(k; G)$.

Our next goal is to show that $\text{Sel}_2(k; G)$ is finite.

Proposition 281. Fix a finite étale group scheme F over a local field k , and let X be a proper variety over k . For any F -torsor $f: Z \rightarrow X$, the map

$$X(k) \rightarrow H^1(k; F)$$

has finite image.

The key input to the proof is Krasner's lemma.

Lemma 282. Let $f: Y \rightarrow X$ be a finite étale scheme over k . For each $x \in X(k)$, there is an open analytic neighborhood of $x \in U$ such that all the fibers $f^{-1}(\{x'\}) \subseteq Y$ are isomorphic for $x' \in U$.

This is a geometric incarnation of the usual Krasner's lemma for roots of polynomials. We will omit the proof.

Proof of Proposition 281. The main point is that $X(k) \rightarrow H^1(k; F)$ is continuous, where the target has been given the discrete topology: indeed, we need to show that the subset of $x \in X(k)$ for which Z_x has the same type in $H^1(k; F)$ is an open subset. But this follows by a G -equivariant version of Lemma 282. (Alternatively, we can apply Lemma 282 directly to the cover $G \times Z \rightarrow X$.) This continuity now completes the proof: X is proper, so $X(k)$ is compact, so its image in the discrete topology $H^1(k; F)$ must be finite! ■

And here is our theorem.

Theorem 283. Let $f: Z \rightarrow X$ be a G -torsor, where X is proper variety over a number field k . Then $\text{Sel}_2(k; G)$ is finite.

Proof. Having a torsor can be described by some equations and some open subset, so one can spread out the G -torsor $f: Z \rightarrow X$ to a $\mathcal{G}$ -torsor $f: \mathcal{Z} \rightarrow \mathcal{X}$, where the latter objects all live over $\mathcal{O}_{K,S}$ for some finite set of places S . Notably, by properness, we see that $X(k_v) = \mathcal{X}(\mathcal{O}_v)$. Thus, we have a commutative diagram as follows.

$$\begin{array}{ccc} \mathcal{X}(\mathcal{O}_v) & \longrightarrow & H^1(\mathcal{O}_v; \mathcal{G}) \\ \parallel & & \downarrow \\ X(k_v) & \longrightarrow & H^1(k_v; G) \end{array}$$

Here, the map on the right is induced by the embedding $\mathcal{O}_v \subseteq k_v$. By definition of $\text{Sel}_2(k; G)$, we know that $\tau \in \text{Sel}_2(k; G)$ automatically comes from $H^1(\mathcal{O}_v; \mathcal{G})$; in other words, $\text{Sel}_2(k; G) \subseteq H_S^1(k; \mathcal{G})$.

From previous talks, one knows that each map in the composite

$$H_S^1(k; \mathcal{G}) \rightarrow H_S^1(k; \mathcal{G}/\mathcal{G}^\circ) \rightarrow \prod_{v \in S} H^1(k_v; \mathcal{G}/\mathcal{G}^\circ)$$

all have finite fibers. Thus, it is enough to show that the image of $\text{Sel}_2(k; G)$ in $\prod_{v \in S} H^1(k_v; \mathcal{G}/\mathcal{G}^\circ)$ is finite. Because S is finite, we may just look at a single place v , and then we want to show that the image of $\text{Sel}_2(k; G)$ in $H^1(k_v; \mathcal{G}/\mathcal{G}^\circ)$ is finite for each $v \in S$. But $\text{Sel}_2(k; G)$ embeds into $X(k_v)$, so this is Proposition 281. ■

Remark 284. As an application, we note that there is a separable extension k' of k which trivializes all elements of $\text{Sel}_2(k; G)$. It follows that $Z_x(k')$ will admit a k' -rational point, so $X(k) \subseteq f(Z(k'))$.

17.2 The Descent Obstruction

We are now ready to define our obstruction. The Hasse “local” obstruction amounts to saying that $X(k)$ embeds in $X(\mathbb{A}_k)$. Given a G -torsor $f: Z \rightarrow X$, we can refine this with the following diagram.

$$\begin{array}{ccc} X(k) & \longrightarrow & X(\mathbb{A}_k) \\ \downarrow & & \downarrow \\ H^1(k; G) & \longrightarrow & \prod_v H^1(k_v; G) \end{array}$$

Here is our refinement.

Notation 285. Fix a G -torsor $f: X \rightarrow Z$ over a number field k . Then we define

$$X(\mathbb{A}_k)^f := \bigcup_{\tau \in \text{Sel}_2(k; G)} f^\tau(Z^\tau(\mathbb{A}_f)).$$

Remark 286. By definition, we see that $X(\mathbb{A}_k)^f$ is nonempty if and only if $\text{Sel}_2(k; G)$ is nonempty.

Then we see that $X(k)$ embeds into $X(\mathbb{A}_k)^f$, which consists of those $x \in X(\mathbb{A}_k)$ which are in the image of

$$H^1(k; G) \rightarrow \prod_v H^1(k_v; G).$$

By letting f vary over all G -torsors $Z \rightarrow X$, we may refine further.

Definition 287 (descent obstruction). Fix a variety X over k . For any smooth algebraic group G over k , we set

$$X(\mathbb{A}_k)^{H^1(k; G)} := \bigcap_{f \in H^1(k; G)} X(\mathbb{A}_k)^f.$$

We then define the *descent obstruction* to be

$$X(\mathbb{A}_k)^{\text{descent}} = \bigcap_{\text{smooth affine } G} X(\mathbb{A}_k)^{H^1(k; G)}.$$

We will say that X has the *descent obstruction* if and only if $X(\mathbb{A}_k) \neq \emptyset$ while $X(\mathbb{A}_k)^{\text{descent}} = \emptyset$.

This sort of refinement holds in rather general situations: above, the only purpose of the group G was to be able to write down $H^1(-; G)$, which for our purposes could really be any functor $\text{Sch}_k^{\text{op}} \rightarrow \text{Set}$.

Notation 288. Suppose we have a functor $F: \text{Sch}_k^{\text{op}} \rightarrow \text{Set}$, and then any $A \in F(X)$ comes with some evaluation maps $\text{ev}_A: X(k) \rightarrow F(k)$: any rational point $x: \text{Spec } k \rightarrow X$ produces a map $F(X) \rightarrow F(k)$, which we denote by $A \mapsto \text{ev}_A(x)$.

Accordingly, we have a commutative diagram

$$\begin{array}{ccc} X(k) & \longrightarrow & X(\mathbb{A}_k) \\ \downarrow & & \downarrow \\ F(k) & \longrightarrow & F(\mathbb{A}_k) \end{array}$$

This gives us an obstruction.

Definition 289. Fix a variety X over a number field k , and choose a functor $F: \text{Sch}_k^{\text{op}} \rightarrow \text{Set}$. For any $A \in F(X)$, we define

$$X(\mathbb{A}_k)^A := \{y \in X(\mathbb{A}_k) : \text{ev}_A(y) \subseteq \text{im}(F(k) \rightarrow F(\mathbb{A}_k))\}.$$

We also define $X(\mathbb{A}_k)^F := \bigsqcup_{A \in F(X)} X(\mathbb{A}_k)^A$.

Example 290. Taking F to be $H^1(-; G)$ recovers the descent obstruction.

Remark 291. One can check this construction is functorial: a natural transformation $\eta: F \Rightarrow G$ produces a map $X(\mathbb{A}_k)^F \rightarrow X(\mathbb{A}_k)^G$. (In short, one sees that $X(\mathbb{A}_k)^A \subseteq X(\mathbb{A}_k)^{\eta(A)}$.) If the natural transformation is surjective, then the map is injective.

Our application is the so-called Brauer–Manin obstruction.

Definition 292 (Brauer–Manin obstruction). Let X be a variety over a number field k . Then we define the *Brauer–Manin obstruction* $X(\mathbb{A}_k)^{\text{Br}}$ to be the obstruction produced by the functor

$$H_{\text{ét}}^2(-; \mathbb{G}_m): \text{Sch}_k^{\text{op}} \rightarrow \text{Set}.$$

It turns out that the Brauer–Manin obstruction can be seen as a version of the descent obstruction!

Theorem 293. Let X be a regular and quasiprojective variety over a number field k . Then

$$X(\mathbb{A}_k)^{\text{Br}} = \bigcap_{n \geq 1} X(\mathbb{A}_k)^{\text{PGL}_n}.$$

Proof. The idea is that elements of the Brauer group are covered by $H^1(X; \text{PGL}_n)$ s: the short exact sequence of sheaves

$$1 \rightarrow \mathbb{G}_m \rightarrow \text{GL}_n \rightarrow \text{PGL}_n \rightarrow 1$$

produces embeddings $H^1(X; \text{PGL}_n) \rightarrow H^2(X; \mathbb{G}_m)$ for each n (by Hilbert’s theorem 90). Under the given assumptions, it turns out that elements of the Brauer group are all represented by Azumaya algebras, so this is surjective, provided that we sum over all n . The result now follows from Remark 291. ■

17.3 Iskovskikh’s Surface

Let’s work out an example of the descent obstruction. Consider the surface given by the equation

$$X: y^2 + t^2 = (x^2 - 2)(3 - x^2),$$

which we embed into a weighted projective space as

$$X: Y^2 + T^2 = (X^2 - 2Z^2)(3Z^2 - X^2),$$

where $\deg X = \deg Z = 1$, and $\deg Y = \deg T = 2$.

Remark 294. One can show that X admits points over $\mathbb{Q}_p$ for all p . The moral is that $y^2 + t^2$ attains every value in $\mathbb{Z}_p$ for all odd primes p , which one sees by computing which things are squares.

To use the descent obstruction, we will need some torsors. Well, for $c \in \mathbb{Q}^\times$, we define Y_c by the equations

$$Y_c: \begin{cases} c(U^2 + V^2) = X^2 - 2Z^2, \\ c(W^2 + S^2) = 3Z^2 - X^2. \end{cases}$$

(Note that Y_c , up to isomorphism, only depends on $\mathbb{Q}^\times / \text{im } N_{\mathbb{Q}(i)/\mathbb{Q}}$.) We claim that the map $f_c: Y_c \rightarrow X$ defined by

$$f_c: (X, Z, Y, V, W, S) \mapsto (X, Z, c(UW - VS), c(US + VW))$$

is a torsor for the circle group $G := \text{Spec } \mathbb{Q}[x, y] / (x^2 + y^2 - 1)$. Indeed, the fiber over some (X, Y, Z, T) will have $Y + iT = c(U + iV)(W + iS)$. Thus, we are hunting for two complex numbers $U + iV$ and $W + iS$, which the equations yield a norm requirement, and they have fixed product $Y + iT$. The locus of such complex numbers is exactly given by the circle group G !

Example 295. At $c = 1$, we see that $Z := Y_1$ is cut out by the equations

$$\begin{cases} U^2 + V^2 = X^2 - 2Z^2, \\ W^2 + S^2 = 3Z^2 - X^2. \end{cases}$$

Now that we have a G -torsors, we should work to classify them.

Lemma 296. Let G be the circle group over $\mathbb{Q}$. Then all G -torsors over $\mathbb{Q}$ take the form

$$T_c := \text{Spec } \mathbb{Q}[u, v] / (u^2 + v^2 - c),$$

where $c \in \mathbb{Q}^\times$.

Proof. We need to compute $H^1(k; G)$. Well, by definition, there is a short exact sequence

$$1 \rightarrow G \rightarrow \text{Res}_{\mathbb{Q}(i)/\mathbb{Q}} \mathbb{G}_m \xrightarrow{N} \mathbb{G}_m \rightarrow 1,$$

where the second map is given by the norm. The long exact sequence in cohomology yields

$$\mathbb{Q}(i)^\times \xrightarrow{N} \mathbb{Q}^\times \rightarrow H^1(\mathbb{Q}; G) \rightarrow H^1(\mathbb{Q}; \text{Res}_{\mathbb{Q}(i)/\mathbb{Q}} \mathbb{G}_m).$$

By unwinding Shapiro's lemma, we see that $H^1(\mathbb{Q}; \text{Res}_{\mathbb{Q}(i)/\mathbb{Q}} \mathbb{G}_m) = H^1(\mathbb{Q}(i); \mathbb{G}_m)$, which is 0 by Hilbert's theorem 90. Thus, $H^1(\mathbb{Q}; G)$ is isomorphic to $\mathbb{Q}^\times / \text{im } N_{\mathbb{Q}(i)/\mathbb{Q}}$. A calculation of the boundary map and then realizing the H^1 as torsors (which basically amounts to taking a pre-image), we find that the torsor corresponding to $c \in \mathbb{Q}^\times / \text{im } N_{\mathbb{Q}(i)/\mathbb{Q}}$ is given by $\text{Spec}[u, v] / (u^2 + v^2 - c)$. ■

To apply our descent obstruction, we now want to do a little twisting. Recall that we set $Z := Y_1$, and one can calculate that the twist Z^c is given by $Z \times^G T_{c^{-1}}$. There is an explicit isomorphism $Z^c \rightarrow Y_c$ given by

$$((X, Z, U + iV, W + iS), (A + iB)) \mapsto (X, Z, (A + iB)(U + iV), (A - iB)(W + iS)).$$

(One merely has to check that this is a G -equivariant map of torsors.) Thus, we see that our torsors $\{Y_c\}_c$ are all the twists of Z . The benefit of these equations is that one can show the following.

Lemma 297. Fix everything as above. If $Y_c(\mathbb{Q})$ is nontrivial, then $c \equiv 1 \pmod{\mathbb{Q}^{\times 2}}$.

Proof. After some care, we may assume that X and Z are coprime: for any prime factor p which is $3 \pmod{4}$, then $p^2 \mid (U^2 + V^2)$ forces p to divide both the summands, so we may divide it out. But for any prime $p \equiv 1 \pmod{4}$, then we may divide $(U + iV)(U - iV)$ out by $(a + bi)(a - bi)$ where $p = a^2 + b^2$. We then do some case-by-case analysis. One can rule out solutions at $X = 0$ or $Z = 0$ by looking over $\mathbb{R}$. Once X and Z are coprime, we know $c = 1$. ■

It remains to study $Y_1(\mathbb{Q})$. Rearranging the second equation, we see that

$$3 = \left(\frac{X}{Z}\right)^2 + \left(\frac{W}{Z}\right)^2 + \left(\frac{S}{Z}\right)^2.$$

If $v_2(X/Z) > 0$, then $3 - \left(\frac{X}{Z}\right)^2 \equiv -1 \pmod{4}$, which creates a contradiction. If $v_2(X/Z) = 0$, then we see that $\left(\frac{X}{Z}\right)^2 - 2 \equiv -1 \pmod{4}$, and we use the first equation. Lastly, if $v_2(X/Z) < 0$, then $3(Z/X)^2 - 1 \equiv -1 \pmod{4}$, and we again have a contradiction.

We conclude that $X(\mathbb{Q}) = \emptyset$. Indeed, we have seen that $Z^c(\mathbb{Q}) = \emptyset$ for all c , so $X(\mathbb{A}_{\mathbb{Q}})^G$ is empty.

Remark 298. There is a faster proof because $Y^2 + T^2$ needs to be a sum of two squares, so it must have an even number of prime factors which are $3 \pmod{4}$.

18 March 17: The Automorphic Category

This talk was given by Kenta Suzuki at MIT for the categorical local Langlands seminar.

18.1 Affine Deligne–Lusztig Induction

Here are our two stories.

- In the finite-dimensional setting, we fix a Borel subgroup $B \subseteq G$ over $\mathbb{F}_q$ and some parabolic B of G . Then there is a Newton map $G/\mathrm{Ad}_{\sigma} B \rightarrow G/\mathrm{Ad}_{\sigma} G$, and its fibers look like G/B . The target is $BG(\mathbb{F}_q)$. Here, there is a Deligne–Lusztig variety $X_{\leq w}(1)$, which is the collection of Borel subgroups $\mathcal{B}$ for which $\mathcal{B} \leq w$ and $\mathcal{B}$ and $\sigma(\mathcal{B})$ are in relative position w . Granted this, we define Deligne–Lusztig induction by

$$R^{\theta}(w) := \sum_i (-1)^i H_c^i(X(w); \mathcal{F}_{\theta}),$$

where θ is a character of the torus, and $\mathcal{F}_{\theta}$ is some specified local system on $X_w(1)$. In terms of the derived category, we may write $R_w^* = H^{\bullet}(X_w(1); \overline{\mathbb{Q}}_{\ell})$ and $R_w^! = H_c^{\bullet}(X_w(1); \overline{\mathbb{Q}}_{\ell})$. Roughly speaking, these are pushforwards of the constant sheaves along the Newton map from a stratum.

One also has a covering $\widetilde{X_w(1)} \rightarrow X_w(1)$ which is a T^{wF} -torsor, defined as follows: note that we can think of $X_w(1)$ as $\{gB \in G/B : g^{-1}\sigma(g) \in BwB\}$, so one can define the cover as

$$\widetilde{X_w(1)} = \{gU \in G/U : g^{-1}\sigma(g) \in UwU\}.$$

- In the affine setting, we use the Iwahori subgroup I of the loop group LG , and we have a parabolic $L^+P \subseteq LG$. In this case, our Newton map looks like $\mathrm{Sht}_P \rightarrow \mathrm{Isoc}_G$, which has fibers which look like LG/L^+P .

Here, the Newton stratification $X_{\leq w}(b)$ of local shtukas gives the affine Deligne–Lusztig varieties. Then we have affine Deligne–Lusztig induction given by $R_w^* = \mathrm{Nt}_{*}(i_{w*}\omega_{\mathrm{Sht}_w}[-\ell(w)])$. There is also a covering $\widetilde{\mathrm{Sht}_w} \rightarrow \mathrm{Sht}_w$.

Perhaps we should explain why affine Deligne–Lusztig induction “recovers” the finite Deligne–Lusztig induction.

Lemma 299. For all $b \in B(G)$, one has

$$i_b^! R_w^* = C_{\bullet}^{\mathrm{BM}}(X_w(b); \Lambda[-\ell(w)]).$$

Proof. One just applies base-change to the following pullback square.

$$\begin{array}{ccc} G_b(F) \backslash X_w(b) & \longrightarrow & \mathrm{Sht}_{P,w} \\ \downarrow & & \downarrow \mathrm{Nt} \\ \mathbb{B}_{\mathrm{pro\acute{e}t}} G(F) & \xrightarrow{i_b} & \mathrm{Isoc}_G \end{array}$$

■

In general, finite Deligne–Lusztig induction will allow us to pushforward general line bundles $\mathcal{L}$ on T^{wF} to $H_c^\bullet(X_w(1); \mathcal{L})$, where the target is in $\mathrm{Rep} G(\mathbb{F}_q)$. One can define similar functors $R_w: \mathrm{Rep} S_k^{w\sigma} \rightarrow \mathrm{Shv}(\mathrm{Isoc}_G)$ by $\mathrm{Nt}_*(i_w)_* \mathrm{pr}_w^\sigma[-\ell(w)]$.

Continuing our analogy, we recall the following result in Deligne–Lusztig theory.

Theorem 300 (Deligne–Lusztig). Choose σ -stable maximal tori T and T' , and choose characters θ and θ' of T^F and $(T')^F$. Suppose that they are not geometrically conjugate, meaning that the pairs $(T, \theta \circ N)$ and $(T', \theta' \circ N)$ are conjugate, for a suitable norm N . Then

$$\mathrm{Hom}(H_c^\bullet(X_w(1); \mathcal{F}_\theta), X_{w'}(1); \mathcal{F}_{\theta'}) = 0.$$

Thus, one now wants to classify pairs (T, θ) up to conjugacy; they turn out to be parameterized by semisimple geometric conjugacy classes in $\check{G}(\mathbb{F}_q)$. In the affine story, one has the following.

Proposition 301. If (w_1, θ_1) and (w_2, θ_2) are not geometrically conjugate, then

$$\mathrm{Hom}_{\mathrm{Shv}}(R_{w_1, \theta_1}^*, R_{w_2, \theta_2}^*) = 0.$$

Thus, again, we want to classify pairs (w, θ) up to geometry conjugacy. They turn out to be given by “tame inertia types,” which are certain homomorphisms $I_F/P_F \rightarrow \check{T}(\overline{\mathbb{Q}}_\ell)$. (Here, P_F is the wild inertia.)

18.2 Categorical Trace

Let’s start by talking about monodromy sheaves. Here is the finite version.

Definition 302. Fix a connected reductive group H over $\overline{\mathbb{F}}_q$. Then we define the *Serre’s fundamental group* as $\pi_1^c(H)$ to be the profinite group

$$\lim_{H' \rightarrow H} \ker(H' \rightarrow H),$$

where $H' \rightarrow H$ is a finite étale connected cover.

Remark 303. By construction, $\pi_1^c(H)$ is covered by $\pi_1^{\acute{e}t}(H)$.

Remark 304. If H is defined over $\mathbb{F}_q$, then $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ naturally acts on these kernels, so it naturally acts on $\pi_1^c(H)$.

Example 305. If $H = \mathrm{PGL}_p$, then $\pi_1^c(H) = 1$. For example, the covering $\mathrm{SL}_p \rightarrow \mathrm{PGL}_p$ is not étale!

Example 306. If H is commutative (such as an abelian variety), then one can take rather general quotients, so one finds that $\pi_1^c(H) = \lim_n H(\mathbb{F}_{q^n})$.

This generality allows us to define a character sheaf.

Definition 307 (character sheaf). Fix a reductive group H over $\overline{\mathbb{F}}_q$. A *character sheaf* is a rank 1 local system $\mathcal{F}$ on H , along with the data of a multiplication

$$m^* \mathcal{F} \cong \mathcal{F} \boxtimes \mathcal{F},$$

and some cocycle condition.

Example 308. If H is connected, then it turns out that the groupoid of character sheaves is discrete. We can represent them as follows; for any continuous homomorphism $\chi: \pi_1^c(H) \rightarrow \Lambda^\times$, we can produce a rank 1 local system via the covering $\pi_1^{\text{ét}}(H) \rightarrow \pi_1^c(H)$. This turns out to give a bijection. One can categorify this construction to a functor

$$\text{Ch}: \text{Coh}(R_{\pi_1^c(H), \mathbb{G}_m})^\heartsuit \rightarrow \text{Shv}(H; \Lambda)^\heartsuit.$$

So here is our definition of a monodromy sheaf.

Definition 309. A *monodromy sheaf* is the full subcategory generated by the heart of the previous example. If X is a prestack with action of H , then the monodromy sheaves on X are the functors $\text{Shv}_{\text{mon}}(H) \rightarrow \text{Shv}(X)$.

Return to finite groups, it turns out that Deligne–Lusztig induction can be viewed as providing a functor $\text{Ch}_{G, \varphi}^{\text{mon}}$ from $\text{Shv}(U \backslash G/U) \rightarrow \text{Rep } G(\mathbb{F}_q)$ by the correspondence

$$U \backslash G/U \leftarrow G / \text{Ad}_\sigma G \xrightarrow{\text{Nt}} G / \text{Ad}_\sigma G.$$

A similar correspondence produces a functor $\text{Ch}_{G, \varphi}^{\text{unip}}: \text{IndShv}_c(B \backslash G/B) \rightarrow \text{Rep } G(\mathbb{F}_q)$. It will turn out that taking the “categorical trace” makes these functors equivalences. (The second functor actually gives an equivalence to $\text{IndRep}_c^{\text{unip}} G(\mathbb{F}_q)$.)

Definition 310. Fix an associative algebra object A in the category of correspondences of some geometric objects, and let F be an A -bimodule. Then we define

$$\text{Tr}(D(A), D(F)) := D(A) \otimes_{D(A) \otimes_{D(\text{pt})} D(A)^{\text{rev}}} D(F),$$

where D is some sheaf theory. There is also a *geometric trace*

$$\text{Tr}_{\text{geo}}(D(A), D(F)) := D(|\text{Bar}(A)_\bullet \otimes_{A \otimes A^{\text{rev}}} F|).$$

We may write this as $D(\text{Tr}(A, F))$

Remark 311. In good situations, the trace and geometric trace agree.

Remark 312. A choice of algebra endomorphism $\varphi: A \rightarrow A$ makes A into an A -bimodule, where A acts on the left by φ and on the right by A^{rev} .

We will discuss the affine situation next time, which is the following result.

Theorem 313. There is an equivalence

$$\text{Tr}(\text{Shv}_{\text{mon}}(\text{Iw}^u \backslash LG / \text{Iw}^u), \varphi) \cong \text{Shv}^{\text{tame}}(\text{Isoc}_G),$$

where tame means that it comes from a depth-zero representation.

19 March 17: Moments of Quadratic L -functions

This talk was given by Adrian Diaconu at Boston College for the BC–MIT number theory seminar.

19.1 The Analytic Conjectures

We are working over function fields today, and we are interested in Dirichlet L -functions (i.e., L -functions attached to automorphic forms of GL_1). Here is the conjecture over $\mathbb{Q}$.

Conjecture 314. Let χ_d be the Kronecker symbol $\left(\frac{\cdot}{d}\right)$. Then

$$\sum_{0 < d < D}^* L(1/2, \chi_d)^r = \frac{3}{\pi} D Q_r(\log D) + o(D).$$

This is known for $r \leq 3$ classically; there is also a second-order term known for $r = 3$, which is similar to the second-order term found when counting cubic number fields. There are some partial results for $r = 4$.

We now turn to function fields.

Definition 315. Fix $d, m \in \mathbb{F}_q[x]$ with m monic, and then we define χ_d to be the d th Kronecker symbol. Then we define

$$L(s, \chi_d) = \sum_f \frac{\chi_d(f)}{|f|^s},$$

where $|f| = q^{\deg f}$.

Conjecture 316. As $g \rightarrow \infty$, we have

$$\frac{1}{q^{2g+1}} \sum_{\substack{\text{monic, squarefree } d \\ \deg d = 2g+1}} L(1/2, \chi_d)^r = Q_r(2g+1) + o(1).$$

We again know about the same: this is known for $r \leq 3$, and there are some asymptotics known for $r = 4$. The reason that we do not see much improvement is because there is not an easy way to relate this problem to geometry.

Let's say something about the polynomial Q_r , which has a conjectural decomposition.

Conjecture 317. Fix $r \geq 3$, and choose integers $D, N \geq 1$ for which $(N+1)^{-1} < \Theta < N^{-1}$. Then

$$\sum_{\substack{\text{monic, squarefree } d \\ \deg d = D}} L(1/2, \chi_d)^r = \sum_{n \leq N} q^{(\frac{1}{2} + \frac{1}{2n})D} Q_{n,r}(D, q) + O_{\Theta, q, r} \left(q^{D(1+\Theta)/2} \right).$$

This is known for $r = 3$ (even for number fields), and there are partial results for $r = 4$.

Our main result is a proof of the asymptotic formula in the “large q ” regime.

Theorem 318. For an odd prime-power q , one has

$$\frac{1}{q^{2g+1}} \sum_{\substack{\text{monic, squarefree } d \\ \deg d = D}} L(1/2, \chi_d)^r = Q_r(2g+1) + O \left(4^{g(r+1)} q^{(17-g)/34} \right).$$

Morally, this is a refinement of the following result.

Theorem 319 (Katz–Sarnak). Fix g . Then

$$\lim_{q \rightarrow \infty} \frac{1}{q^{2g}(q-1)} \sum_{\substack{\text{monic, squarefree } d \\ \deg d = D}} L(1/2, \chi_d)^r = \int_{\mathrm{USp}(2g)} \det(1 - A)^r dA.$$

Remark 320. The method has been extended to handle ratios of products of L -functions by Victor Wang.

19.2 Configuration Spaces

We are interested in the following spaces.

Definition 321. Let $\mathrm{Conf}_n(\mathbb{C})$ be the configuration space of n distinct ordered points, which is equivalent to the space of degree- n monic polynomials with no repeated roots.

Remark 322. It turns out that $\pi_1 \mathrm{Conf}_n(\mathbb{C})$ is Artin’s braid group on n strands, denoted β_n . For $n \geq 2$, it turns out that $\mathrm{Conf}_n(\mathbb{C})$ is a $K(\beta_n, 1)$.

Notation 323. There is a canonical local system $d \mapsto H^1(\mathrm{Conf}_n(\mathbb{C}); \mathbb{C})$, which we denote by V .

It will turn out that we want to understand $H^*(\mathrm{Conf}_{2g+1}(\mathbb{C}); V) = H^*(\beta_{2g+1}; V)$. We can also replace V with other representations V_λ of Sp_{2g} . This must be a hard problem because it solves our question, as we will explain.

Remark 324. One can identify $\mathrm{Conf}_{2g+1}(\mathbb{A}^1)/\mathbb{G}_a$ with the moduli space $\mathcal{H}_{2g+1}$ of hyperelliptic curves with $2g + 1$ branch points, along with a marked Weierstrass point and a nonzero tangent vector there.

The moral is that

$$\mathrm{tr}(\mathrm{Frob}_q; H_\bullet^{\mathrm{ét}}(\beta_{2g+1}; V_\lambda(|\lambda|/2))) = \frac{1}{q^{2g+1}} \sum_{\substack{\text{monic, squarefree } d \\ \deg d = D}} s_{\langle \lambda \rangle}(q^{-1/2} \Theta_d),$$

where $s_{\langle \lambda \rangle}$ is a symplectic Schur function, and Θ_d is the multiset of our Frobenius eigenvalues for the hyperelliptic curve $y^2 = d(x)$. On the other hand, one can show that

$$\frac{1}{q^{2g+1}} \sum_{\substack{\text{monic, squarefree } d \\ \deg d = D}} L(t_1; \chi_d) \cdots L(t_r; \chi_d) = (t_1 \cdots t_r)^g \sum_{\lambda} \mathrm{tr}_{\lambda}(g) s_{\lambda^\dagger}(q^{\pm 1/2} t_1^{\pm 1}, \dots, q^{\pm 1/2} t_r^{\pm 1}),$$

where $\mathrm{tr}_{\lambda}(g)$ is the Frobenius trace from the previous equation. Thus, our goal has become to compute some homology groups with uniformity in λ , so the problem is now one in homotopy theory.

Remark 325. We remark that the “source” of our homological stability is from Borel’s result in automorphic forms that there is homological stability for algebraic representations of arithmetic groups. One uses homotopy theory to pull back the stability from the arithmetic groups to the braid groups.

20 March 17: Elliptic Curves of Small Rank

This talk was given by David Zywin at Boston College for the BC–MIT seminar.

20.1 The Lower Bound

Fix a number field K , like $\mathbb{Q}$, and we choose an elliptic curve E over K . It is known that $E(K)$ is a finitely generated abelian group, and we are interested in understanding its rank. For example, it is not known which ranks occur (e.g., are they bounded?) or how often (e.g., which ranks can occur infinitely often?).

Example 326. For $K = \mathbb{Q}$, the highest known rank is 29. The consensus seems to change on whether ranks should be bounded.

Example 327. There is a heuristic that says that ranks up to 21 will occur infinitely often (from elliptic curves over $\mathbb{Q}$), and there are only finitely many elliptic curves with ranks more than 21.

Here is our main result.

Theorem 328. Fix a number field K and integer $r \in \{0, 1, 2, 3, 4\}$. Then there are infinitely many elliptic curves (up to geometric isomorphism) over K with rank r .

Remark 329. Mazur and Rubin showed the case $r = 0$. The case of $r = 1$ was independently proven by Koymans–Pagano. There were also special cases known for some fields.

Remark 330. One can find infinitely many elliptic curves of lower-bounded rank. For example, Elkies found infinitely many elliptic curves of rank at least 19. Thus, the main content of our result is to be able to upper-bound ranks.

Remark 331. The Birch and Swinnerton-Dyer conjecture is only known for finitely many elliptic curves of rank greater than 1. Here, there is a difficulty that we cannot compute these analytic ranks without careful calculation.

We will focus on the $r = 4$ case today. Basically, the only way to compute a rank is to lower-bound by finding explicit points, upper-bound by computing some Selmer groups, and then hope for an equality. Thus, let's start by choosing a special elliptic surface, which we engineer to have many points.

Example 332. We define the elliptic surface $\mathcal{E}$ over $K(T)$ by $y^2 = x^3 + ax^2 + bx$, where

$$\begin{cases} a = -70(T^2 - 25), \\ b = 2^4 \cdot 7^2(T^2 - 11^2)(T^2 - 25^2). \end{cases}$$

For example, it turns out that the discriminant splits completely as

$$2^4 \cdot 3^2 \cdot 7^6(T - 11)^2(T + 11)^2(T - 25)^3(T + 25)^3(T - 39)(T + 39).$$

(This is something very remarkable!) One can calculate that $\mathcal{E}(K(T)) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}^4$, and one can explicitly compute what generates the $\mathbb{Z}^4$.

The idea is to plug in various values of T to produce an elliptic curve $\mathcal{E}_t$ over K ; note that we should avoid $t \in \{\pm 11, \pm 25, \pm 39\}$ to continue to an elliptic curve. We are able to lower-bound the rank thanks to the following.

Theorem 333 (Silverman). Fix an elliptic surface $\mathcal{E}$ over $K(T)$. For all but finitely many $t \in K$, one has

$$\text{rank}(\mathcal{E}/K(T)) \leq \text{rank}(\mathcal{E}_t/K).$$

The moral is that one can “plug in” the generating points from $\mathcal{E}$ and (generically) receive linearly independent points in $\mathcal{E}_t$.

20.2 The Upper Bound

It remains to upper-bound the ranks of the $\mathcal{E}_t$ for infinitely many t , for which we will use Selmer groups. (One also knows that each j -invariant only appears in $\mathcal{E}_t$ finitely many times, so we receive different isomorphism classes of elliptic curves.)

For our set-up, we choose a finite set S of prime ideals of $\mathcal{O}_K$ and a nonempty open set $U_{\mathfrak{p}} \subseteq K_{\mathfrak{p}} \times K_{\mathfrak{p}}$ for each $\mathfrak{p} \in S$. We also choose open $U_v \subseteq K_v$ for each $v \mid \infty$. In this setting, we have the following existence result.

Theorem 334 (Kai). Fix everything as in the previous paragraph. There are nonzero $m, n \in \mathcal{O}_{K,S}$ satisfying the following.

- (i) For each $e \in \{\pm 11, \pm 25, \pm 39\}$, the elements $m + en$ generate distinct prime ideals of $\mathcal{O}_{K,S}$.
- (ii) We have $(m, n) \in U_{\mathfrak{p}}$ for each $\mathfrak{p} \in S$.
- (iii) We have $m/n \in U_v$ for each $v \mid \infty$.

This talk has no interest in the proof of this theorem, but we will use to select our ts . For example, let's assume that S contains enough primes to make $\mathcal{O}_{K,S}$ a principal ideal domain, and we assume that it contains all primes above 2, 3, and 7. Then we receive $m, n \in \mathcal{O}_{K,S}$, and we set $t := m/n$ so that $\Delta(t)$ has a nice prime factorization

$$2^{14} \cdot 3^2 \cdot 7^6 \cdot (m - 11n)^2(m + 11n)^2(m - 25n)^3(m + 25n)^3(m - 39n)(m + 39n)n^{-17}.$$

In particular, $\mathcal{E}_t$ has bad reduction only at some explicit primes $\mathfrak{p}$, which are either in S , in B , or one of the primes $(m + en)$ for $e \in B$. (It turns out that a prime dividing n has good reduction, which we can see by adjusting our model.) By adjusting the local conditions $U_{\mathfrak{p}}$, we can more or less control if the primes in S are actually bad reduction.

Let's now review how Selmer groups will be able to give us upper bounds on rank. Set $E := \mathcal{E}_t$, which we recall has an element $(0, 0)$ of order 2, so it admits a quotient $\varphi: E \rightarrow E'$ with kernel given by this 2-torsion element. Now, taking cohomology of the short exact sequence

$$0 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow E \rightarrow E' \rightarrow 0$$

gives an exact sequence

$$E(K) \rightarrow E'(K) \rightarrow H^1(K; \mathbb{Z}/2\mathbb{Z}).$$

Kummer theory allows us to compute the target as $K^{\times}/K^{\times 2}$; it turns out that the connecting map sends $(x, y) \in E'(K)$ to $xK^{\times 2}$.

Definition 335. Fix a 2-isogeny $\varphi: E \rightarrow E'$. Then we define the *Selmer group* $\text{Sel}_{\varphi}(E) \subseteq K^{\times}/K^{\times 2}$ to fit into the following pullback square.

$$\begin{array}{ccc} \text{Sel}_{\varphi}(E) & \longrightarrow & K^{\times}/K^{\times 2} \\ \downarrow & \lrcorner & \downarrow \\ \prod_v E'(K_v) & \longrightarrow & \prod_v K_v^{\times}/K_v^{\times 2} \end{array}$$

It turns out that the Selmer group is finite and computable for a given elliptic curve.

Remark 336. Let's give the local conditions at good primes: for primes $p \nmid 2$ at which E has good reduction, then the image of $E'(K_v) \rightarrow K_v^\times / K_v^{\times 2}$ is exactly given by $\mathcal{O}_p^\times$. Thus, computing the Selmer group amounts to computing something about

This discussion helps us compute ranks because $E'(K)/\varphi(E(K))$ embeds into $\text{Sel}_\varphi(E)$ by construction. Indeed, there is a short exact sequence

$$0 \rightarrow \frac{E[2]}{\varphi(E(K)[2])} \rightarrow \frac{E'(K)}{\varphi(E(K))} \rightarrow \frac{E(K)}{2E(K)} \rightarrow \frac{E(K)}{\varphi^\vee(E'(K))} \rightarrow 0,$$

so the rank of E is found as

$$\dim_{\mathbb{F}_2} E'(K)/\varphi(E(K)) + \dim_{\mathbb{F}_2} E(K)/\varphi^\vee(E'(K)) - 2,$$

which can be upper-bounded as

$$\dim_{\mathbb{F}_2} \text{Sel}_\varphi(E) + \dim_{\mathbb{F}_2} \text{Sel}_{\varphi^\vee}(E') - 2.$$

We now hope that each Selmer group has rank 3. Our remaining task is to reverse-engineer local conditions U_v s and U_p s in order to be able to compute these Selmer groups. Notably, this requires control of the bad primes in order to explicitly compute the Selmer groups.

Example 337. Choosing $t \in K \setminus B$ to have the $v_p(t - e) = 1$ for $e \in \{\pm 11, \pm 39\}$, we find that the image of $E'(K_v) \rightarrow K_v^\times / K_v^{\times 2}$ is

$$\begin{cases} K_p^\times / K_p^{\times 2} & \text{if } e \in \{\pm 11\}, \\ 1 & \text{if } e \in \{\pm 39\}. \end{cases}$$

Using conditions like this can achieve $\dim_{\mathbb{F}_2} \text{Sel}_\varphi(E') = 3$, which is as small as possible.

However, we should not just focus on bounding $\text{Sel}_\varphi(E')$.

Theorem 338 (Cassels). Fix everything as above. Then

$$\frac{|\text{Sel}_\varphi(E)|}{|\text{Sel}_{\varphi^\vee}(E')|} = \prod_v \frac{1}{2} |\text{im}(E'(K_v) \rightarrow K_v^\times / K_v^{\times 2})|.$$

We would like for our left-hand side to be 1 (indeed, both Selmer ranks should be three), so we need to be very sparing on the local conditions that we impose. If these conditions are chosen to be minimal, then it turns out to be okay, and we prove our theorem.

Remark 339. One gets fairly lucky here: it turns out that the average size of $\text{Sel}_2 \mathcal{E}_t$ is infinite as t varies. Also, other choices of families have nontrivial $\text{III}(\mathcal{E})[2]$, which prevents us from finding the correct upper bounds.

21 March 19: $x^2 + y^3 = z^7$

This talk was given by Arav Karighattam at MIT for the STAGE seminar.

21.1 Generalized Fermat Equations

We have the following results about generalized Fermat equations.

Theorem 340. Fix positive integers p, q , and r , and choose nonzero integers A, B , and C . Set $\chi := \frac{1}{p} + \frac{1}{q} + \frac{1}{r} - 1$, and we will consider the curve $X: Ax^p + By^q = Cz^r$.

- (a) Beukers: if $\chi > 0$, then this equation has either zero or infinitely many primitive solutions in a finite collection of two-parameter families.
- (b) If $\chi = 0$, then this equation reduces to torsors of elliptic curves.
- (c) Darmon–Granville: if $\chi > 0$, then there are finitely many primitive solutions.

Remark 341. Part (b) has little content: there are finitely many (p, q, r) with $\chi = 0$, and one can check that they yield elliptic curves.

We are only going to focus on showing (c) of Theorem 340. It will follow from the following analogous trichotomy.

Theorem 342. Let C be a projective curve over $\mathbb{Q}$ of genus g , and set $\chi := 2 - 2g$.

- (a) Hasse–Minkowski: if $\chi > 0$, then $C(\mathbb{Q})$ is either empty or infinite and can be explained by the local-to-global principle.
- (b) Mordell–Weil: if $\chi = 0$, then $C(\mathbb{Q})$ is either empty or a finitely generated abelian group.
- (c) Faltings: if $\chi > 0$, then $C(\mathbb{Q})$ is finite.

It turns out that Faltings’s theorem implies Theorem 340, but one has to be careful because $Ax^p + By^q = Cz^r$ is not a projective curve in general.

For convenience, we will assume that p, q , and r are pairwise coprime, and for simplicity, we will assume that $A = B = C = 1$. Then one can certainly use the equation $x^p + y^q = z^r$ to define some surface in $\mathbb{A}_{\mathbb{Q}}^3 \setminus \{0\}$. In the case where $p = q = r$, we can take a quotient by the $\mathbb{G}_m$ -action given by $(x, y, z) \mapsto (\lambda x, \lambda y, \lambda z)$, so we receive a closed subscheme of $\mathbb{P}^2$, which is our projective curve. In general, we are only stable under the $\mathbb{G}_m$ -action given by

$$\lambda: (x, y, z) \mapsto (\lambda^{qr}x, \lambda^{pr}y, \lambda^{pq}z)$$

Accordingly, we may pass to the quotient stack $[S/\mathbb{G}_m]$ in weighted projective space. Up to rearranging (p, q, r) , we find that all stabilizers of points in $[S/\mathbb{G}_m]$ are trivial, except for a stabilizer μ_q at $(1, 0, 1)$, a stabilizer μ_r at $(1, -1, 0)$, and a stabilizer μ_p at $(0, 1, 1)$.

We will now hunt for a finite étale cover X of $[S/\mathbb{G}_m]$, where X is actually a scheme. We will then apply Faltings’s theorem to X to produce our finiteness. The existence of this X will be conjured using complex analysis: finding such a cover $X \rightarrow [S/\mathbb{G}_m]$ is equivalent to finding a cover $X \rightarrow \mathbb{P}^1$ (of complex algebraic varieties) which is ramified above 0, 1, and ∞ , with degrees p, q , and r , respectively. We will use the following tool.

Theorem 343. Fix a smooth complex variety Y . Then the category of finite étale covers of Y is equivalent to the category of finite covering spaces of $Y(\mathbb{C})$.

We will apply this theorem to $Y := [S/\mathbb{G}_m] \setminus \{0, 1, \infty\}$, which is a genuine complex affine curve. Then $\pi_1(Y) = \langle \gamma_0, \gamma_1, \gamma_\infty : \gamma_0\gamma_1\gamma_\infty = 1 \rangle$. It turns out that $\pi_1(Y)$ admits a finite-index subgroup for which γ_0, γ_1 , and γ_∞ have orders p, q , and r in the quotient. Thus, we receive a finite étale cover $X^\circ \rightarrow Y$ with the corresponding fundamental group. Then there is a smooth projective curve X containing X° so that $X^\circ \rightarrow Y$ extends uniquely to $X \rightarrow \mathbb{P}^1$ with the desired ramifications. This then passes to some étale cover $X \rightarrow [S/\mathbb{G}_m]$. We remark that (a priori) X is only defined over $\mathbb{C}$, but it surely can descend to $\overline{\mathbb{Q}}$ and then to a number field K .

We now apply Faltings’s theorem to X , for which we will use the following tool.

Theorem 344 (Beckmann). Let $\pi: X \rightarrow [S/\mathbb{G}_m]$ be the above covering. For each $t \in S(\mathbb{Z})$, the fiber over t in X is defined over an extension L of K of degree at most $\deg \pi$, where L is unramified away from pqr .

Thus, we see that $\#S(\mathbb{Z}) \leq \#X(L)$, where L is the composite of all extensions L of K with degree bounded by $\deg \pi$ and unramified away from pqr . There are only finitely many such extensions, so L is finite over $\mathbb{Q}$, so $X(L)$ is finite by Faltings's theorem, so we are done! (Note $\chi(X) < 0$ by a calculation with Riemann–Hurwitz.)

21.2 A Sketch

We would like to solve the equation $S: x^2 + y^3 = z^7$. The above discussion has motivated us to find a covering $X \rightarrow \mathbb{P}^1$ which is ramified at three points with degrees 3, 2, and 7. It will turn out to be convenient to move the ramified points to $\{0, 1728, \infty\}$.

Over the complex numbers, one knows that the universal cover $\tilde{X}$ of X is the universal cover of $\mathbb{P}^1$ which is ramified over three points with degrees 3, 2, and 7. Such a thing should still be a complex curve, so it should be one of $\mathbb{C}$, $\mathbb{P}^1$, or the upper-half plane $\mathcal{H}$. Depending on its curvature. The Euler characteristic of X is negative, so we see that we need to get the upper-half plane $\mathcal{H}$. Thus, we see that we are hunting for a quotient of $\mathcal{H}$.

Example 345. The natural action of $\mathrm{PSL}_2(\mathbb{R})$ on $\mathcal{H}$ (by fractional linear transformations) induces an analytic map $\mathrm{PSL}_2(\mathbb{Z}) \backslash \mathcal{H} \rightarrow \mathbb{P}^1(\mathbb{C})$.

The map $\mathrm{PSL}_2(\mathbb{Z}) \backslash \mathcal{H} \rightarrow \mathbb{P}^1(\mathbb{C})$ fails to be surjective. To fix this, we add in cusps.

Notation 346. We let $\mathcal{H}^* := \mathcal{H} \sqcup \mathbb{P}^1(\mathbb{Q})$ denote the completion of $\mathcal{H}$.

We thus receive a surjective map $\mathrm{PSL}_2(\mathbb{Z}) \backslash \mathcal{H}^* \rightarrow \mathbb{P}^1$. Magically, $\mathrm{PSL}_2(\mathbb{Z}) \backslash \mathcal{H}^*$ has precisely three points with stabilizers.

- The point ∞ has stabilizer given by $\mathbb{Z} \subseteq \mathrm{PSL}_2(\mathbb{Z})$.
- The point i has stabilizer μ_2 .
- The point ω has stabilizer μ_3 .

In order to get the correct ramification, we take a quotient by the smaller subgroup $\Gamma(7) \subseteq \mathrm{SL}_2(\mathbb{Z})$, which turns out to give ∞ the correct stabilizer μ_7 . Thus, $X(7) := \Gamma(7) \backslash \mathcal{H}^*$ will be our desired curve.

Remark 347. Note that $X(7)$ descends to a curve over $\mathbb{Q}(\mu_7)$ via its moduli description: it is the moduli space of elliptic curves E , together with an isomorphism $(\mathbb{Z}/7\mathbb{Z})^2 \rightarrow E[7]$. In order to descend to a curve over $\mathbb{Q}$, we define $X(7)$ as a curve over $\mathbb{Q}$ which classifies elliptic curves E , together with an isomorphism $\varphi: \mu_7 \times \mathbb{Z}/7\mathbb{Z} \rightarrow E[7]$ which is an isomorphism of symplectic spaces.

We would now like to use this cover to compute points on $S: x^2 + y^3 = z^7$. The approach of the previous subsection would have us compute points on $X(7)$ in some field extensions of degree equal to the degree of the map $X(7) \rightarrow \mathbb{P}^1$, which is the huge value $\frac{1}{2}[\Gamma(1) : \Gamma(7)] = 168$. We will not do this. Instead, recall that the descent obstruction informs us that points on $S(\mathbb{Z})$ come from points on the twists of $X(7) \rightarrow \mathbb{P}^1$.

Thus, we are motivated to find such twists. Twists come from $H^1(\mathbb{Q}; \mathrm{Aut}(X(7)/X(1)))$. This automorphism group is basically given by symplectic automorphisms of $\mu_7 \times \mathbb{Z}/7\mathbb{Z}$ (up to $\{\pm 1\}$). For example, any choice of elliptic curve E_0 can be equipped with an isomorphism $E_0[7] \cong \mathbb{Z}/7\mathbb{Z} \times \mu_7$, at least over the algebraic closure. This data is enough to define a twist of $X(7)$, which is the moduli space of elliptic curves E equipped with a symplectic isomorphism $E[7] \cong E_0[7]$.

22 March 31: The Automorphic Category

This talk was given by Hao Peng at MIT for the categorical local Langlands seminar.

22.1 Iwahori–Weyl Theory

Fix a reductive group G over F , and let A be a maximal split torus. We also choose S to be a torus containing A so that $S_{\check{F}}$ is a maximal split torus of $G_{\check{F}}$, and we let T be the centralizer of S . We now set $W_0 := N_G(T)(\check{F})/T(\check{F})$ and $\widetilde{W} := N_G(T)(\check{F})/T(\check{O})$ to be some Weyl groups. More or less by construction, we receive a short exact sequence

$$1 \rightarrow X_*(T)_{I_F} \rightarrow \widetilde{W} \rightarrow W_0 \rightarrow 1.$$

Now, let W_{aff} be the subgroup generated by the affine reflections; it turns out that $\widetilde{W}/W_{\text{aff}}$ is isomorphic to $\pi_1(G)_{I_F}$. Accordingly, there is a map $\widetilde{W} \rightarrow \pi_1(G)_{I_F}$ given by the projection. There is also a map $\widetilde{W} \rightarrow X_*(T)_{\mathbb{Q}}^{I_F}$ given by sending w to

$$\frac{1}{n} (w \cdot \sigma w \cdots \sigma^{n-1} w),$$

where n is chosen large enough so that this element is in $X_*(T)_{\mathbb{Q}}^{I_F}$.

Now, choose a parabolic $\check{P}$ containing $T_{\check{F}}$, whose Levi decomposition we denote by $\check{M}\check{U}$.

There is a length function ℓ on $\widetilde{W}$ induced by its decomposition as a semidirect product $X_*(T)_{I_F} \rtimes W_0$.

Definition 348. We say that $\sigma \in \widetilde{W}$ is σ -straight if and only if $\ell(w) = \langle 2\rho, \nu_w \rangle$, where ν_w is the image of w in $X_*(T)_{\mathbb{Q}}^{I_F}$.

Remark 349. It is equivalent to ask for

$$\ell(w \cdot \sigma w \cdots \sigma^{n-1} w) = n\ell(w)$$

for all $n > 0$. Thus, we are morally asking for w and its conjugates to have no cancellation when we multiply them together.

We also need a little combinatorics.

Definition 350. We write $w \xrightarrow{t}_{\sigma} w'$ if and only if $w' = tw\sigma(t)^{-1}$ and $\ell(t) \leq 1$ and $\ell(w') \leq \ell(w)$. Furthermore, we may just write $w \rightarrow_{\sigma} w'$ if there is a suitable path between the two Weyl elements. If $w \rightarrow_{\sigma} w'$ and $w' \rightarrow_{\sigma} w$, we say that w and w' are *conjugate by cyclic shift*.

Theorem 351 (He–Nie). Fix a conjugacy class C in $\widetilde{W}$.

- (a) If C is σ -straight, then $C_{\min}$ is the set of σ -straight elements.
- (b) For all $\nu \in C$, there is a path $\nu \rightarrow \nu_n$ given by simple reflections, and there is a decomposition with $\nu_n = uw$ where w is σ -straight, and u is in the Weyl group $W_{\check{f}}$ of some alcove $\check{f}$, and $w \cdot \sigma W_{\check{f}} \cdot w^{-1} = W_{\check{f}}$.

We are interested in classifying the σ -conjugacy classes in $\widetilde{W}$. Let $B(\widetilde{W})$ denote this set.

Theorem 352 (He). The set $B(\widetilde{W})$ is in bijection with the collection Q of quadruples $(M, x, \check{f}_M, \mathfrak{c})$, where

- M is a Levi subgroup of a standard parabolic P for which $T \subseteq M$,
- $x \in \widetilde{W}_M$ has $\ell_M(x) = 0$, and $\tilde{\nu} - x$ is dominant, and $Z_G(\nu_x) = M$,
- $\check{f}_M$ is a facet of the alcove which is $\text{Ad}_x \sigma$ -stable, and
- $\mathfrak{c} \in W_{\check{f}_M}$ is an elliptic $\text{Ad}_x \sigma$ -conjugacy class.

The bijection sends a quadruple $(M, x, \check{f}_M, \mathfrak{c})$ to the class of $\mathfrak{c} \cdot x$.

Example 353. Consider $G = \text{SL}_2(F)$, where $\widetilde{W}$ is an infinite dihedral group, generated by two simple reflections s_0 and s_1 . This group is split, so $B(\widetilde{W})$ is simply the conjugacy classes.

- There is the conjugacy class of 1, which is the quadruple $(G, 1, e, 1)$, where e is an edge of the building.
- There is the conjugacy class of s_0 , which is $(G, 1, v_0, s_0)$. (Note $W_{v_0} = \langle s_0 \rangle$.)
- The quadruple for s_1 is similar.
- There is lastly the conjugacy class of translations t_n (by $2n$, for $n \geq 1$). Here, our Levi must be T , so $\widetilde{W}_T$ is just $\mathbb{Z}$. We find that we need to take $x = t_n$, and the rest of the quadruple has no content.

22.2 Deligne–Lusztig Theory

Recall that for an algebraic k -group H , we have a category of character sheaves, which we can refine to a category of monodromic sheaves $\text{Shv}_{\text{mon}}(H)$. The inclusion $\text{Shv}_{\text{mon}}(H) \subseteq \text{Shv}(H)$ admits a right adjoint $\text{Avg}: \text{Shv}(H) \rightarrow \text{Shv}_{\text{mon}}(H)$.

Notation 354. We set $\widetilde{\text{Ch}} := \text{Avg}^{\text{mon}}(\delta_1)$, where δ_1 is the pushforward of the trivial sheaf along the inclusion $\{1\} \hookrightarrow H$.

Over finite fields, we already understand our representation theory.

Theorem 355 (Deligne–Lusztig). Fix a reductive group G over a finite field k . Then the category $\text{Rep } G(k)$ is generated by the representations $\widetilde{R}_w^*$.

Notation 356. We let $\text{Rep}^{\text{unip}} G(k)$ be the subcategory generated by $\widetilde{R}_{w, \check{u}}^{\text{mon}, *}$, and we let $\text{Rep}_c^{\text{unip}} G(k)$ be the subcategory generated by R_w^* .

In the affine setting, we let Iw^u be the kernel of the map $\text{Iw} \rightarrow S_k$ (where S_k lives over the finite field). Note that there is an action of $S_k \times S_k$ on $\text{Iw}^u \backslash LG / \text{Iw}^u$, so we receive a sheaf category $\text{Shv}_{\text{mon}}(\text{Iw}^u \backslash LG / \text{Iw}^u)$. One can define functors $R_w^?: \text{Rep } S_k \rightarrow \text{Shv}(\text{Isoc}_G)$ given by sending M to a shift of the Newton pushforward. We let $\widetilde{R}_w^?$ be the image of the trivial sheaf.

Proposition 357. If $w \in \widetilde{W}$ has minimal length in its conjugacy class, then there is b and u

$$\widetilde{R}_w^? \cong i_{b?} \text{cInd}_{P_b}^{G_b} \widetilde{R}_u^?[-\langle 2\rho, \nu_b \rangle].$$

Proof. Use Theorem 351 to put w into a standard form, and then we compute. ■

23 March 31: An Equivariant Tamagawa Number Formula

This talk was given by Cristian D. Popescu at MIT for the MIT number theory seminar.

23.1 Drinfeld Modules

The Tamagawa number formula is intended to generalize the Dirichlet class number formula. The equivariant Tamagawa number formula adds in some Galois action.

Conjecture 358 (Equivariant Tamagawa number). Fix an abelian extension K/F of number fields with Galois group G . Define an L -function $\Theta_{K/F} : \mathbb{C} \rightarrow \mathbb{C}[G]$ to be given by

$$\Theta_{K/F}(s) = \sum_{\chi} L(\chi, s) e_{\chi^{-1}}.$$

Then

$$\Theta_{K/F}^*(0) \mathbb{Z}[G] = \text{triv}_{\mathbb{R}[G]} \det_{\mathbb{Z}[G]} C_F^{\bullet},$$

where $C_F^{\bullet}$ is some perfect complex of $\mathbb{Z}[G]$ -modules.

There is a Tamagawa number formula for Drinfeld t -modules, and there is even an equivariant version. This equivariant version is our target.

Let q be a power of p , set $A := \mathbb{F}_q[t]$ and $k := \mathbb{F}_q(t)$. Let K/F be an abelian extension of function fields over k , with Galois group G , and we will assume that their field of constants is $\mathbb{F}_q$.

Definition 359 (Drinfeld). A Drinfeld module E over $\mathcal{O}_F$ is a morphism $\varphi_E : A \rightarrow \mathcal{O}_F\{\tau\}$ (where $\tau(x) = x^q$) of $\mathbb{F}_q$ -algebras such that $\varphi_E(t)$ has constant term t . The degree of φ_E is the rank of E .

Remark 360. Extension of scalars upgrades a Drinfeld module to a functor.

Remark 361. For each place v , there is a Tate module

$$T_{v_0} E := \lim_n E(\overline{F})[v_0^n],$$

which has a natural Galois action.

As with elliptic curves, these Tate modules assemble into a family of representations

$$H^1(E; A_{v_0}[G]) := T_{v_0} E^* \otimes_{A_{v_0}} A_{v_0}[G],$$

which has the natural diagonal Galois action.

Remark 362. There are compatibility conditions on these Galois representations.

We now work to define an equivariant L -function. Our analogue of $\mathbb{R}$ is $k_{\infty} := \mathbb{F}_q((1/t))$, and our analogue of $\mathbb{C}$ is $\widehat{k_{\infty}}$. Note that k_{∞} has a subgroup of monic elements, which is our analogue of $\mathbb{R}^+$.

Notation 363. Set $\mathbb{S}_{\infty} := \mathbb{C}_{\infty}^{\times} \times \mathbb{Z}_p$. Then we define $\Theta_{K/F}^E$ to be

$$\prod_v^* P_v^G(N v^{-s}),$$

where P_v^G is a characteristic polynomial. (Here, the $*$ means that we are only taking a product over the tamely ramified places which have good reduction for E .)

Here is our first special value result.

Proposition 364. In $k_\infty[G]^+$, one has

$$\Theta_{K/F}^E(0) = \prod_v \frac{|\mathcal{O}_K/v|_G}{|E(\mathcal{O}_K/v)|_G}.$$

To continue, we need to complete our Euler product. The idea is to plug in $\mathcal{M}/v$ into E for some projective module $\mathcal{M}$, instead of $\mathcal{O}_K/v$; then we go ahead and define $\Theta_{K/F}^{E,\mathcal{M}}$ to be the full product. After defining an Arakelov class, one is able to state and prove an equivariant Tamagawa number formula.

24 April 6: Geometry of the Spectral Side

This talk was given by Zhuoni Chi at MIT for the categorical local Langlands seminar.

24.1 Local Systems for cG

Let's start by defining the c -group.

Definition 365. Fix a reductive group G over F . Then we define

$${}^cG := \widehat{G} \rtimes (\mathbb{G}_m \times \text{Gal}(\check{F}/F)).$$

Here, the Galois action on the dual group $\widehat{G}$ is standard, and the action of $\mathbb{G}_m$ is given by

$$\mathbb{G}_m \rightarrow \widehat{G}_{\text{ad}} \rightarrow \text{Aut } \widehat{G},$$

where the left map is the character given by the half-sum ρ . We then let d

Example 366. For $G = \text{GL}_n$ or $\text{U}(2n+1)$ or there is a $\sqrt{q}$, one has ${}^cG \cong {}^L G \times \mathbb{G}_m$ (indeed, the point is that we can undo the half in the $\mathbb{G}_m$ -action), where ${}^L G = \Gamma$. Thus, in these cases, the stack of local systems will be the same. Thus, cG intuitively prevents us from having to choose $\sqrt{q}$.

Remark 367 (Bertolini Meli). This group already appeared in the global situation, where it is used to fix some square roots in some normalizations at real places.

The c -group allows us to define our stack of local systems.

Definition 368. We define $\text{Loc}_{{}^cG,F}$ over $\mathbb{Z}_\ell$ to be $\text{Loc}_{{}^cG,F}^\square / \widehat{G}$, which classifies parameters $\rho: W_F \rightarrow {}^cG$ up to suitable conjugation by $\widehat{G}$. A parameter is *tame* if and only if it factors through W_F^t .

Remark 369. There is a similar definition of $\text{Loc}_{{}^cG,L/F}^\square$ and $\text{Loc}_{{}^cG,L/F}$ by replacing W_F with W_F/W_L , where L is a finite Galois extension of F^t (which is the maximal tamely ramified extension of F). This is a reduced affine scheme which is flat and finite type of relative equi-dimension $\dim \widehat{G}$. Taking the colimit and then the quotient gives an algebraic stack.

Remark 370 (Zhang). It turns out that $\text{Loc}_{{}^cG,L/F}^\square$ is a locally complete intersection. For F^t , one can basically prove this by twisting the fact that the space $\{(X,Y) : XY = YX^q\}$ is a locally complete intersection.

24.2 The Bernstein Center

Here is our definition.

Definition 371 (Bernstein center). For each finite extension L of F , we define

$$Z_{cG,L/F} := H^0(\mathrm{Loc}_{cG,L/F}, \mathcal{O}),$$

and we define $Z_{cG,F}$ to be the limit over all L (which has an analogous presentation as above).

Remark 372. Given a parabolic subgroup $P \subseteq G$, one can make a similar definition of $\mathrm{Loc}_{cP,F}$. It admits an induction to $\mathrm{Loc}_{cG,F}$ and an induction and restriction to $\mathrm{Loc}_{cM,F}$, where M is a Levi subgroup. The construction of the Bernstein center also commutes.

Remark 373. One can also define $\mathrm{Loc}_{LcG,F}$ similarly, which agrees with $\mathrm{Loc}_{cG,F}$ up to choice of $\sqrt{q}$. There is an explicit isomorphism, which looks like it is given by fixing the $1/2$ in ρ , which is where the $\sqrt{q}$ is used.

We would now like to extract some geometry by descending from $\tilde{F}$. In addition to our usual notation (e.g., F is a nonarchimedean local field and so on), we let Γ_q be the group generated by τ and σ with the relation $\sigma\tau\sigma^{-1} = \tau^q$. One can choose a dense embedding $\iota: \Gamma_q \rightarrow W_F^t$ by sending τ to a geometric generator of I_F^t and σ to a lift of the Frobenius. Accordingly, we define $\mathrm{Loc}_{cG,F,\iota}$ to be parameters of the group $\Gamma_{F,\iota} := W_F \times_{W_{F,\iota}^t} \Gamma_q$.

Remark 374. It turns out that $\mathrm{Loc}_{cG,F,\iota}^{\mathrm{tame}}$ admits an explicit description

$$\lim_i \widehat{G} / \tau_i \widehat{G},$$

where $\tau_i := \sigma^{-i} \tau \sigma^i$.

Notation 375. There is an endomorphism ϕ of $\mathrm{Loc}_{cG,\tilde{F}}$ given by sending $\check{\varphi}$ to the parameter

$$\gamma \mapsto \bar{\sigma} \check{\varphi} (\sigma^{-1} \gamma \sigma).$$

Remark 376. Roughly speaking, ϕ looks like the Frobenius, and one can basically take fixed points of ϕ to recover $\mathrm{Loc}_{cG,F}$.

Remark 377. On the Bernstein center, one finds that $\mathrm{Spf} Z_{cG,F}$ embeds into ϕ -fixed points of $\mathrm{Spf} Z_{cG,\tilde{F}}$; the latter turns out to be a scheme which is finite over $\mathbb{Z}_\ell$.

24.3 The Inertia Type

Here is our definition.

Definition 378 (inertia type). An *inertia type* is a Λ -point of $(\mathrm{Spf} Z_{cG,\tilde{F}})^\phi$. It is *tame* if it comes from $(\mathrm{Spf} Z_{cG,F}^{\mathrm{tame}})^\phi$.

Notation 379. Given a locally closed substack $Z \rightarrow \mathrm{Loc}_{c,G,\tilde{F}}$, we set

$$\mathrm{Loc}_{c,G,F}^Z := Z \times_{\mathrm{Loc}_{c,G,\tilde{F}}} \mathrm{Loc}_{c,G,\tilde{F}}.$$

One can make a similar definition of $\mathrm{Loc}_{c,G,F}^{\widehat{Z}}$, where $\widehat{Z}$ is the formal completion of Z .

In particular, for a choice of inertia type $\widehat{\zeta}$, there is a substack $\mathrm{Loc}_{c,G,F}^{\widehat{\zeta}}$, and there is an explicit parameterization of the geometric irreducible components of this stack as given by $\widehat{G}$ -orbits in $\mathrm{Loc}_{c,G,\tilde{F}}^{\square}$ (under ζ) together with something related to a centralizer.

Example 380. Suppose ζ is unipotent, where $G = \mathrm{PGL}_2$ and $\widehat{G} = \mathrm{SL}_2$. Then there are two options for ζ , which are the identity and $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. It is not too hard to enumerate the parameters.

Each inertia type may have a distinguished element.

Definition 381 (essentially discrete). A parameter $\varphi \in \mathrm{Loc}_{c,G,F}$ is *essentially discrete* if and only if

$$C_{\widehat{G}}(\varphi)/Z_{\widehat{G}}^{\mathrm{Gal}(\tilde{F}/F)}$$

is finite.

Remark 382. It turns out to be equivalent to ask for $\varphi(\sigma)$ to be semisimple.

Remark 383. It turns out that $\mathrm{Loc}_{c,G,F}$ is smooth at φ .

Remark 384. There is at most one essential discrete parameter over each inertia type.

25 April 13: Galois Action on Higher Étale Homotopy Groups

This talk was given by Alexander Petrov at MIT for the Algebraic topology seminar.

25.1 Linear Groups

Here is a motivating result.

Theorem 385 (Borel). The natural map

$$\lim_n H_{\mathrm{cts}}^*(\mathrm{SL}_n(\mathbb{R}); \mathbb{R}) \rightarrow \lim_n H^*(\mathrm{SL}_n(\mathbb{Z}); \mathbb{R})$$

given by restriction of cocycles is an isomorphism.

The target can be computed by analytic techniques (with Lie algebra cohomology) as $\wedge^*(b_5, b_9, \dots)$ where $\deg b_i = i$, so we find that

$$K_i(\mathbb{Z})_{\mathbb{Q}} = \begin{cases} \mathbb{Q} & \text{if } i \equiv 1 \pmod{4}, \\ 0 & \text{otherwise.} \end{cases}$$

Namely, K -theory is just primitive elements in the group cohomology.

We are interested in finding a p -adic analogue.

Conjecture 386. The natural map

$$\lim_n H^*(SL_n(\mathbb{Z}_p); \mathbb{Q}_p) \rightarrow \lim_n H^*(SL_n(\mathbb{Z}); \mathbb{Q}_p)$$

is surjective.

Again, one can compute the left-hand side as $\wedge^*(\ell_3, \ell_5, \dots)$ where $\deg b_i = i$, again using some Lie algebra cohomology techniques. This was done by Lazard.

Remark 387. These natural maps are maps of Hopf algebras, so it turns out to be enough to merely send primitive elements to primitive elements. Namely, the archimedean calculation is enough to show that the natural restriction map on primitive elements from $K_{4k+1}(\mathbb{Z})_{\mathbb{Q}_p}$ to $K_{4k+1}^{\text{cont}}(\mathbb{Z}_p)_{\mathbb{Q}}$ is an isomorphism. This is interesting to number theorists because it is equivalent to the natural map

$$H^1(\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}); \mathbb{Q}_p(2k+1)) \rightarrow H^1(\text{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p); \mathbb{Q}_p(2k+1)).$$

We have nothing new to say about this conjecture.

Remark 388. For $n \geq 3$, every finite-index subgroup of $SL_n(\mathbb{Z})$ contains a congruence subgroup. Thus, the profinite completion of $SL_n(\mathbb{Z})$ is the limit of the groups $SL_n(\mathbb{Z}/N\mathbb{Z})$ (using some strong approximation), which is $SL_n(\widehat{\mathbb{Z}}) \cong \prod_p SL_n(\mathbb{Z}_p)$. For continuity reasons, it follows that

$$H^*(SL_n(\mathbb{Z}); \mathbb{Q}_p) \cong H^*(\widehat{SL_n(\mathbb{Z})}; \mathbb{Q}_p)$$

Thus, we see that we can take a profinite completion on the right-hand side of Conjecture 386. It will turn out that there are some interesting homotopy groups at play. To see this, we recall the definition of the profinite completion of a space.

Definition 389 (profinite completion). Fix a space X . Then the *profinite completion* of X is the limit over all functors $X \rightarrow F$, where F is a finite space.

Remark 390. One has $\pi_1(\widehat{X}; x) = \widehat{\pi_1(X; x)}$, but this might not be true for higher homotopy groups.

Remark 391. Profinite completion does not change cohomology with finite coefficients. If one defines

$$R\Gamma(\widehat{X}; \mathbb{Z}_p) := R\lim R\Gamma(\widehat{X}; \mathbb{Z}/p^n\mathbb{Z}),$$

then this can again be read off of the cohomology of X .

Example 392. Set $X := B\Gamma$, where Γ is some discrete group. Then

$$H_{\text{cts}}^*(\widehat{\Gamma}; \mathbb{Q}_p) = H_{\text{cts}}^*(\pi_1(\widehat{B\Gamma}); \mathbb{Q}_p).$$

This then maps to $H^*(\widehat{B\Gamma}; \mathbb{Q}_p)$, which is the interesting map. If Γ has $H^i(\Gamma; \mathbb{Z}/m\mathbb{Z})$ finite for all i and m (for example, if Γ is finite already), then we can descend to $H^*(B\Gamma; \mathbb{Q}_p)$, which is $H^*(\Gamma; \mathbb{Q}_p)$. Thus, the general question of whether this long composite is an isomorphism should be measured by $\widehat{B\Gamma}$; for example, if it is not an isomorphism, then $\pi_i(\widehat{B\Gamma}) \neq 0$ for some i .

Remark 393. Conversely, one can use Conjecture 386 to compute homotopy groups of $B\widehat{SL_n(\mathbb{Z})}$.

25.2 The Étale Homotopy Type

Here is our definition.

Definition 394 (étale homotopy type). There is a functor $(-)_\text{ét}$ which takes schemes and outputs profinite spaces.

Remark 395. It turns out that $Y_\text{ét}$ remembers cohomology, and it remembers the (étale) fundamental group.

Example 396. If Y is finite type over a subfield F of $\mathbb{C}$, then the étale homotopy type is immune to base-change, so we may assume $F = \mathbb{C}$. Then we define $X_{\mathbb{C},\text{ét}}$ as the profinite completion of $X(\mathbb{C})$. For example, $X_\text{ét}$ automatically has an action by $\text{Gal}(\overline{F}/F)$, which recovers the Galois action on the étale cohomology groups.

Example 397. For projective space, the Galois action factors through the cyclotomic character.

Example 398. Let F be a number field, and let X be an elliptic curve over F . Then the Tate module $H_\text{ét}^1(X_{\overline{F}}; \mathbb{Z}_p)$ is enough to recover X up to isogeny, by a theorem of Siegel.

Example 399. Let Y be a smooth proper curve over a number field F , and define X to be the scheme Y glued together at two rational points x and y . Then there is an extension

$$0 \rightarrow \mathbb{Z}_p \rightarrow H_\text{ét}^1(X_{\overline{F}}; \mathbb{Z}_p) \rightarrow H_\text{ét}^1(Y_{\overline{F}}; \mathbb{Z}_p) \rightarrow 0.$$

Even though the left and right terms do not depend on the choice of points, the extension class depends exactly on $[x] - [y] \in \text{Pic}(Y)/\text{Pic}(Y)_\text{tors}$.

25.3 Symplectic Groups

Now, we are interested in the homotopy type $\widehat{\text{BGL}}_n(\mathbb{Z})$. But this turns out to not be the étale homotopy type of any smooth algebraic variety as soon as $n \geq 3$. To fix this, we avoid working with linear groups and instead work with symplectic groups.

Notation 400. Let $\mathcal{A}_g$ be the moduli stack of principally polarized abelian varieties of dimension g .

Remark 401. We can realize $\mathcal{A}_g$ as the orbifold $\mathbb{H}_g/\text{Sp}_{2g}(\mathbb{Z})$, where $\mathbb{H}_g$ is the Siegel space. Thus, $\mathcal{A}_g(\mathbb{C})$ is $\text{BSp}_{2g}(\mathbb{Z})$. It follows that we receive an interesting Galois action on $\widehat{\text{BSp}}_{2g}(\mathbb{Z})$.

Theorem 402. There is a non-split extension

$$0 \rightarrow \mathbb{Q}_p(-1) \rightarrow V \rightarrow \mathbb{Q}_p \rightarrow 0$$

of Galois representations such that V embeds into $\text{Hom}_{\text{cts}}(\pi_2^\text{ét}(\mathcal{A}_{g,\overline{\mathbb{Q}}}), \mathbb{Q}_p)$.

Remark 403. This V is a genuinely new Galois representation: one cannot view V as a subquotient of any “geometric” Galois representation, namely coming from $H_{\text{ét}}^n(X_{\overline{\mathbb{Q}}}; \mathbb{Q}_p)$. The reason is due to Hodge–Tate weights: the quotient $\mathbb{Q}_p$ of V has weight 0 while the source $\mathbb{Q}_p(-1)$ has weight -1 , and there are no such non-split Galois representations coming from geometry.

Let’s give retell the story of our motivation for symplectic groups. One has

$$\lim_n H^*(\mathrm{Sp}_{2n}(\mathbb{Z}); \mathbb{Q}_p) \cong \mathbb{Q}_p[x_2, x_6, x_{10}, \dots],$$

where the subscripts denote degrees. These come from some Chern classes. On the other hand,

$$H_{\text{cts}}^*(\mathrm{Sp}_{2n}(\mathbb{Z}); \mathbb{Q}_p) = H_{\text{cts}}^*(\mathrm{Sp}_{2n}(\widehat{\mathbb{Z}}); \mathbb{Q}_p) \cong \wedge^*(b_3, b_7, \dots).$$

Thus, the natural map from the second algebra to the first one must vanish for degree reasons! It follows that we expect many interesting homotopy groups. Here is what we expect.

Conjecture 404. For each i , the direct system

$$\pi_i^{\text{ét}}(\mathcal{A}_{g, \overline{\mathbb{Q}}}) \rightarrow \pi_i^{\text{ét}}(\mathcal{A}_{g+1, \overline{\mathbb{Q}}}) \rightarrow \pi_i^{\text{ét}}(\mathcal{A}_{g+2, \overline{\mathbb{Q}}}) \rightarrow \dots$$

stabilizes to 0 if $i \not\equiv 2 \pmod{4}$; otherwise, if $i = 4k + 2$, it stabilizes to a Galois representation V_k fitting in an extension

$$0 \rightarrow \mathbb{Q}_p \rightarrow V_k \rightarrow \mathbb{Q}_p(2k + 1) \rightarrow 0.$$

Remark 405. Again, these extensions V_k fail to be geometric.

Here is what we can prove.

Theorem 406 (Pan–S). Fix a smooth algebraic variety X over $\mathbb{C}$, and suppose that we have a representation $\pi_1(X(\mathbb{C})) \rightarrow \mathrm{Sp}_{2n}(\mathbb{Z}_p)$. Then the map

$$H_{\text{cts}}^*(\mathrm{Sp}_{2n}(\mathbb{Z}_p); \mathbb{Q}_p) \rightarrow H^*(X(\mathbb{C}); \mathbb{Q}_p)$$

vanishes.

Remark 407. There is an analogous statement for the group SL_n instead of Sp_{2n} .

Instead of explaining anything about this proof, we give the following example.

Example 408. Let’s prove this for $X = \mathcal{A}_{g, \mathbb{C}}$, without using the calculations from earlier. Then there is a natural monodromy representation $\rho: \pi_1(X(\mathbb{C})) \rightarrow \mathrm{Sp}_{2n}(\mathbb{Z}_p)$, so we get a natural map

$$H_{\text{cts}}^*(\mathrm{Sp}_{2n}(\mathbb{Z}_p); \mathbb{Q}_p) \rightarrow H_{\text{ét}}^*(\mathcal{A}_{g, \overline{\mathbb{Q}}}; \mathbb{Q}_p).$$

This map is Galois-equivariant because the representation is defined over $\mathbb{Q}$. It then follows by looking at some Hodge–Tate weights that the target has no nonzero invariants (e.g., one can use the Weil conjectures), but the Galois action on the source is trivial, so the actual map must vanish.

While we’re here, let’s explain how we can say anything about the Galois action.

Theorem 409. The natural map

$$\mathrm{R}\Gamma(\mathrm{Sp}_{2n}(\mathbb{Z}_p); \mathbb{Q}_p) \rightarrow \mathrm{R}\Gamma(\mathrm{Sp}_{2n}(\mathbb{Z}); \mathbb{Q}_p)$$

is Galois-equivariant and nonzero in the derived category. In fact, this map is given by certain explicit classes from V_k s as constructed in the conjecture.

26 April 14: The Spectral Category

This talk was given by Mohit at MIT for the categorical local Langlands seminar.

26.1 Working with IndCoh

We are interested in proving an equivalence of categories like the following.

Theorem 410. There is an equivalence

$$\mathrm{Shv}^{\widehat{\mathrm{unip}}}(\mathrm{Isoc}_G, \overline{\mathbb{Q}}_\ell) \rightarrow \mathrm{IndCoh}(\mathrm{Loc}_{cG,F}^{\widehat{\mathrm{unip}}} \otimes \overline{\mathbb{Q}}_\ell).$$

The idea is to write both sides as the categorical trace of some categories; we did this for the left-hand side for the past few weeks, so it is now time to do it for the right-hand side. Notably, we have yet to receive a good definition of the right-hand side.

Let's start by defining coherent sheaves.

Notation 411. For today, a *stack* is an algebraic stack of almost finite presentation over a coefficient ring Λ . Similarly, a *space* is an algebraic space which is almost finite presentation over Λ .

To start, note that we already have a functor $\mathrm{Mod}: \mathrm{CAlg}_\Lambda \rightarrow \mathrm{LinCat}_\Lambda$. One can extend this to quasicoherent sheaves by taking a Kan extension.

Definition 412 (quasicoherent sheaf). For a prestack X , we define its category of *quasicoherent sheaves* as

$$\mathrm{QCoh}(X) := \lim_{\mathrm{Spec} A \rightarrow X} \mathrm{Mod}(A).$$

Remark 413. It follows from the construction that $\mathrm{QCoh}(X)$ has a t -structure, an exterior product $\boxtimes$, and $*$ -pullbacks and $*$ -pushforwards for morphisms representable in algebraic spaces.

Similarly, we have perfect sheaves.

Definition 414 (perfect). For an algebra A , we let $\mathrm{Perf}(A)$ be the thick subcategory of $\mathrm{Mod}(A)$ generated by A . For a prestack X , we define

$$\mathrm{QCoh}(X) = \lim_{\mathrm{Spec} A \rightarrow X} \mathrm{Perf}(A).$$

Remark 415. When X is an algebraic space, $\mathrm{IndPerf}(X) = \mathrm{QCoh}(X)$. In general, $\mathrm{Perf}(X)$ is the dualizable objects in $\mathrm{QCoh}(X)$.

This allows us to define coherent sheaves.

Definition 416 (coherent). For a stack X , we define $\mathrm{Coh}(X)$ to be those quasicoherent sheaves $\mathcal{F}$ such that $\mathcal{H}^i(\mathcal{F})$ is coherent on the underlying classical stack of X and zero almost everywhere. We then define $\mathrm{IndCoh}(X)$ to be the ind-objects of $\mathrm{Coh}(X)$.

Example 417. One has that $\mathcal{H}^0(\mathcal{O}_X)$ is just the structure sheaf on the underlying classical stack.

Remark 418. There is an inclusion $\mathrm{Coh}(X) \rightarrow \mathrm{QCoh}(X)$, so one gets an embedding $\mathrm{IndCoh}(X) \rightarrow \mathrm{QCoh}(X)$. This turns out to give an equivalence on bounded objects.

Remark 419. There is an action of $\mathrm{Perf}(X)$ on $\mathrm{Coh}(X)$, which extends to an action for the ind-categories. If $\mathrm{Perf}(X) \subseteq \mathrm{Coh}(X)$ (this amounts to requiring that $\mathcal{O}_X$ is coherent), then we actually receive a map $\mathrm{IndPerf}(X) \rightarrow \mathrm{IndCoh}(X)$.

We will want some functoriality for IndCoh .

Remark 420. Given a map $f: X \rightarrow Y$, the map $f_*: \mathrm{Coh}(X) \rightarrow \mathrm{QCoh}(Y)$ actually maps to objects with bounded homotopy groups, which is $\mathrm{QCoh}(Y)^+$. This is then equivalent to $\mathrm{IndCoh}(Y)^+$, which embeds into $\mathrm{IndCoh}(Y)$. Taking ind-completion, we receive a functor

$$f_*: \mathrm{IndCoh}(X) \rightarrow \mathrm{IndCoh}(Y).$$

If f has “finite tor amplitude,” then there is an adjoint functor f^* ; this means that $f^*: \mathrm{QCoh}(Y)^{\geq 0} \rightarrow \mathrm{QCoh}(X)^{\geq n}$ for some n .

The previous remark motivates us to upgrade IndCoh to a functor.

Theorem 421. There is a functor IndCoh^* from the category of correspondences on ind-stacks (where the pullback morphism needs to exist) to $\mathrm{LinCat}_\Lambda(X)$. Here,

$$\mathrm{IndCoh}^*(\mathrm{colim} X_\bullet) := \mathrm{colim} \mathrm{IndCoh}^*(X_\bullet),$$

where the latter colimit is taken for $*$ -pushforwards of closed embeddings.

Here is some more functoriality.

Remark 422. Given X and Y , the exterior product $\boxtimes: \mathrm{QCoh}(X) \times \mathrm{QCoh}(Y) \rightarrow \mathrm{QCoh}(X \times Y)$ restricts to coherent sheaves and then completes to

$$\boxtimes \mathrm{IndCoh}(X) \otimes_\Lambda \mathrm{IndCoh}(Y) \rightarrow \mathrm{IndCoh}(X \times Y).$$

This is fully faithful, and one can check that it is an equivalence by stratifying. The idea is to use some excision for IndCoh .

With some $*$ -functors, we will want some $!$ -functors.

Remark 423. Every morphism $f: X \rightarrow Y$ of algebraic spaces admits a compactification $X \hookrightarrow \overline{X} \rightarrow Y$. One shows this by passing to schemes. An adjoint functor theorem shows that $(-)^!$ exists for the proper morphisms, so taking the composite defines $f^!$.

Theorem 424. There is a functor $\mathrm{IndCoh}^!$ from the category of correspondences of ind-stacks (where the morphisms are representable in ind-spaces) to LinCat_Λ . It sends the correspondence $X \xleftarrow{f} Z \xrightarrow{g} Y$ to $g_* f^!$. It turns out that $\mathrm{IndCoh}^!(X) = \mathrm{IndCoh}^*(X)$.

Remark 425. There is content in the statement $\mathrm{IndCoh}^!(X) = \mathrm{IndCoh}^*(X)$. Indeed, one should define this for ind-stacks to

$$\lim \mathrm{IndCoh}(X_\bullet),$$

where the latter limit is taken over $!$ -pullbacks.

We now have enough functors to do some interesting things.

Remark 426. The category $\mathrm{IndCoh}^!$ is symmetric monoidal with tensor product

$$\mathcal{F} \otimes^! \mathcal{G} := \Delta^!(\mathcal{F} \boxtimes \mathcal{G}).$$

The unit is the dualizing sheaf. This turns out to give a duality $\mathbb{D}$ on $\mathrm{IndCoh}(X)$, recovering Grothendieck duality.

Remark 427. If $f: X \rightarrow Y$ is representable in ind-spaces, then $\mathrm{IndCoh}(X \times_Y X)$ admits a convolution structure. If $\boxtimes$ is an equivalence for the product $X \times_Y X$ (with itself), then the convolution has some duality.

26.2 The Steinberg Stack

We are now allowed to write down the expression $\mathrm{IndCoh}(\mathrm{Loc}_{c_{G,F}}^{\mathrm{unip}})$. Eventually, we will want to show an equivalence with this category, so we need to be able to find some interesting objects.

Example 428. Here are some “constant objects.” There is a natural map $f: \mathrm{Loc}_{c_{G,F}}^{\mathrm{unip}} \rightarrow B\widehat{G}$, so we can pull back perfect objects in the target to perfect objects in the source. These turn out to give objects in IndCoh .

Let’s find some more objects. We will go through the following intermediate object.

Definition 429 (Steinberg stack). We define the *Steinberg stack* as

$$\mathcal{S}_{c_{G,F}}^{\mathrm{unip}} := \mathrm{Loc}_{c_{B,\widehat{F}}}^{\mathrm{unip}} \times_{\mathrm{Loc}_{c_{G,\widehat{F}}}} \mathrm{Loc}_{B,\widehat{F}}^{\mathrm{unip}}$$

There is also a tame variant.

Remark 430. There is a resolution

$$\mathcal{S}_{c_{G,\widehat{F}}}^{\mathrm{unip}} \cong (\widetilde{U}_{\widehat{G}} \times_{\widehat{G}} \widetilde{U}_{\widehat{G}}) / \widehat{G},$$

where $\widetilde{U}_{\widehat{G}}$ is the unipotent variety. This gives many nice geometric properties of the Steinberg stack; for example, it admits a stratification from a stratification of the unipotent variety.

We still need a way to mediate from the Steinberg stack.

Definition 431. We define

$$\widetilde{\mathrm{Loc}}_{cG,F}^{\mathrm{tame}} := \mathrm{Loc}_{cB,\tilde{F}}^{\mathrm{tame}} \times_{\mathrm{Loc}_{cG,\tilde{F}}^{\mathrm{tame}}} \mathrm{Loc}_{cG,F}^{\mathrm{tame}}.$$

There is a slightly more complicated definition in the unipotent case.

Remark 432. There is a natural projection $\widetilde{\mathrm{Loc}}_{cG,F}^{\mathrm{tame}} \rightarrow \mathrm{Loc}_{cG,F}^{\mathrm{tame}}$. It is harder to define, but there is also a map δ to $\mathcal{S}_{G,\tilde{F}}^{\mathrm{tame}}$. For brevity, define $X := \mathrm{Loc}_{B,\tilde{F}}^{\mathrm{tame}}$ and $Y := \mathrm{Loc}_{G,\tilde{F}}^{\mathrm{tame}}$, and then

$$\widetilde{\mathrm{Loc}}_G^{\mathrm{tame}} = X \times_Y \mathcal{L}_\varphi(Y) = X \times_Y (Y \times_Y Y) = X \times_{Y \times_Y Y} Y,$$

which then maps via diagonal to $(X \times X) \times_{Y \times_Y Y} Y = X \times_Y X$, which is $\mathcal{S}_G^{\mathrm{tame}}$. In short, this map δ can be given as $x \mapsto (x, \varphi(x))$.

Thus, we see that $\pi_* \delta^!$ can give us many interesting objects.

Definition 433 (Deligne–Lusztig induction). We define *Deligne–Lusztig induction* $\mathrm{Ch}_c^{\mathrm{tame}} := \pi_* \delta^!$ to be the functor

$$\pi_* \delta^! : \mathrm{IndCoh}(\mathcal{S}_{cG,\tilde{F}}^{\mathrm{tame}}) \rightarrow \mathrm{IndCoh}(\mathrm{Loc}_{cG,F}^{\mathrm{tame}}).$$

Remark 434. It turns out that the essential image of Ch is contained in $\mathrm{IndCoh}(\mathrm{Loc}_{cG,F}^{\mathrm{unip}})$ and stable under the action of $\mathrm{IndPerf}$.

We are now interested in understanding the Steinberg stack better.

Proposition 435. Let S be the Steinberg stack. Then

$$\boxtimes : \mathrm{IndCoh}(S) \otimes \mathrm{IndCoh}(Y) \rightarrow \mathrm{IndCoh}(S \times Y).$$

One shows this by reducing to the case of $Y = S$, where one can check it via explicit calculation on tangent spaces and things.

Remark 436. Taking $Y = S$ reveals that $\mathrm{IndCoh}(S)$ admits a duality $\mathbb{D}^*$ coming from convolution (because then $\mathrm{IndCoh}(S)$ becomes semi-rigid with a unit). It turns out that this is the “swap” of the original duality $\mathbb{D}$.

Remark 437. The dualities almost commute with Ch : one has $\mathbb{D} \circ \mathrm{Ch} = \mathrm{Ch} \circ \mathrm{swap} \circ \mathbb{D}^*$.

27 April 14: p -adic Families and Arithmetic

This talk was given by Giada Grossi at MIT for the BC–MIT number theory seminar.

27.1 Motivation

Here is our motivating result.

Theorem 438 (Gross–Zagier). Fix an elliptic curve E over $\mathbb{Q}$, and choose a large prime p . Then if the analytic rank of E is in $\{0, 1\}$, then the algebraic rank equals the analytic rank, and $\#\mathrm{III}(E/\mathbb{Q})[p^\infty]$ is finite.

This was shown in two steps. Choose an imaginary quadratic extension $K/\mathbb{Q}$ satisfying some Heegner hypothesis on ramification.

- Gross–Zagier showed that $L'(E/K, 1) \neq 0$ implies that a particular Heegner point has infinite order.
- Kolyvagin showed that the Heegner point having infinite order implies that $\text{Sel}_{p^\infty}(E/K)$ has rank 1.

We are interested in showing some cases of the Bloch–Kato conjecture for groups of higher rank. The idea is to allow our objects vary p -adically and combine an Iwasawa main conjecture with a calculation of p -adic L -functions. The problem is that the value of a p -adic L -value may have a difficult time actually recovering an L -value at the end. However, taking further p -adic variation allows us to find the L -values.

28 April 21: Spectral Category as Categorical Trace

This talk was given by Alexaner Bertoloni Meli at MIT for the categorical local Langlands seminar.

28.1 Back to Categorical Trace

We work with $\Lambda = \overline{\mathbb{Q}_\ell}$ for simplicity. Here is some formalism.

Definition 439. Let A be an algebra object in a stable category R , and let F be an A -bimodule. Then we define the trace

$$\text{Tr}(A, F) := A \otimes_{A \otimes A^{\text{op}}} F,$$

which is basically some Hochschild cohomology. For an endomorphism φ of A , then $\text{Tr}(A, \varphi) := \text{Tr}(A, {}^\varphi A)$, where ${}^\varphi A$ is the A -bimodule whose left action is twisted by φ .

Example 440. If A is an ordinary k -algebra, then the trace is the quotient of M by the submodule generated by the elements $am - ma$.

Definition 441. Let A be an algebra object in a stable category R , and let F be an A -bimodule. Then we define the center as $\text{Hom}_{A \otimes A^{\text{op}}}(A, F)$.

Here is our target.

Theorem 442. There is a commutative diagram

$$\begin{array}{ccc} \text{IndCoh}(\mathcal{S}_G^t) & \xrightarrow{\delta^!} & \text{IndCoh}(\widetilde{\text{Loc}}_G^t) \\ \downarrow & & \downarrow \pi_* \\ \text{Tr}(\text{IndCoh}(\mathcal{S}_G^t), \varphi) & \longrightarrow & \text{IndCoh}(\text{Loc}_G^t) \end{array}$$

where the bottom arrow is an isomorphism.

Here, this categorical trace makes sense because the IndCoh lives in the stable category of stable categories, so taking its trace will output some other stable category.

Remark 443. Note that

$$\mathrm{Tr}(\mathrm{IndCoh}(\mathcal{S}_G^t), \varphi) = \mathrm{Tr}(\mathrm{IndCoh}(\mathcal{S}_G^t), {}^\varphi \mathrm{IndCoh}(\mathcal{S}_G^t)),$$

so it is desirable to rewrite the second category more geometrically. For brevity, set $X := \mathrm{Loc}_{B, \tilde{F}}^t$ and $Y := \mathrm{Loc}_{G, \tilde{F}}^t$. Then the second category is $\mathrm{IndCoh}(X \times_Y Y \times_Y X)$, where the left fiber product is given by $\varphi: Y \rightarrow Y$, and the right fiber product is given by $\mathrm{id}: Y \rightarrow Y$. (Morally, the middle Y is a graph of the Frobenius.) There are some correspondences which explain why $\mathrm{IndCoh}(X \times_Y Y \times_Y X)$ has the correct bimodule structure.

We will spend the rest of this subsection saying something (but very little) about the proof of Theorem 442. The factorization in the bottom is roughly automatic from writing down some correspondences. The fact that the bottom functor is fully faithful follows from some general nonsense, so let's focus on the essential surjectivity. This will come from something about singular support.

Definition 444 (singular support). Let X be a quasi-smooth algebraic stack. Then we define the *singular support* to be

$$\mathrm{Sing} X := \mathrm{Spec}_{\mathcal{O}_X} \mathrm{Sym} \mathcal{H}^1 \mathbb{T}_X.$$

Remark 445. If $X = \mathrm{Spec} B$, it turns out that $H^1(\mathbb{T}_X) = \mathrm{HH}^2(B)$.

Remark 446. One can show that $\mathrm{Sing}(\mathrm{Loc}_G)$ naturally contains the space of Arthur parameters.

Definition 447 (singular support). Fix a coherent sheaf $\mathcal{F}$. Then we define the *singular support* of $\mathcal{F}$ to be the support of the corresponding sheaf on $\mathrm{Sing}(X)$. For $S \subseteq X$, we define $\mathrm{Coh}_S(X)$ to be the category of coherent sheaves supported on S .

It now turns out that a quasi-smooth map $f: X \rightarrow Y$ in characteristic zero induces maps f^* and f_* on some categories of coherent sheaves with some support, and the images of these functors generate. By some singular support calculation, one can use

28.2 Applications

For our first application, let's recall Bezrukavnikov's equivalence.

Theorem 448. There is an equivalence

$$\mathrm{Shv}(\mathrm{Iw}^u \backslash LG / \mathrm{Iw}^u) \rightarrow \mathrm{IndCoh}(\mathcal{S}_G^t).$$

Taking categorical trace on both sides shows the following upon recognizing our earlier work.

Theorem 449. There is an equivalence

$$\mathrm{Shv}^t(\mathrm{Isoc}_G) \cong \mathrm{IndCoh}(\mathrm{Loc}_G^t).$$

One may hope that this equivalence preserves some structure on both sides. For example, it should preserve some monoidal units; on the left, this is some compact induction of the trivial representation, and on the right, this is the pushforward of the structure sheaf. One can play this game more generally by taking various categorical traces of sheaves.

Remark 450. It is also possible to recover a semisimple tame local Langlands correspondence: for $b \in B(G)$, one has functors

$$\mathrm{Rep}^t(G_b(F)) \rightarrow \mathrm{Shv}^t(\mathrm{Isoc}_G) \cong \mathrm{IndCoh}(\mathrm{Loc}_G^t),$$

which induces functors on the centers in the opposite direction. We can now de-categorify: for irreducible π , the center of $\mathrm{Rep}^t(G_b(F))$ acts by some central character. On the other hand, the center of $\mathrm{IndCoh}(\mathrm{Loc}_G^t)$ is $\mathcal{O}_{\mathrm{Spf} Z}$ for a suitable algebra Z , so we find that irreducibles correspond to points on this formal object.

Remark 451. We can recover a matching of supercuspidals: for basic b and depth-zero supercuspidal representation $\pi \in \mathrm{Rep}^t(G_b(F))$, then π corresponds to a pushforward of some vector bundle from some “Vogan stack.”

29 April 21: Excursions on p -Adic Groups

This talk was given by Jared Weinstein at MIT for the MIT number theory seminar.

29.1 The Bernstein Center

Here is a general construction.

Definition 452 (Bernstein center). Let $\mathcal{C}$ be a rigid $\otimes$ -category. Then we define its *center* $Z(\mathcal{C})$ as the endomorphisms of the identity natural transformation.

Example 453 (Bernstein center). For a p -adic group G , we define $Z(G)$ to be the center of $\mathrm{Rep}^{\mathrm{sm}}(G)$. It decomposes into a sum of “Bernstien blocks.”

Example 454. Note that $Z(G)$ acts by a scalar on each irreducible representation. Thus, for a finite group G , we find that $Z(G)$ is a product of $\mathbb{C}$ s, with a number of factors equal to the number of irreducible representations of G .

One can also view the Bernstein center coming from distributions.

Definition 455 (distribution). Fix a p -adic group G . Then we define $C_c(G)$ to be the space of compactly supported functions on G . The space of *distributions* $\mathrm{Dist}(G)$ is the space of continuous linear functionals on $C_c(G)$. We say that μ is *essentially compact* if and only if $(f * \mu) \in C_c(G)$ for all $f \in C_c(G)$.

Remark 456. Note that G admits an adjoint action on $C_c(G)$ and thus on $\mathrm{Dist}(G)$.

Remark 457. The box product gives $\mathrm{Dist}_c(G)$ an algebra structure.

Theorem 458 (Bernstein). Fix a p -adic group G . Then there is a map $\mathrm{Dist}_c(G)^G \rightarrow Z(G)$ given by sending a distribution μ to the endomorphism defined as follows: for smooth π , we define $z_\mu \in \mathrm{End}(\pi)$ to have

$$z_\mu(\pi)(v) := \pi(1_K dk * \pi)(v).$$

This map $\mu \mapsto z_\mu$ is an isomorphism.

Example 459. The “identity” distribution δ_1 goes to the unit $1 \in Z(G)$.

Example 460 (Varma). One can see something about the local Langlands correspondence in this case: given a smooth irreducible representation π of $\mathrm{GL}_2(F)$, let $\varphi_\pi: W_F \rightarrow \mathrm{GL}_2(\mathbb{C})$ be the associated Langlands parameter. Then given $r \geq 1$, one can compute

$$z_r(\pi) := \mathrm{tr}(\mathrm{Frob}; (\mathrm{Sym}^r \varphi_\pi)^{I_F}).$$

29.2 Excursion Operators

Here is a toy model for our excursion operators.

Definition 461. Fix a p -adic reductive group G with complex points $\widehat{G}$. Given a pair (ρ, τ) of a smooth algebraic representation ρ of $\widehat{G}$ and $\tau \in W_F$, we define $\mathrm{Exc}(V, \tau) \in Z(G)$ by

$$\mathrm{tr}(\tau; \rho \circ \varphi_\pi^{\mathrm{FS}}),$$

where $\varphi_\pi^{\mathrm{FS}}: W_F \rightarrow {}^L G$ is the Fargue–Scholze parameter.

We will not define the Fargue–Scholze parameters. In fact, we remark that our entire exposition is backwards: Fargue–Scholze define the excursion operators separately using p -adic geometry, and then they go back and use this property to define the parameters.

Here is our main result.

Theorem 462 (Bakkeberg). Set $G := \mathrm{SL}_{2,F}$, choose a smooth irreducible representation π , and let φ_π be the Langlands parameter, which is the Fargue–Scholze parameter in this case. For any (V, τ) , there is an explicit sequence $\{\mathrm{Exc}_r(V, \tau)\}_{r \geq 1}$ of locally constant functions $G(F) \rightarrow \mathbb{Z}[1/p]$ satisfying the following.

- (a) One has $\mathrm{Exc}_r(V, \tau) \rightarrow \mathrm{Exc}(V, \tau)$ in distribution.
- (b) On a dense open subset $U(V, \tau)$ of the regular semisimple locus of G , the distributions Exc_r arise from functions converging pointwise.

Remark 463. It is not too hard to describe $U(V, \tau)$: it is the set of non-collision locus, where we say that regular semisimple g collides if and only if the centralizer $S := C_G(g)$ admits $\chi \in X_*(S)$ for which χ is defined over a finite extension E/F for which

$$\chi(\mathrm{rec}_E \tau) \sim_{\overline{F}} g$$

for some $\tau \in W_E$.

The main ingredient in our proof is the Plancherel formula.

Theorem 464 (Harish-Chandra). Fix a p -adic reductive group G . For any $f \in C_c(G)$, one has

$$f(e) = \int_{\mathrm{Temp}(G)} \Theta_\pi(f) d\mu(\pi),$$

where μ is some measure on the space of tempered distributions.

Roughly speaking, our new excursion operators are defined by

$$\mathrm{Exc}'(\rho, \tau)(f) := \int_{\mathrm{Temp}(G)} \mathrm{tr}(\tau; \rho \circ \varphi_\pi) \Theta_\pi(f) d\mu(\pi).$$

One can then compute this integral.

30 April 28: The Cohomology of Shimura Varieties

This talk was given by Hao Peng at MIT for the categorical local Langlands seminar.

30.1 Stuff

As usual, F is a finite extension of $\mathbb{Q}_p$, and we choose a reductive group G over F . Our coefficients Λ will be either $\overline{\mathbb{Q}}_\ell$ or $\overline{\mathbb{F}}_\ell$, where $\ell \neq p$. In this setting, the local Langlands conjectures take the following form.

Conjecture 465. Suppose there is an unramified reductive group G^* over F equipped with a pinning $T^* \subseteq B^*$ and an isomorphism $\eta: G_{\tilde{F}} \cong G_{\tilde{F}}^*$ admitting $b \in G^*(\tilde{F})$ has $\eta\sigma\eta^{-1} = \text{Ad}_b \sigma^*$. Writing $\tilde{w}$ for b , where $w \in \widetilde{W}$, we then have an isomorphism $\eta_{\tilde{w}}: \text{Isoc}_G \rightarrow \text{Isoc}_{G^*}$, and the composite

$$\text{IndShv}_{\text{fg}}^{\text{unip}}(\text{Isoc}_G) \rightarrow \text{IndShv}_{\text{fg}}^{\text{unip}}(\text{Isoc}_{G^*}) \rightarrow \text{Coh}\left(\widehat{\text{Loc}}_{c,G,F}^{\text{unip}}\right)$$

is an equivalence.

The point of G^* is to put some unramified condition on G . The main issue is the construction of the map.

Let (G, X) be a Shimura datum. We choose level structures $K_p \subseteq G(\mathbb{Q}_p)$ and neat $K^p \subseteq G(\mathbb{A}_f^p)$, and we set $K := K_p K^p$. Then we have a Shimura variety $\text{Sh}_K(G, X)$ defined over the reflex field E . We also go ahead and choose an isomorphism $\overline{\mathbb{Q}}_\ell \cong \mathbb{C}$, which induces a place v of E over p .

Let μ be the Hodge cocharacter of the Shimura variety, and we let $\mu^* := -w_0\mu$ be its opposite.

Notation 466. We set

$$\text{Adm } \mu^* := \{w \in \widetilde{W} : w \leq \bar{\lambda} \text{ for some } \bar{\lambda} \in W_0 \cdot \overline{\mu^*}\}.$$

Here, $\bar{\lambda}$ is the image of λ along the embedding $X_*(T) \subseteq \widetilde{W}$.

I am a bit too tired to take notes, but let me learn something about the Igusa stack.

Theorem 467. There is an Igusa stack $\text{Igs}(G, X)$ over $\overline{\mathbb{F}}_p$ fitting into the following Cartesian square.

$$\begin{array}{ccc} \text{Sh}_{\mu,f} & \xrightarrow{\text{loc}_p} & \text{Sh}_{P,\mu^*}^{\text{loc}} \\ \text{Nt} \downarrow & & \downarrow \text{Nt} \\ \text{Igs} & \xrightarrow{\text{loc}_p} & \text{Isoc}_G \end{array}$$

Idea. The right-hand side is given by descending the Hecke stack. The left-hand side can thus be defined by descending the Hecke stack. ■

31 April 30: Descent Obstructions for Curves

Here we go.

31.1 Comparing Descents

Throughout, k is a number field, and X is a smooth projective geometrically connected variety.

Notation 468. Fix a variety X over a number field k . Then we define

$$X(\mathbb{A}_k)_\bullet := \prod_{v \nmid \infty} X(k_v) \times \prod_{v \mid \infty} \pi_0(X(k_v)).$$

Here is our target.

Theorem 469. Fix a smooth projective geometrically connected curve X over a global field k . Then the finite abelian descent obstruction and the Brauer–Manin obstructions are the same.

Let's start by reviewing the descent obstruction.

Proposition 470. Fix a finite abelian group scheme G over a field k . For any variety X over a field k , the group $H^1(X; G)$ classifies G -torsors over X .

Definition 471. Fix a G -torsor $Y \rightarrow X$. Then $p \in X(\mathbb{A}_k)_\bullet$ survives Y if and only if it lifts to a point in some twist $Y_\xi(\mathbb{A}_k)_\bullet$ for some $\xi \in H^1(k; G)$. We define $X(\mathbb{A}_k)^{\text{fab}}$ to be the set of points which survive all torsors for all finite abelian groups.

Remark 472. One can show that $X(k) \subseteq X(\mathbb{A}_k)^{\text{fab}}$.

And we should also review the Brauer–Manin obstruction. We need the following result from class field theory.

Theorem 473. Fix a local field k_v . Then there is a natural embedding $\text{inv}_v: \text{Br } k_v \rightarrow \mathbb{Q}/\mathbb{Z}$, which is surjective if v is finite, has image $\frac{1}{2}\mathbb{Z}/\mathbb{Z}$ if v is real, and has image 0 if v is complex.

Theorem 474. Fix a global field k . Then localizations provide the exact sequence

$$0 \rightarrow \text{Br } k \rightarrow \bigoplus_v \text{Br } k_v \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0,$$

where the last map sums inv_v .

Notation 475. We define a pairing $X(\mathbb{A}_k)_\bullet \times \text{Br } k \rightarrow \mathbb{Q}/\mathbb{Z}$ by

$$\langle x, b \rangle := \sum_v \text{inv}_v \text{ev}_{x_v}(b),$$

where ev_{x_v} is the torsor induced by the composite map $\text{Spec } k_v \rightarrow X \rightarrow \text{BG}$.

Remark 476. For any $x \in X(k)$, we have $\text{ev}_x: \text{Br } X \rightarrow \text{Br } k$, and we can see that $\langle x, b \rangle = 0$ for all b because $\text{ev}_x(b) \in \text{Br } k$.

Definition 477. We define the *Brauer–Manin set* to be $X(\mathbb{A}_k)^{\text{Br}}$ to be the set of $x \in X(\mathbb{A}_k)$ for which $\langle x, b \rangle = 0$ for all $b \in \text{Br } X$.

We can now restate our main theorem.

Theorem 478 (Stoll). Fix a smooth projective geometrically connected curve X over a global field k . Then $X(k)^{\text{fab}} = X(k)^{\text{Br}}$.

One of the inclusions does not need to use the curve hypothesis.

Theorem 479 (Skorobogatov). Fix a smooth projective geometrically connected curve X over a global field k . Then

$$X(\mathbb{A}_k)^{\text{Br}} \subseteq X(\mathbb{A}_k)^{\text{Br}_1} \subseteq X(\mathbb{A}_k)^{\text{fab}}.$$

Here, Br_1 is the kernel of the natural map $\text{Br } X \rightarrow \text{Br } X_{\bar{k}}$.

Proof. Choose $x \in X(\mathbb{A}_k)^{\text{Br}_1}$. For any finite abelian group scheme S over k and S -torsor $Y \rightarrow X$, we would like to find $\sigma \in H^1(k; S)$ such that x lifts to $Y_\sigma(\mathbb{A}_k)$. To this end, we define $M := \text{Hom}_{\bar{k}}(S, \mathbb{G}_m)$ to be the dual group, which is a finite discrete $\text{Gal}(\bar{k}/k)$ -module.

By spreading out, we note that there is a finite set place S of places for which M is unramified outside S , so we can find an integral model $\mathbb{S} := \text{Hom}(M, \mathbb{G}_m)$ of S . By further expanding T , we may assume that $Y \rightarrow X$ becomes an S -torsor $\mathcal{Y} \rightarrow \mathcal{X}$, and we may assume that $x_v \in \mathcal{X}(\mathcal{O}_v)$ for all $v \notin T$. Now, we can pull back $\mathcal{Y} \rightarrow \mathcal{X}$ along $x_v: \text{Spec } k_v \rightarrow \mathcal{X}$ to produce an S -torsor $\mathcal{Y}_{x_v} \rightarrow \text{Spec } k_v$. By construction, the collection of torsors $(\mathcal{Y}_{x_v})_v$ provide a class in

$$P^1(k; S) := \prod_{v \in T} H^1(k_v; S) \times \prod_{v \notin T} H^1(\mathcal{O}_v; S_v).$$

It is enough to show that the collection of torsors $(\mathcal{Y}_{x_v})_v$ come from a single class $\sigma \in H^1(k; S)$, for then we can simultaneously trivialize all these torsors by twisting by σ . To this end, recall from global duality that there is an exact sequence

$$H^1(k; S) \rightarrow P^1(k; S) \rightarrow H^1(k; M)^*,$$

where the last map is the composite of the sum of the local maps

$$H^1(k_v; S) \cong H^1(k_v; M)^* \xrightarrow{\text{res}^*} H^1(k; M)^*.$$

Thus, we would like to show that the image of $(\mathcal{Y}_{x_v})_v$ vanishes in $H^1(k; M)^*$, which is just some calculation with invariants. ■

31.2 Working with Brauer Groups

We will want some facts about the Brauer group.

Proposition 480. Fix a smooth projective geometrically connected curve X over a field k . Then there is an exact sequence

$$\text{Br } k \rightarrow \text{Br}_1 k \rightarrow \text{Br}_1 X \rightarrow H^1(k; \text{Pic } X) \rightarrow 0.$$

Proof. This follows from the spectral sequence for the composition of taking Galois invariants and étale cohomology. ■

Here is some more notation.

Notation 481. Fix a smooth projective geometrically connected curve X over a field k . Then there is a canonical map $H^1(k; \text{Pic}_X^\circ) \rightarrow H^1(k; \text{Pic}_X)$. We define $\text{Br}_{1/2} X$ to be the set of $\sigma \in \text{Br}_1 X$ whose image in $H^1(k; \text{Pic}_X)$ is in the image of $H^1(k; \text{Pic}_X^\circ)$. We define $\text{Br}_{1/2, n} X$ to be the same thing but now asking to be in the image of $H^1(k; \text{Pic}_X^\circ)[n]$.

Let's now give the other inclusion.

Theorem 482. Fix a curve X . Then $X(\mathbb{A}_k)^{n,ab} \subseteq X(\mathbb{A}_k)^{Br_{1/2},n}$, where the former n means that we only consider n -torsion abelian groups.

Proof. For convenience, we assume that Pic_C^1 admits a rational point, and we may as well assume that X has positive genus.

Choose $x \in X(\mathbb{A}_k)^{n,ab}$. Then for any $b \in Br_{1/2,n} X$, we must show that $\langle x, b \rangle = 0$. By definition of b , the image b' of b in $H^1(k; \text{Pic}_X)$ comes from some $b'' \in H^1(k; \text{Pic}_C^\circ)[n]$ mapping to b . On the other hand, x maps to an element $\varphi(x)$ of $\text{Pic}_X^\circ(\mathbb{A}_k)^{n,ab}$. Thus, the torsor

$$[n]: \text{Pic}_X^\circ \rightarrow \text{Pic}_X^\circ$$

admits a twist W of this torsor so that $\varphi(x)$ lifts to W . We now let $\tilde{C}$ be the pullback fitting into the diagram

$$\begin{array}{ccc} \tilde{C} & \longrightarrow & W \\ \pi \downarrow & & \downarrow \\ C & \xrightarrow{\varphi} & \text{Pic}_X^\circ \end{array}$$

so that π is some covering. This gives a commutative diagram on Pic° which looks like

$$\begin{array}{ccc} \tilde{C} & \longleftarrow & \text{Pic}_W^\circ \\ \pi^* \uparrow & & \uparrow n \\ \text{Pic}_C^\circ & \xlongequal{\quad} & \text{Pic}_C^\circ \end{array}$$

but now $\text{Pic}_C^\circ \cong \text{Pic}_W^\circ$ (certainly they are isomorphic over the algebraic closure, and changing the choice of isomorphism only adjusts by a translation, which vanishes on Pic°). This gives a commutative diagram on H^1 , which can be used to show that $\langle x, b \rangle = 0$. ■

Thus, we have inclusions

$$C(\mathbb{A}_k)^{Br} \subseteq C(\mathbb{A}_k)^{Br_1} \subseteq C(\mathbb{A}_k)^{fab} \subseteq C(\mathbb{A}_k)^{Br_{1/2}}.$$

To complete the proof, we recall the following.

Proposition 483 (Tsen). Fix a curve C over an algebraically closed field k . Then $Br C = 0$.

Proof. The main point is to check that $k(C)$ is C_1 , meaning that all homogeneous polynomials of degree d in n variables with $n > d$ admit a nontrivial root. Let's explain why this is enough: we must show that every division algebra D over $k(C)$ equals k . Well, with $\dim D = n^2$, the reduced norm has degree n , so the reduced norm has a nonzero root whenever $n^2 > n$, so $n = 1$ is the only possibility, meaning $D = k$.

It remains to show that $k(C)$ is C_1 . Choose F of degree d over $k(C)$ in n variables with $n > d$. Now, for any choice of an ample divisor H , we may view F as a map

$$\Gamma(C; \mathcal{O}(mH))^n \rightarrow \Gamma(C; \mathcal{O}(dmH + D)),$$

where D has the poles in the coefficients of F . By Riemann–Roch, the left-hand side has dimension about mn , and the right-hand side has dimension about dm , so for m large enough, the above map has a kernel. Both sides include into $k(C)$, so we have found a zero of F in $k(C)$. ■

Thus, $Br C = Br_1 C$, so we will be done if we can show that $Br_1 X = Br_{1/2} X$. To this end, we note that taking Galois cohomology of the short exact sequence

$$1 \rightarrow \text{Pic}_C^\circ \rightarrow \text{Pic}_C \xrightarrow{\deg} \mathbb{Z} \rightarrow 0,$$

which gives

$$H^1(k; \text{Pic}_C^\circ) \rightarrow H^1(k; \text{Pic}_C) \rightarrow H^1(k; \mathbb{Z}).$$

However, $H^1(k; \mathbb{Z}) = 0$, so we see that $Br_{1/2} X = Br_1 X$.

32 May 7: Descent Obstruction and Modularity

This talk was given by Mohit Hulse at MIT for STAGE.

32.1 Eichler–Shimura Theory

We will need to know something about elliptic curves.

Definition 484 (Tate module). Fix an elliptic curve E over a field K , and fix a prime ℓ not divisible by $\text{char } K$. Then its *Tate module* is the inverse limit

$$T_\ell E := \lim E[\ell^\bullet](\overline{K}).$$

We let $\rho_{\ell,E}: \text{Gal}(\overline{K}/K) \rightarrow \text{GL}(T_\ell E)$ denote the induced Galois representation. We may also write $\bar{\rho}_{\ell,E}$ for the action on $E[\ell]$.

Remark 485. In characteristic zero, one always has that $E[n](\overline{K}) \cong (\mathbb{Z}/n\mathbb{Z})^2$, which one can see by base-changing to $\mathbb{C}$. Thus, $T_\ell E$ is a free module over $\mathbb{Z}_\ell$ of rank 2.

It turns out that the Tate module can read certain properties of the elliptic curve.

Notation 486. Fix an elliptic curve E over a field K . For a prime p , we define

$$a_p := 1 - \#E(\mathbb{F}_p) + p.$$

Remark 487. By the Lefschetz trace formula (or even earlier), one has $a_p = \text{tr}(\text{Frob}_p; T_\ell E)$, where p has good reduction and $p \neq \ell$.

Remark 488. It turns out that $T_\ell E$ as a Galois representation uniquely determines the isogeny class of E . This is about as hard as the Tate conjecture for abelian varieties.

We will also need to know something about modular forms.

Definition 489 (cusp form). Fix a “level” positive integer N , a “weight” positive integer $k \geq 2$, and a character $\varepsilon: (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \overline{\mathbb{Z}}^\times$. Then a *cusp form* is a holomorphic function $f: \mathcal{H} \rightarrow \mathbb{C}$ admitting a q -expansion

$$f(z) = \sum_{n \geq 1} q_{n,f} q^n,$$

where $q := e^{2\pi iz}$, and satisfying

$$f\left(\frac{az+b}{cz+d}\right) = \varepsilon(d)(cz+d)^k f(z)$$

whenever $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is in $\Gamma_0(N)$. We also require that f vanishes at the other cusps of $\Gamma_0(N) \backslash \mathcal{H}$. The space is denoted by $S(N, k, \varepsilon)$.

It may be convenient to take moduli.

Definition 490. Fix a prime ℓ . Then a *mod- ℓ cusp form* is a power series of the form

$$f(q) := \sum_n a_{n,f} q^n$$

where $a_{n,f} \in \overline{\mathbb{F}}_\ell$ and lifts to a cusp form with coefficients in $\overline{\mathbb{Z}}$. The space is denoted by $\overline{S}(N, k, \varepsilon, \ell)$.

Lifting to $\overline{\mathbb{Z}}$ is a rather bizarre condition because modular forms appear to be analytic objects. One can get coefficients in $\overline{\mathbb{Z}}$ by taking Hecke eigenforms.

Definition 491 (Hecke operators). There is an algebra $\mathbb{T}$ of commuting endomorphisms given by $\{T_p\}_{p \text{ prime}}$ acting on $\overline{S}(N, k, \varepsilon, \ell)$. If $p \nmid N$, then the Hecke operator lifts to $S(N, k, \varepsilon)$.

Remark 492. If f is a Hecke eigenform, then it can be normalized so that $a_{1,f} = 1$. It then turns out that $T_p f = a_{p,f} f$, and one finds that these are all elements of $\overline{\mathbb{Z}}$, so we produce a mod- ℓ modular form. In fact, mod- ℓ cusp forms $\overline{f}$ with $a_{1,\overline{f}} \equiv 1$ can all be lifted to

The last ingredient we will need is something about Eichler–Shimura.

Definition 493 (modular curve). The *modular curve* is some curve $X_0(N)$ defined over $\mathbb{Q}$ which is isomorphic to $\mathcal{H}/\Gamma_0(N)$ over $\mathbb{C}$. It basically parameterizes elliptic curves with a choice of cyclic subgroup of size N .

Remark 494. By definition, a cusp form f of weight two induces a holomorphic differential form on $X_0(N)$.

Notation 495. Set $J_0(N) := \text{Jac } X_0(N)$. For any Hecke eigenform f , let $\lambda_f: \mathbb{T} \rightarrow \mathbb{C}$ denote the eigenvalue of f , and we define

$$A_f := J_0(N)/(\ker \lambda_f).$$

Here is the main theorem.

Theorem 496 (Eichler–Shimura). Fix a Hecke eigenform f , and let $\lambda_f: \mathbb{G} \rightarrow \mathbb{C}$ be the eigenvalue. If λ_f outputs to $\mathbb{Z}$, then A_f is an elliptic curve. In fact,

$$\text{tr}(\text{Frob}_p; T_\ell A_f) = \lambda_f(T_p)$$

for all primes p of good reduction, distinct from ℓ .

Remark 497. It turns out that $\overline{\rho}_{\ell,\overline{f}}$ only depends on $\overline{f}$.

32.2 Serre's Conjecture

The following was conjectured by Serre before 1987 and proven only in the 2000s.

Definition 498 (odd). Fix a modular Galois representation $\overline{\rho}_\ell: \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}_n(\mathbb{F}_\ell)$. Let $c \subseteq \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ be the conjugacy class of complex conjugation. Then we say that $\overline{\rho}_\ell$ is *odd* if and only if $\det \overline{\rho}_\ell(c) = -1$.

Example 499. If f is a Hecke eigenform, then $\bar{\rho}_{f,\ell}$ is odd. Indeed, for an elliptic curve $\mathbb{C}/\Lambda$, complex conjugation acts with determinant -1 on the lattice $\Lambda \subseteq \mathbb{C}$, so it acts by -1 on the torsion and thus also on the Tate module. Alternatively,

Theorem 500 (Khare–Winterberger). Fix a modular Galois representation $\bar{\rho}_\ell: \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}_2(\mathbb{F}_\ell)$. If $\bar{\rho}_\ell$ is absolutely irreducible and odd, then there is a mod- ℓ Hecke eigenform $\bar{f}$ such that $\bar{\rho}_\ell = \bar{\rho}_{\bar{f},\ell}$.

We won't prove this; it's very hard. Here is an application.

Corollary 501. Fix a prime $\ell \geq 5$. Then

Proof. Suppose (a, b, c) is a solution. Define the elliptic curve E by

$$y^2 = x(x + a^\ell)(x - b^\ell).$$

We now consider the Galois representation $\bar{\rho}_{E,\ell}$. Its discriminant is some power of 2 times $a^\ell b^\ell c^\ell$, but any primes appearing in $a^\ell b^\ell c^\ell$ needs to appear with multiplicity of ℓ . Work of Serre is able to remove these extra primes from the conductor, so $N(\rho_{\ell,E}) = 2$ and $k = 2$, but there are no modular forms with these criteria. ■

For our main application, we will need the following definition.

Definition 502 (locally elliptic family). Fix a finite set S of primes. A *locally elliptic family of Galois representations* is a tuple $\{\rho_\ell\}_\ell$ of Galois representations $\rho_\ell: \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}_2(\mathbb{Z}_\ell)$ such that each $p \notin S$ admits an elliptic curve $E_p/\mathbb{Q}_p$ with good reduction and for which

$$\rho_\ell|_{\text{Gal}(\bar{\mathbb{Q}}_p/\mathbb{Q}_p)} \cong \rho_{\ell,E_p}$$

for all ℓ .

Theorem 503. Fix a finite set S of primes and a locally elliptic family of Galois representations $\{\rho_\ell\}_\ell$. Then there is an elliptic curve E over $\mathbb{Q}$ such that $\rho_{E,\ell} = \rho_\ell$ for all ℓ .

Proof. For each prime ℓ , let N_ℓ , k_ℓ , and ε_ℓ be the usual invariants for $\bar{\rho}_\ell$.

1. Let's start by showing that $k_\ell = 2$ for $\ell \notin S$. Well, for $\ell \notin S$, Serre's work shows that k_ℓ only depends on $\bar{\rho}_\ell|_{\text{Gal}(\bar{\mathbb{Q}}_\ell/\mathbb{Q}_\ell)}$, but this is $\bar{\rho}_{\ell,E_\ell}$, which then has $k_\ell = 2$ already.
2. We claim there is some N for which there is a positive density set of $\ell \notin S$ such that $N_\ell \mid N$. This is a rather lengthy calculation. Having good reduction everywhere implies that N_ℓ is always supported in S . Accordingly, fix some $p \in S$, and we will bound the exponent of p dividing N_ℓ ; we would like to show that it is uniformly bounded in ℓ and almost always zero.
Well, define G_1 to be the image of ρ_ℓ along the image of wild inertia $I_p^1 \subseteq \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$. By work of Serre, the exponent of p dividing N_ℓ is bounded in terms of $\#G_1$, but $G_1 \subseteq \text{GL}_2(\mathbb{F}_\ell)$, which is some function of ℓ . Thus, we are hunting for p -subgroups of $\text{GL}_2(\mathbb{F}_\ell)$, but if p fails to divide $\#\text{GL}_2(\mathbb{F}_\ell)$, then G_1 must be trivial, and we get our uniform bound. We can ensure that $p \nmid \#\text{GL}_2(\mathbb{F}_\ell)$ by adding some congruence conditions to ℓ , so we are done!
3. For all but finitely many ℓ , we claim that $\bar{\rho}_\ell$ is irreducible. Let's take $\ell \notin S$. Well, $\bar{\rho}_\ell$ is irreducible if and only if E_ℓ has supersingular good reduction because we need to kill ℓ -torsion to get irreducibility.

Suppose that $\bar{\rho}_\ell$ reduces for infinitely many primes ℓ . Then $\bar{\rho}_\ell$ divides up into two characters, and by the Weil pairing, these two characters look like $\psi \oplus \chi\psi^{-1}$, where χ is the cyclotomic character. Thus,

for $p \equiv 1 \pmod{N}$ where N is the conductor, we can find that a_{p,E_ℓ} can be calculated as the trace of the Frobenius, which is $p + 1 \pmod{\ell}$ (because ψ is trivial on Frob_p). This cannot vanish for infinitely many ℓ , so we are done.

4. Now, we can use Serre's conjecture on the previous two steps to find a Hecke eigenform f with $a_{p,\bar{f}} = a_{p,E_p}$ for $p \notin S$, which will complete the proof.

Indeed, Serre's conjecture gives f_ℓ for infinitely many primes ℓ for which $a_{p,E_\ell} = a_{p,\rho_\ell}$. Some f appears infinitely often, and this is enough to conclude by making an argument with the Euler product. ■

32.3 Descent Obstruction

We now apply the descent obstruction.

Lemma 504. Fix a curve X over $\mathbb{Q}$ admitting a smooth model $\mathcal{X}_S$. Further, choose a Galois cover $\pi: Y \rightarrow X$. Then

$$\mathcal{X}(\mathbb{Z}_S) = \bigcup_{\pi'} \pi'(\mathcal{Y}'(\mathbb{Z}_S)),$$

where $\pi': Y' \rightarrow X$ varies over finitely many twists of Y .

Proof. This is the usual descent. One needs something about Chevalley–Warning to explain why there are only finitely many twists. ■

We can now define the descent sets $\mathcal{X}(\widehat{\mathbb{Z}}_S)^{\text{fet}}$. We remark that one can even require that the various representatives are all compatible.

33 May 12: Galois Orbit Lower Bounds

This talk was given by David Urbanik at MIT for the MIT number theory seminar.

33.1 Boundedness of Special Points

Throughout, we fix a smooth projective family $f: X \rightarrow S$ of algebraic varieties over a number field K ; in fact, in our applications, S will always be a curve, and X will be a family of surfaces. We would like to understand the set of points $\mathcal{S} \subseteq S(\overline{K})$ over which the fibers have some extra structure.

Example 505. Let $X \rightarrow S$ be the Legendre family $y^2 = x(x-1)(x-\lambda)$ of elliptic curves over $S := \mathbb{Z}[\lambda, 1/\lambda, 1/(\lambda-1)]$. Then we could let $\mathcal{S}$ be the locus where the elliptic curve has complex multiplication.

Example 506. Let $f: X \rightarrow S$ be a family of projective surfaces over a curve S . One could consider the locus $\mathcal{S}_k$ of points $\bar{s}$ where the fiber has

$$\text{rank NS}(X_{\bar{\eta}}) = \text{rank NS}(X_{\bar{s}}) - k.$$

Namely, we are studying exceptional points with k “extra” line bundles. When this is a family of K3 surfaces, it is known that $\mathcal{S}_1$ is dense and even equidistributed (by viewing these as variations of Hodge structures), and one expects that $\mathcal{S}_2$ is usually finite.

The fact that K is a number field does not play a role in the statements, but there are other methods available if K is larger.

Anyway, in light of the possible density of $\mathcal{S}$, our goal is to find positive constants a and b for which

$$h(s) \leq a[K(s) : K]^b$$

for all $s \in S$, where h is the log Weil height. This basically means that S may be infinite, but the degree of the field has to grow.

To begin, let's fix a semistable compactification

$$\begin{array}{ccc} X & \hookrightarrow & \overline{X} \\ f \downarrow & & \downarrow \overline{f} \\ S & \hookrightarrow & \overline{S} \end{array}$$

of $X \rightarrow S$, meaning $\overline{f}$ is projective, $X \subseteq \overline{X}$ is open, and each $x_0 \in \overline{X}$ and $s_0 := \overline{f}(x_0)$ grants $\overline{f}$ local formal coordinates that look like monomials. Namely, if the local coordinates of X are (x_1, x_2, x_3) , then $\overline{f}$ will look like x_1 or x_1x_2 or $x_1x_2x_3$, depending on the normal crossings divisor of the semistable compactification.

There is also some local monodromy going on: for each loop y around some $s_0 \in \overline{S}$, there is an induced unipotent operator T_y on $H^2(X_{s_0}; \mathbb{Q})$, and the corresponding nilpotent operator $N := \log T_y$, and it has index in $\{1, 2, 3\}$. The larger indexes are really only possible if the normal crossing divisors is non-smooth.

Theorem 507. Suppose that f has large monodromy. Then there are positive constants a and b for which

$$h(s) \leq a[K(s) : K]^b$$

for all $s \in S$ under one of the following assumptions.

- (a) There is a degeneration with the large monodromy of index $k = 3$ and $S = S_2$.
- (b) There is a degeneration with the large monodromy of index $k = 2$ and $S = S_\ell$ for any ℓ .

Corollary 508. If $f: X \rightarrow S$ is a family of K3 surfaces, then there is a degeneration with large monodromy of index $k \in \{2, 3\}$, so there are positive constants a and b for which

$$h(s) \leq a[K(s) : K]^b$$

for all $s \in S_{5-k}$.

33.2 G -functions

Our main arithmetic input comes from G -functions.

Definition 509. Fix a number field K . A G -function is a formal power series $G \in K[[s]]$ satisfying the following.

- (a) Differential equation: the power series G satisfies a linear differential equation

$$a_\ell \frac{d^\ell G}{ds^\ell} + \cdots + a_1 \frac{dG}{ds} = 0,$$

where $a_i \in K(s)$ for each i .

- (b) Integrality: if $G = \sum_{i \geq 0} b_i s^i$, then there are integers $\{c_i\}$ such that $\{c_i b_i\}_{i \geq 0}$ is integral, and $|c_i| \leq \kappa^i$ for some fixed κ .

We are interested in the relationship between G -functions.

Definition 510 (global relation). Fix a G -function G over a number field K . Then a place v of K is *relevant* for $\xi \in \overline{K}$ and G if and only if $|\xi|_v$ is less than the v -adic radius of convergence of G . Then a polynomial P is a *global relation* at a fixed $\xi \in \overline{K}$ of G -functions $\{G_1, \dots, G_m\}$ if and only if

$$P(G_1(\xi), \dots, G_m(\xi)) = 0$$

for all relevant places v .

Remark 511. The G -functions tend to have radius of convergence 1 by the integrality hypotheses, so one expects to have only finitely many relevant places.

Theorem 512 (Bombieri). Fix G -functions $\{G_1, \dots, G_m\}$ over a number field K . There are positive constants a and b such that all $\xi \in \overline{K}$ admitting a nontrivial global relation P_ξ have

$$h(\xi) \leq a(\log P_\xi)^b.$$

In our application, G -functions arise as follows; using the local coordinates of our compactification, one can build a diagram

$$\begin{array}{ccccc} X & \longrightarrow & \overline{X} & \longleftarrow & \mathrm{Spf} K[[x_1, x_2, x_3]] \\ f \downarrow & & \downarrow \overline{f} & & \downarrow \\ S & \longrightarrow & \overline{S} & \longleftarrow & \mathrm{Spf} K[[s]] \end{array}$$

and then try to push algebraic forms from $\overline{X}$ down to $\mathrm{Spf} K[[s]]$.

34 May 14: Optimal Bounds on Sums of Arithmetic Functions

This talk was given by Harald Helfgott at MIT for the MIT number theory seminar.

34.1 Some Explicit Bounds

For today, we let $a_\bullet : \mathbb{Z}^+ \rightarrow \mathbb{C}$ be an arithmetic function, and we are interested in bounding the partial sums

$$\sum_{n \leq x} a_n.$$

Usually, our arithmetic functions will have Dirichlet series

$$A(s) := \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

with some meromorphic continuation.

Example 513. With

$$a_n := \begin{cases} \log p & \text{if } n = p^k \text{ for some prime } p, \\ 0 & \text{otherwise.} \end{cases}$$

Then $A(s) = -\zeta'(s)/\zeta(s)$. Zero-free regions of $\zeta(s)$ prove the Prime number theorem. In the sequel, we will denote $\psi(x) := A(x)$.

Example 514. With $a_n := \mu(n)$, we have $A(s) = 1/\zeta(s)$. In the sequel, we will denote $M(x) := A(X)$. The Prime number theorem is equivalent to $M(x) = o(x)$.

The bounds on $|\psi(x) - x|$ are much better than those on $|M(x)|$. Here are some reasons why.

1. Estimating $|\psi(x) - x|$ basically needs to know where the zeroes are (to control $-\zeta'/\zeta$), but estimating $M(x)$ requires one need to know some residues (to control $1/\zeta$). If zeroes are close together (which is difficult to control), then such residues can explode.
2. It is difficult to only use finite information about residues in order to produce bounds on $M(x)$.

The main result of the present talk is to explain how to use finite informatino about residues to produce bounds.